

SWE202 - Software Modeling and Design Project

Topic:Property Rental System

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Property Rental System

Requirements Specification Version 3.0

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Use this Requirements Specification template to document the requirements for your product or service, including priority and approval. Tailor the specification to suit your project, organizing the applicable sections in a way that works best, and use the checklist to record the decisions about what is applicable and what isn't.

The format of the requirements depends on what works best for your project.

This document contains instructions and examples which are for the benefit of the person writing the document and should be removed before the document is finalized.

To regenerate the TOC, select all (CTL-A) and press F9.

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1. Executive Summary

1.1 Project Overview

The **Property Rentals System** is a software platform designed to simplify and manage the process of renting properties between tenants, property owners, and administrative staff. The system provides a centralized environment where property owners can list rental properties, tenants can search and book accommodations, and administrators can monitor and manage overall system operations.

The platform supports several core services including:

- Property listing and management
- Property search and filtering
- Rental request and booking management
- Online payment processing
- Messaging and notifications
- Complaint and dispute management
- Reporting and administrative monitoring

The intended audience of the system includes:

- **Tenants** searching for rental accommodations
- **Property Owners** managing rental properties
- **Administrators and Managers** supervising operations and disputes
- **Financiers** handling transactions and refunds
- **HR Staff** managing complaints and user support

The main goal of the system is to improve efficiency, transparency, accessibility, and communication throughout the property rental process by digitizing rental operations and allowing users to access services online from anywhere.

2. Product/Service Description

The Property Rentals System is designed as a multi-user rental management platform that connects tenants, property owners, and administrative personnel through a single integrated system. The platform automates major rental activities such as property publishing, booking requests, payment processing, communication, and issue resolution.

The development of this system is motivated by the increasing need for efficient digital rental management solutions that reduce manual paperwork, improve communication, and provide better accessibility for users. Traditional rental management processes often involve delays, fragmented communication, and difficulties in maintaining accurate records. This system addresses those challenges through centralized data management and automated workflows.

Several general factors influence the design and requirements of the system:

- The need for secure authentication and protection of user and payment data
- Support for multiple user roles with different permissions
- Integration with external payment and notification services
- Efficient handling of large numbers of property listings and concurrent users
- Easy-to-use interfaces for both technical and non-technical users
- Real-time communication and booking updates
- Scalable architecture to support future growth and additional services

The system is intended to operate primarily as a web-based application, although the architecture can later support mobile application integration.

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2.1 Product Context

The Property Rentals System is a web-based rental management platform that operates as a centralized system for managing rental activities between tenants, property owners, administrators, managers, financiers, and HR personnel. The system is primarily designed as a standalone application, but it also interfaces with external services such as payment and notification systems. In the future, the platform can be extended to support mobile application integration in addition to the web-based environment.

The system provides a centralized database that stores and manages all important information related to:

- User accounts and authentication
- Property listings and availability
- Rental requests and bookings
- Payment transactions and refunds
- Messages and notifications
- Complaints, disputes, and issue resolutions
- Reviews, ratings, and reports

The centralized structure ensures that all modules and users work with consistent and updated information across the platform.

The Property Rentals System interacts with several external systems and internal subsystems:

External Systems

- **Payment System**
 - Processes rental payments, refunds, and transaction confirmations.
 - Used during booking payment operations and financial management.
- **Notification System**
 - Sends booking confirmations, approval/rejection alerts, reminders, payment updates, and complaint notifications to users.

Internal System Modules

The platform is divided into multiple functional modules that work together:

- Property Management Module
- Booking & Rental Request Module
- Messaging & Communication Module
- Payment Management Module
- Complaint & Issue Management Module
- User & Role Management Module
- Reporting & Monitoring Module
- Notification Management Module

Each module communicates through the centralized system architecture and shares data through the common database.

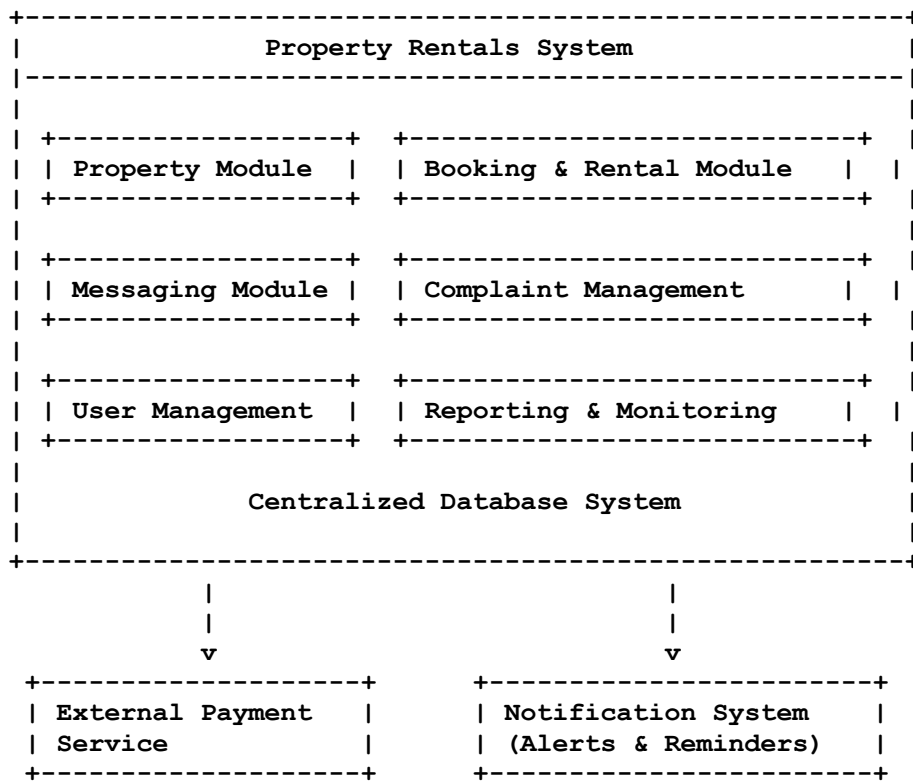
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The platform supports multiple user roles, each with different permissions and responsibilities:

- Tenant
- Property Owner
- Administrator
- Manager
- Financier
- HR Staff

These users interact with the system through role-specific functionalities such as booking management, listing management, payment handling, complaint resolution, and system monitoring.

The overall system context can be represented as follows:



The system architecture also incorporates several software design patterns to improve maintainability and communication between components:

- The **Facade Pattern** simplifies booking operations by coordinating rental requests, bookings, payments, and notifications through a unified interface.
- The **Observer Pattern** automatically sends notifications when booking or payment statuses change.
- The **Adapter Pattern** allows the system to communicate with external payment providers without modifying the internal payment logic.

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2.2 User Characteristics

The Property Rental System will be used by several categories of users with different responsibilities and technical abilities. The product should therefore be designed with a clear, user-friendly interface, role-based access, secure login, and simple navigation so both technical and non-technical users can use it effectively.

User Type	Category	Experience	Technical Expertise	General Characteristics
Tenant	Other / Customer	May have basic experience using websites or mobile apps	Basic	Needs a simple interface to search properties, view details, send rental requests, make payments, and communicate with owners.
Property Owner	Other / Service Provider	May have experience managing rental properties	Basic to intermediate	Needs tools to add, edit, remove, and manage listings, upload images, respond to requests, and view booking history.
Administrator	Staff	Experienced in managing system operations	Intermediate to advanced	Responsible for managing users, listings, disputes, reports, and platform control.
Manager	Staff	Experienced in supervision and coordination	Intermediate	Oversees rental requests, complaints, reports, discounts, and coordination between users and administrators.
Financier	Staff	Experienced in payment and transaction handling	Intermediate	Handles payment monitoring, refunds, transaction issues, and financial reports.
HR Staff	Staff	Experienced in user support and communication	Basic to intermediate	Manages complaints, reminders, user satisfaction, and issue-resolution records.

2.3 Assumptions

The following assumptions affect the requirements of the Property Rentals System:

- Users will have access to a device such as a computer, tablet, or smartphone.
- Users will have a stable internet connection to access the web-based system.
- Users have basic knowledge of using websites or mobile applications.
- Property owners will provide accurate property information, images, prices, and availability.
- Tenants will provide correct personal, booking, and payment information.
- The external payment system will be available to process rental payments and refunds.

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- The notification system will be available to send alerts, reminders, and booking updates.
- The system will use a centralized database to store users, properties, bookings, payments, messages, complaints, and reports.
- Administrators, managers, financiers, and HR staff will have the required permissions to perform their tasks.
- Secure authentication and authorization will be used before users access role-specific functions.

2.4 Constraints and Dependencies

The following constraints and dependencies affect the system requirements:

Constraints

- The system must protect user information, passwords, booking records, and payment data.
- Access must be controlled using role-based permissions.
- Only authorized users should access sensitive features such as user management, reports, refunds, and dispute handling.
- The system should maintain logs or records for important actions such as payments, refunds, complaints, booking approvals, and account changes.
- The system must support multiple users at the same time without major performance issues.
- The interface must remain simple enough for users with basic technical skills.
- Payment processing must be secure and reliable.
- The application depends on accurate and updated property availability information.
- The system must prevent invalid actions, such as deleting listings with active bookings or submitting reviews without completed rentals.

Dependencies

- The booking module depends on the property search and property listing modules.
- The payment module depends on approved bookings before payments can be processed.
- The refund process depends on payment and booking cancellation records.
- The notification module depends on booking, payment, complaint, and request status changes.
- The complaint management module depends on user accounts, property records, and issue-resolution records.
- The reporting module depends on stored data from users, bookings, payments, complaints, and listings.
- The system depends on the external payment service for transaction processing.
- The system depends on the notification service for sending alerts and reminders.
- User role management must be completed before role-specific modules can be fully implemented.

3. Requirements

The Property Rentals System (PRS) is a web-based application designed to simplify and digitalize the process of managing rental properties between tenants, property owners, administrators, managers, financiers, and HR staff. The system allows users to search and rent properties, manage bookings, process payments, handle complaints, exchange messages, and monitor rental operations efficiently.

The platform supports role-based access control, ensuring that each type of user can only access the functions relevant to their responsibilities. The system also integrates payment and notification services to improve communication, transaction management, and operational efficiency.

• System Inputs

Input	Description
User login credentials	Username/email and password entered during authentication
Property search queries	Keywords, location, price range, or property type used to search properties
Rental requests	Requests submitted by tenants to rent a property
Property listing information	Property details entered by owners including title, description, price, location, and availability
Property images	Photos uploaded by property owners for listings
Payment information	Payment details entered by tenants for booking payments
Messages and conversations	Messages exchanged between tenants and property owners
Reviews and ratings	Feedback and ratings submitted by tenants after rentals
Complaint records	Complaints submitted by users regarding issues or disputes
Refund requests	Refund information submitted for canceled bookings
Promotion details	Discount and promotional information created by managers
Notification data	Alerts and reminders generated by the system

• System Outputs

Output	Description
Search results	List of properties matching the search criteria
Property details page	Complete information about a selected property
Booking confirmation	Confirmation message for approved bookings
Rental request status	Approval, rejection, or pending status notifications
Payment confirmation	Confirmation of successful rental payments
Refund confirmation	Confirmation message for approved refunds
Property listing updates	Updated property availability and listing information
User notifications	Alerts for bookings, payments, complaints, and reminders
Messages and chat history	Display of communication between users
Booking history	Record of completed and active bookings
Reviews and ratings display	Published tenant reviews and property ratings
Complaint resolution reports	Status and resolution details of complaints
Management reports	Reports related to rentals, payments, and system activities
Financial reports	Payment, refund, and transaction summaries

3.1 Functional Requirements

Priority Definitions

The following definitions are intended as a guideline to prioritize requirements.

- **Priority 1** – The requirement is a “must have” as outlined by policy/law
- **Priority 2** – The requirement is needed for improved processing, and the fulfillment of the requirement will create immediate benefits
- **Priority 3** – The requirement is a “nice to have” which may include new functionality. It may be helpful to phrase the requirement in terms of its priority, e.g., "The value of the employee status sent to DIS must be either A or I" or "It would be nice if the application warned the user that the expiration date was 3 business days away". Another approach would be to group requirements by priority category.

A good requirement is:

- **Correct**
- **Unambiguous** (all statements have exactly one interpretation)
- **Complete** (where TBDs are absolutely necessary, document why the information is unknown, who is responsible for resolution, and the deadline)
- **Consistent**
- **Ranked for importance and/or stability.**
- **Verifiable** (avoid soft descriptions like “works well”, “is user friendly”; use concrete terms and specify measurable quantities)
- **Modifiable** (evolve the Requirements Specification only via a formal change process, preserving a complete audit trail of changes)
- **Does not specify any particular design**
- **Traceable** (cross-reference with source documents and spawned documents).

Property Rental System Requirement Specification

Req#	Requirement	Comments	Priority	Date Rvwd	Done By
BR_PRS_1	The tenant shall be able to update personal information because profile information must remain accurate and updated.	Business Process = User Profile Management	2	18/03/2026	Anisa Domi
BR_PRS_2	The tenant shall be able to search properties because tenants need to find suitable rental accommodations.	Business Process = Property Search	2	18/03/2026	Anisa Domi
BR_PRS_3	The tenant shall be able to filter properties because search results need to be narrowed based on preferences.	Business Process = Property Filtering	2	18/03/2026	Anisa Domi
BR_PRS_4	The tenant shall be able to view property details and booking history because tenants need complete rental information and history records.	Business Process = Property Viewing & Booking History	2	18/03/2026	Anisa Domi
BR_PRS_5	The tenant shall be able to send rental requests and messages to property owners because communication and booking requests are required for rentals.	Business Process = Rental Request Management	1	18/03/2026	Anisa Domi
BR_PRS_6	The tenant shall be able to cancel rental requests because tenants may change booking decisions before approval.	Business Process = Rental Request Cancellation	2	18/03/2026	Anisa Domi
BR_PRS_7	The tenant shall be able to receive messages BR_PRS_om property owners because communication between users must be supported.	Business Process = Messaging System	2	18/03/2026	Haidi Toto
BR_PRS_8	The tenant shall be able to submit rental payments because approved bookings require payment processing.	Business Process = Payment Processing	1	18/03/2026	Haidi Toto
BR_PRS_9	The tenant shall be able to save favorite properties because users may want to review properties later.	Business Process = Favorites Management	3	18/03/2026	Haidi Toto
BR_PRS_10	The tenant shall be able to review and rate properties because feedback improves transparency and user trust.	Business Process = Review & Rating System	3	18/03/2026	Haidi Toto
BR_PRS_11	The property owner shall be able to update profile information and property availability because listing information must remain accurate.	Business Process = Property Owner Profile Management	2	18/03/2026	Haidi Toto
BR_PRS_12	The property owner shall be able to add property listings because rental properties need to be published in the system.	Business Process = Property Listing Management	1	18/03/2026	Haidi Toto
BR_PRS_13	The property owner shall be able to upload property images because tenants need visual information about properties.	Business Process = Property Media Management	2	18/03/2026	Ledio Tafciu
BR_PRS_14	The property owner shall be able to edit property listings because property details may change over time.	Business Process = Property Listing Management	2	18/03/2026	Ledio Tafciu

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BR_PRS_15	The property owner shall be able to delete property listings because unavailable or invalid listings must be removed.	Business Process = Property Listing Management	2	18/03/2026	Ledio Tafciu
BR_PRS_16	The property owner shall be able to view rental requests because owners need to manage booking requests.	Business Process = Rental Request Monitoring	1	18/03/2026	Ledio Tafciu
BR_PRS_17	The property owner shall be able to accept or reject rental requests because booking approvals are required before rentals proceed.	Business Process = Rental Approval Workflow	1	18/03/2026	Ledio Tafciu
BR_PRS_18	The property owner shall be able to send messages to tenants because communication between both parties must be supported.	Business Process = Messaging System	2	18/03/2026	Ledio Tafciu
BR_PRS_19	The property owner shall be able to view booking history because owners need access to previous rental records.	Business Process = Booking History Management	2	18/03/2026	Desard Lybeshari
BR_PRS_20	The property owner shall be able to mark properties as unavailable or rented because listing availability must remain accurate.	Business Process = Property Availability Management	1	18/03/2026	Desard Lybeshari
BR_PRS_21	The administrator shall be able to edit user accounts because user information and permissions may require updates.	Business Process = User Account Administration	1	18/03/2026	Desard Lybeshari
BR_PRS_22	The administrator shall be able to delete user accounts because invalid or inappropriate accounts must be removed.	Business Process = User Account Administration	1	18/03/2026	Desard Lybeshari
BR_PRS_23	The administrator shall be able to monitor rental transactions because financial and booking activities require supervision.	Business Process = Transaction Monitoring	1	18/03/2026	Desard Lybeshari
BR_PRS_24	The administrator shall be able to view system reports and property listings because system operations must be monitored.	Business Process = Reporting & Monitoring	2	18/03/2026	Desard Lybeshari
BR_PRS_25	The administrator shall be able to remove inappropriate listings because platform content must comply with system policies.	Business Process = Content Moderation	1	18/03/2026	Matias Bali
BR_PRS_26	The administrator shall be able to manage disputes and system settings because system stability and issue resolution must be maintained.	Business Process = System Administration	1	18/03/2026	Matias Bali
BR_PRS_27	The administrator shall be able to restore user accounts because suspended or deleted accounts may require recovery.	Business Process = User Account Recovery	2	18/03/2026	Matias Bali
BR_PRS_28	The administrator shall be able to suspend user accounts because policy violations may require temporary restrictions.	Business Process = User Account Control	1	18/03/2026	Matias Bali
BR_PRS_29	The administrator shall be able to verify and approve property listings because listings must be validated before publication.	Business Process = Listing Verification	1	18/03/2026	Matias Bali
BR_PRS_30	The manager shall be able to monitor rental requests because booking operations require supervision.	Business Process = Rental Monitoring	2	18/03/2026	Matias Bali

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BR_PRS_31	The manager shall be able to generate management reports because operational performance must be evaluated.	Business Process = Management Reporting	2	18/03/2026	Ornelt Lemellari
BR_PRS_32	The manager shall be able to approve large discounts and promotions because special pricing actions require authorization.	Business Process = Promotion Approval	3	18/03/2026	Ornelt Lemellari
BR_PRS_33	The manager shall be able to review complaints and forward unresolved issues because escalated problems require supervision.	Business Process = Complaint Management	1	18/03/2026	Ornelt Lemellari
BR_PRS_34	The financier shall be able to view transaction history because financial records must be monitored and verified.	Business Process = Financial Monitoring	1	18/03/2026	Ornelt Lemellari
BR_PRS_35	The financier shall be able to monitor payment processing and security because secure transactions are required.	Business Process = Payment Security Management	1	18/03/2026	Ornelt Lemellari
BR_PRS_36	The financier shall be able to communicate payment issues with tenants and owners because payment problems require resolution.	Business Process = Payment Issue Resolution	2	18/03/2026	Ornelt Lemellari
BR_PRS_37	The financier shall be able to approve refunds for cancelled rental transactions because financial adjustments may be required.	Business Process = Refund Management	1	18/03/2026	Kevin Muho
BR_PRS_38	HR staff shall be able to manage complaints because user issues and disputes must be handled properly.	Business Process = Complaint Handling	1	18/03/2026	Kevin Muho
BR_PRS_39	HR staff shall be able to send notifications and reminders because users must receive important updates and alerts.	Business Process = Notification Management	2	18/03/2026	Kevin Muho
BR_PRS_40	HR staff shall be able to maintain records of user issues and resolutions because complaint history must be documented.	Business Process = Issue Resolution Tracking	2	18/03/2026	Kevin Muho
BR_PRS_41	Users shall be able to register accounts because authenticated access is required to use the system.	Business Process = User Registration	1	18/03/2026	Kevin Muho
BR_PRS_42	Users shall be able to log into the system because secure account access is required.	Business Process = User Authentication	1	18/03/2026	Kevin Muho
BR_PRS_43	Users shall be able to log out of the system because secure session termination is required.	Business Process = Session Management	1	18/03/2026	Kevin Muho

3.2 Non-Functional Requirements

3.2.1 Product Requirements

Requirements which specify that the delivered product must behave in a particular way such as execution speed, reliability, usability, and security.

3.2.1.1 Usability Requirements

The Property Rentals System must provide a user-friendly and intuitive interface that allows tenants, property owners, administrators, managers, financiers, and HR staff to interact with the system efficiently without requiring advanced technical knowledge.

Ease of Learning (Learnability)

The system should be easy for first-time users to understand. New users should be able to learn basic operations such as searching properties, sending rental requests, managing listings, and processing payments within a short period of use.

Clear and Intuitive Interface

The interface should present information in a clear and organized manner. Menus, buttons, and navigation elements should be clearly labeled so users can easily perform tasks such as viewing properties, managing bookings, handling complaints, and processing payments.

Consistent Navigation

Navigation across all modules of the system should remain consistent. Similar operations such as adding listings, managing requests, or updating profiles should follow the same interaction patterns throughout the platform.

Accessible Design

The system should be accessible from desktop computers, tablets, and mobile devices using standard web browsers. The user interface should adapt properly to different screen sizes and resolutions.

Error Prevention and Feedback

The system should validate user inputs and provide informative error messages. For example, if a tenant submits incomplete payment information or a property owner enters invalid listing data, the system should clearly explain the issue and guide the user to correct it.

Help and Documentation

The system should include basic help information, tooltips, or guidance pages to assist users in understanding key functions such as booking properties, processing refunds, managing complaints, and updating listings

Efficiency of Use

Frequent users such as administrators, managers, and property owners should be able to perform common tasks quickly with minimal steps. Examples include approving rental requests, updating property availability, generating reports, and resolving complaints.

3.2.1.2 Performance Requirements

This section specifies measurable requirements for the performance of the Property Rentals System.

• Static numerical requirements:

- The system must support a minimum of 500 concurrent users.
- The system should support multiple user roles simultaneously, including tenants, property owners, administrators, managers, financiers, and HR staff.
- The system should handle at least 20,000 property listings and related booking records.
- The database should support storage of property images, booking histories, reviews, complaints, and payment transactions efficiently.

• Dynamic numerical requirements:

- Property search and filtering operations must return results within 2 seconds under normal load.
- Rental request processing and booking approval operations should complete within 3 seconds.
- Payment transactions should be processed within 5 seconds under normal operating conditions.
- The system should maintain stable performance during peak usage periods, such as holidays or high booking seasons.
- Notifications and messages should be delivered to users within a few seconds after system events occur.

3.2.1.3 Availability

Level of availability required

- The system shall maintain at least 99% uptime during normal operating hours.
- The system shall be accessible to authorized users whenever the platform is operational.

Coverage for geographic areas

- The system shall be accessible through the internet from different geographic locations.
- Authorized users should be able to access the system remotely using valid login credentials.

Impact of downtime on users and business operations

- If the system becomes unavailable, users shall not be able to process bookings, payments, property updates, or complaint management operations.

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- Downtime shall be minimized to reduce interruptions to rental operations and user communication.

Impact of scheduled and unscheduled maintenance

- Scheduled maintenance should be performed outside peak operating hours whenever possible.
- Users shall be notified in advance about planned maintenance periods.
- In the event of unexpected maintenance or system failure, the system should be restored as quickly as possible and users should be informed about service interruptions.

Reliability

- The system shall be designed to minimize failures during normal operation.
- The system should aim for a high Mean Time Between Failures (MTBF).
- The system shall recover quickly from failures to restore normal functionality and maintain data integrity.

3.2.1.4 Security

Encryption

- The system shall encrypt sensitive information such as passwords and payment details during transmission.
- Sensitive user information stored in the database shall be protected from unauthorized access.

Activity logging and historical data sets

- The system shall maintain logs of important user activities including logins, booking approvals, payments, refunds, complaint handling, and administrative actions.
- Activity logs shall be stored for auditing and monitoring purposes.

Restrictions on intermodule communications

- The system shall restrict communication between modules to authorized interactions only.
- Each module shall only access the data and operations necessary for its functionality.

Data integrity checks

- The system shall validate all user input before storing it in the database.
- The system shall prevent incomplete, invalid, or corrupted data from being stored.
- The system shall maintain consistency and accuracy of booking, payment, and property records.

Role-Based Access Control

- The system shall implement role-based access control to ensure users can only access features and information relevant to their assigned roles.
- Administrative functions shall only be accessible to authorized staff members.

3.2.2 Organizational Requirements

Organizational requirements are requirements that result from organizational policies, procedures, and implementation standards.

- The system development process should follow standard software engineering and UML modeling practices.
- The project should maintain proper documentation for requirements, diagrams, use cases, and design patterns.
- The system should support maintainability and future scalability for additional features and modules.
- Developers should follow consistent coding standards and database naming conventions throughout implementation.
- The system should maintain backup procedures for important booking, payment, and complaint records.

3.2.3 External Requirements

External requirements are conditions and constraints that arise from factors outside the system and its development process.

Interoperability Requirements

The Property Rentals System should be capable of integrating with external payment systems and notification services for processing transactions and sending alerts to users.

Legislative Requirements

The system must comply with data protection and privacy regulations regarding the storage and management of personal user information and payment data.

Security Requirements

The system should provide secure authentication using usernames and passwords. Access to system functionality should be restricted based on user roles and permissions.

Platform Requirements

The system will be developed as a web-based application and should function properly on modern web browsers such as Chrome, Firefox, Edge, and Safari without requiring additional software installation.

Communication Requirements

The system should support communication between tenants and property owners through messaging and notification features integrated within the platform.

4. Use Case Tables

Done by: Anisa Domi (Responsible for UC_PRS_1 until UC_PRS_6)

UC Name	UC_PRS_1 - Update Personal Information
Summary	Edit personal profile data such as name, phone number, or contact details.
Dependency	None
Actors	Primary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The tenant logs in. 2. The tenant opens the profile settings page. 3. The tenant edits the required personal information. 4. The tenant saves the changes. 5. The tenant profile is updated and he reads the success message on screen.
Description of alternative sequence	1. If data is invalid, tenant reads error message on screen.
Non functional requirements	The system shall validate input data securely and save profile updates within a few seconds.
Postconditions	The tenant profile is updated.

UC Name	UC_PRS_2 - Search Properties
Summary	Search for rental properties that match general preferences.
Dependency	None
Actors	Primary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The tenant opens the property search page. 2. The tenant enters search keywords or basic preferences. 3. The tenant starts the search. 4. The system retrieves matching properties. 5. The system displays the search results.
Description of alternative sequence	1. If no properties match the search, the system shows an empty-results message.
Non functional requirements	Search results should be displayed quickly and support large numbers of property records.
Postconditions	Matching properties are displayed or the tenant is informed that no results were found.

UC Name	UC_PRS_3 - Filter Properties
Summary	Filter available properties by location, price, and property type to narrow search results.
Dependency	UC_PRS_2 - Search Properties
Actors	Primary: Tenant
Preconditions	The tenant is logged into the system and the property search page is accessible.
Description of main sequence	1. The tenant opens the property search page. 2. The tenant selects filter options such as location, price range, or property type. 3. The tenant applies the selected filters. 4. The system processes the selected criteria. 5. The system displays properties that match the selected filters.
Description of alternative sequence	1. If no properties match the selected filters, the system displays a message indicating that no results were found.
Non functional	The filtering process should respond efficiently and accurately under normal system load.

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requirements	
Postconditions	Filtered property results are displayed to the tenant or the tenant is informed that no matching properties were found.

UC Name	UC_PRS_4 - View Property Details and Booking History
Summary	View detailed information about a property and the tenant's booking history.
Dependency	UC_PRS_2 - Search Properties
Actors	Primary: Tenant
Preconditions	The tenant is logged into the system and has access to property listings or booking history.
Description of main sequence	1. The tenant selects a property from the search results. 2. The Tenant sees property details 3. The tenant navigates to their booking history section. 4. If history exists, tenant reads the booking history
Description of alternative sequence	1. If the tenant has no booking history, he will read the no history message on screen
Non functional requirements	Property details and booking history should load securely and within acceptable response time.
Postconditions	The tenant views property details and/or booking history successfully.

UC Name	UC_PRS_5 - Send Rental Request and Message to Property Owner
Summary	Allow the tenant to send a rental request or message to a property owner regarding a property listing.
Dependency	UC_PRS_4 - View Property Details and Booking History
Actors	Primary: Tenant Secondary: Property Owner
Preconditions	The tenant is logged into the system and viewing the details of a property listing.
Description of main sequence	1. The tenant selects a property listing. 2. The tenant chooses the option to send a rental request or message the property owner. 3. The tenant enters the request or message details. 4. The tenant submits the request or message. 5. If valid, tenant sends request to owner and then stores the request.
Description of alternative sequence	1. Tenant reads the error on screen. 2. If the message field is empty or invalid, the system prompts the tenant to correct the input before sending.
Non functional requirements	The system shall securely store requests and deliver messages reliably.
Postconditions	The rental request or message is successfully sent to the property owner and recorded in the system.

UC Name	UC_PRS_6 - Cancel Rental Request
Summary	Cancel a previously submitted rental request before it is finalized.
Dependency	UC_PRS_5 - Send Rental Request and Message to Property Owner
Actors	Primary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The tenant logs into the system. 2. The tenant opens the list of submitted rental requests. 3. The tenant selects a pending request. 4. The tenant chooses the cancel option. 5. The system updates the request status and confirms the cancellation.

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Description of alternative sequence	1. If the request has already been approved, rejected, or processed, the system may deny cancellation and show the cant cancel message.
Non functional requirements	Cancellation requests should update system records immediately and maintain data consistency.
Postconditions	The selected request is canceled when cancellation is allowed.

Done by: Haidi Toto (Responsible for UC_PRS_7 until UC_PRS_12)

UC Name	UC_PRS_7 - Receive Messages from Property Owner
Summary	Read messages sent by property owners.
Dependency	UC_PRS_5 - Send Rental Request and Message to Property Owner
Actors	Primary: Tenant Secondary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The tenant logs in. 2. The tenant opens inbox/notifications. 3. The tenant opens a conversation. 4. The tenant checks the received message content.
Description of alternative sequence	1. If there are no new messages, the tenant sees the current conversation history or an empty inbox message.
Non functional requirements	Messages should load securely and be delivered in real time with minimal delay.
Postconditions	The tenant can view messages received from property owners.

UC Name	UC_PRS_8 - Submit Rental Payment
Summary	Pay for an approved rental booking through the system.
Dependency	UC_PRS_17 - Accept or Reject Rental Request
Actors	Primary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The tenant logs into the system. 2. The tenant opens an approved booking or payment page. 3. The tenant enters payment details or chooses a saved payment method. 4. The tenant confirms the payment. 5. The system sends the payment request to the payment service. 6. The system receives the payment result and displays confirmation.
Description of alternative sequence	1. If payment fails, the system displays the failure reason and asks the tenant to retry or choose another method.
Non functional requirements	Payment processing should be secure, encrypted, and completed within a few seconds.
Postconditions	The payment is recorded if successful, and the booking status is updated.

UC Name	UC_PRS_9 - Save Favorite Properties for later viewing
Summary	Allow tenants to save properties they are interested in for later viewing.
Dependency	UC_PRS_2 - Search Properties
Actors	Primary: Tenant
Preconditions	The tenant is logged into the system and viewing a property listing.
Description of main sequence	1. The tenant browses available property listings. 2. The tenant selects a property they want to save. 3. The tenant clicks the "Save to Favorites" option. 4. The tenant stores the selected property.

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	5. The tenant saves the property.
Description of alternative sequence	1. If the property is already saved in favorites, the tenant sees that the property is already in the favorites list.
Non functional requirements	The system should store favorite properties reliably and retrieve them quickly when requested.
Postconditions	The property is successfully added to the tenant's favorites list.

UC Name	UC_PRS_10 - Review and Rate Property
Summary	Allow tenants to review and rate a property after completing a rental.
Dependency	UC_PRS_19 - View Booking History
Actors	Primary: Tenant Secondary: Property Owner
Preconditions	The tenant has completed a rental transaction for the property and is logged into the system.
Description of main sequence	1. The tenant opens their booking history. 2. The tenant selects a completed rental. 3. The tenant chooses the option to review the property. 4. The tenant enters a rating and optional review comment. 5. The tenant submits the review. 6. The tenant saves and publishes the review for other users to see.
Description of alternative sequence	1. If the tenant has not completed a rental for the property, he prevents the review from being submitted.
Non functional requirements	Reviews and ratings should be stored securely and displayed accurately to users.
Postconditions	The property review and rating are stored in the system and visible to other users.

UC Name	UC_PRS_11 - Update Profile Information and availability of rental properties
Summary	Allow property owners to update their personal profile information and manage the availability status of their rental properties.
Dependency	UC_PRS_12 - Add Property Listing
Actors	Primary: Property Owner
Preconditions	The property owner is logged into the system and has an existing account and property listings.
Description of main sequence	1. The property owner logs into the system. 2. The property owner navigates to their profile or property management section. 3. The property owner selects the option to edit profile information or property availability. 4. The property owner updates the necessary information (contact details, availability status, etc.). 5. The property owner saves the changes. 6. The system updates and stores the new information.
Description of alternative sequence	1. If the entered information is incomplete or invalid, the system displays an error message and asks the owner to correct the data.
Non functional requirements	The system should validate updated information and process changes without data loss.
Postconditions	The property owner's profile information and/or property availability status are successfully updated in the system.

UC Name	UC_PRS_12 - Add Property Listing
Summary	Create a new rental property listing.
Dependency	None
Actors	Primary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or

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	module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens the add-listing page. 3. The property owner enters property information such as title, description, price, location, and availability. 4. The property owner submits the listing. 5. The system validates and stores the listing.
Description of alternative sequence	1. If required listing information is missing, the system asks the owner to complete it before submission.
Non functional requirements	Property listings should be stored securely and support image and data scalability.
Postconditions	A new property listing is stored in the system.

Done by:Ledio Tafciu (Responsible for UC_PRS_13 until UC_PRS_18)

UC Name	UC_PRS_13 - Upload Property Images
Summary	Attach photos to a property listing.
Dependency	UC_PRS_12 - Add Property Listing
Actors	Primary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs in. 2. The property owner opens an existing or draft listing. 3. The property owner selects image files to upload. 4. The property owner uploads and attaches the images to the property listing.
Description of alternative sequence	1. If a file format is unsupported or upload fails, the system shows an error message.
Non functional requirements	Uploaded images should support common file formats and upload efficiently.
Postconditions	Property images are linked to the listing.

UC Name	UC_PRS_14 - Edit Property Listing
Summary	Modify an existing property listing.
Dependency	UC_PRS_12 - Add Property Listing
Actors	Primary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens one of their listings. 3. The property owner edits the listing details. 4. The property owner saves the changes. 5. The system validates and updates the listing.
Description of alternative sequence	1. If validation fails, the system shows an error and keeps the previous saved version until corrected.
Non functional requirements	Listing updates should be processed reliably without affecting stored property data.
Postconditions	The selected listing is updated.

UC Name	UC_PRS_15 - Delete Property Listing
Summary	Remove an existing property listing from the platform.
Dependency	UC_PRS_12 - Add Property Listing

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Actors	Primary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens one of their listings. 3. The property owner chooses the delete option. 4. The system asks for confirmation. 5. The property owner confirms deletion. 6. The system removes or deactivates the listing.
Description of alternative sequence	1. If the property is linked to an active booking, the system may block deletion and inform the owner.
Non functional requirements	The system should maintain database consistency when listings are deleted or deactivated.
Postconditions	The listing is removed or marked inactive.

UC Name	UC_PRS_16 - View Rental Requests
Summary	See rental requests sent by tenants.
Dependency	UC_PRS_5 - Send Rental Request and Message to Property Owner
Actors	Primary: Property Owner Secondary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens the rental requests section. 3. The property owner views pending request. 4. The system displays the request list.
Description of alternative sequence	1. If no requests exist, the system shows an empty list message.
Non functional requirements	Rental requests should be retrieved quickly and displayed accurately.
Postconditions	The owner can view rental requests for their properties.

UC Name	UC_PRS_17 - Accept or Reject Rental Request
Summary	Review a tenant request and decide whether to approve or reject it.
Dependency	UC_PRS_16 - View Rental Requests
Actors	Primary: Property Owner Secondary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens a specific rental request. 3. The property owner reviews tenant and property request details. 4. The property owner chooses Accept or Reject. 5. The property owner updates the request status. 6. The property owner notifies the tenant about the decision.
Description of alternative sequence	1. If the property is no longer available, the system may block approval and ask the owner to resolve availability first.
Non functional requirements	Approval decisions should update the request status immediately and securely.
Postconditions	The rental request is approved or rejected and the tenant is informed.

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UC Name	UC_PRS_18 - Send Message to Tenant
Summary	Send a message to a tenant through the system.
Dependency	UC_PRS_16 - View Rental Requests
Actors	Primary: Property Owner Secondary: Tenant
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The property owner logs into the system. 2. The property owner opens a conversation or request. 3. The property owner writes a message. 4. The property owner sends the message. 5. The property owner stores and delivers the message to the tenant.
Description of alternative sequence	1. If the message is empty, the system asks for message content before sending.
Non functional requirements	Messaging functionality should support reliable delivery and secure communication.
Postconditions	The message is stored and delivered to the tenant.

Done by:Desard Lybeshari (Responsible for UC_PRS_19 until UC_PRS_24)

UC Name	UC_PRS_19 - View Booking History
Summary	Allow property owners to view the booking history of their listed properties.
Dependency	UC_PRS_12 - Add Property Listing
Actors	Primary: Property Owner Secondary: Tenant
Preconditions	The property owner is logged into the system and has at least one property listed in the system.
Description of main sequence	1. The property owner logs in. 2. The property owner opens the property management section. 3. The property owner selects one of their listed properties. 4. The property owner chooses the option to view booking history. 5. Current and past bookings are displayed
Description of alternative sequence	1. If the selected property has no booking history, a message no bookings found is shown
Non functional requirements	Booking history data should be stored accurately and retrieved efficiently.
Postconditions	The booking history of the selected property is displayed to the property owner.

UC Name	UC_PRS_20 - Mark Property as Unavailable or Rented
Summary	Allow property owners to update the status of a property as unavailable or rented.
Dependency	UC_PRS_14 - Edit Property Listing
Actors	Primary: Property Owner
Preconditions	The property owner is logged into the system and has an existing property listing.
Description of main sequence	1. The property owner logs in. 2. The property owner opens the property management section. 3. The property owner selects a listed property. 4. The property owner chooses the option to update the property status. 5. The property owner marks the property as unavailable or rented. 6. Status is saved and reflected in listing
Description of alternative sequence	1. If status cannot be updated, an error message is displayed and the change is not saved.
Non functional requirements	Property availability updates should synchronize immediately across the platform.
Postconditions	The selected property status is updated to unavailable or rented in the system.

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UC Name	UC_PRS_21 - Edit User Accounts
Summary	Modify user account information when administrative action is required.
Dependency	UC_PRS_41 - Register Account
Actors	Primary: Administrator
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The administrator logs in. 2. The administrator opens user management. 3. The administrator selects a user account. 4. The administrator edits the account details or status. 5. The administrator saves the changes. 6. The system validates entered data. 7. If data valid, update user account.
Description of alternative sequence	1. If data is invalid, the system shows an error and requests account details.
Non functional requirements	Administrative account updates should be processed securely and logged for auditing purposes.
Postconditions	The selected user account is updated.

UC Name	UC_PRS_22 - Delete User Accounts
Summary	Remove a user account from the system.
Dependency	UC_PRS_21 - Edit User Accounts
Actors	Primary: Administrator
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The administrator logs in. 2. The administrator opens user management. 3. The administrator selects a user account. 4. The administrator chooses Delete. 5. The system asks for confirmation. 6. The administrator confirms the action. 7. The system deletes or deactivates the account.
Description of alternative sequence	1. If the account is linked to unresolved transactions or disputes, the system may block deletion and warn the administrator.
Non functional requirements	The system should maintain user data integrity and log deletion activities securely.
Postconditions	The user account is removed or deactivated.

UC Name	UC_PRS_23 - Monitor Rental Transactions
Summary	Track rental transactions for control and oversight.
Dependency	UC_PRS_8 - Submit Rental Payment
Actors	Primary: Administrator
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The administrator logs in. 2. The administrator opens the transactions section. 3. The system retrieves transaction records. 4. Administrator displays transaction details and statuses.
Description of alternative	1. If no transactions are found for the selected period, an empty-results message is shown.

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sequence	
Non functional requirements	Transaction monitoring should support real-time updates and accurate financial records.
Postconditions	The administrator can review rental transaction records.

UC Name	UC_PRS_24 - View System Reports and property listings
Summary	Allow the administrator to view system reports and monitor property listings available on the platform.
Dependency	UC_PRS_31 - Generate Management Reports
Actors	Primary: Administrator Secondary: Property Owner
Preconditions	The administrator is logged into the system and has administrative access privileges.
Description of main sequence	1. The administrator logs in. 2. The administrator navigates to the admin dashboard. 3. The administrator selects the option to view system reports or property listings. 4. The administrator retrieves the relevant data from the database. 5. The administrator displays reports and property listings.
Description of alternative sequence	1. Error message is displayed.
Non functional requirements	Reports and listings should load efficiently and display accurate data.
Postconditions	The administrator successfully views system reports and property listings for monitoring and management purposes.

Done by:Matias Bali (Responsible for UC_PRS_25 until UC_PRS_30)

UC Name	UC_PRS_25 - Remove Inappropriate Listings
Summary	Take down listings that violate rules or platform standards.
Dependency	UC_PRS_29 - Verify and approve property listings
Actors	Primary: Administrator Secondary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor. The administrator is logged into the system and has administrative privileges.
Description of main sequence	1. The administrator reviews flagged or reported listings. 2. The administrator selects a listing. 3. The administrator removes or disables the listing. 4. The system records the action.
Description of alternative sequence	1. If the listing is already inactive, the system informs the administrator.
Non functional requirements	Listing moderation actions should be logged and processed securely.
Postconditions	The inappropriate listing is removed or disabled.

UC Name	UC_PRS_26 - Manage Disputes and System Settings
Summary	Allow the administrator to manage user disputes and update system settings.
Dependency	UC_PRS_38 - Manage Complaints
Actors	Primary: Administrator
Preconditions	The administrator is logged into the system and has administrative privileges.
Description of main sequence	1. The administrator accesses the admin dashboard. 2. The administrator selects the dispute management or system settings option. 3. The administrator reviews disputes submitted by users or views current system settings. 4. The administrator updates dispute status or modifies the required system settings.

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	5. The system saves the changes made by the administrator.
Description of alternative sequence	1. If the selected dispute or system setting cannot be updated, the system displays an error message and does not save the changes.
Non functional requirements	System settings changes should be restricted to authorized administrators only.
Postconditions	The dispute information and/or system settings are updated successfully in the system.

UC Name	UC_PRS_27 - Restore User Accounts
Summary	Reactivate a previously suspended or deactivated user account.
Dependency	UC_PRS_28 - Suspend User Accounts
Actors	Primary: Administrator
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor. The administrator is logged into the system and has administrative privileges.
Description of main sequence	1. The administrator opens user management. 2. The administrator selects a suspended or inactive account. 3. The administrator chooses Restore. 4. The system updates the account status and confirms the action.
Description of alternative sequence	1. If the account cannot be restored due to policy or unresolved issues, the system notifies the administrator.
Non functional requirements	Account restoration actions should maintain account history and security restrictions.
Postconditions	The selected account is restored when restoration is allowed.

UC Name	UC_PRS_28 - Suspend User Accounts
Summary	Temporarily block a user account.
Dependency	UC_PRS_21 - Edit User Accounts
Actors	Primary: Administrator
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor. The administrator is logged into the system and has administrative privileges.
Description of main sequence	1. The administrator opens user management. 2. The administrator selects a user account. 3. The administrator chooses Suspend. 4. The system asks for confirmation or a reason. 5. The administrator confirms the suspension. 6. The system updates the account status.
Description of alternative sequence	1. If the user is already suspended, the system informs the administrator.
Non functional requirements	Account suspension operations should be secure and recorded in activity logs.
Postconditions	The selected account is suspended.

UC Name	UC_PRS_29 - Verify and approve property listings
Summary	Allow the administrator to verify newly submitted property listings and approve them before publication.
Dependency	UC_PRS_12 - Add Property Listing
Actors	Primary: Administrator Secondary: Property Owner
Preconditions	The administrator is logged into the system and there are pending property listings awaiting

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	approval. The administrator is logged into the system and has administrative privileges.
Description of main sequence	1. The administrator accesses the admin dashboard. 2. The administrator selects the pending property listings section. 3. The system displays newly submitted property listings awaiting verification. 4. The administrator reviews the property details and uploaded information. 5. The administrator approves the listing. 6. The system publishes the approved property listing.
Description of alternative sequence	1. If the property listing contains missing, invalid, or inappropriate information, the administrator rejects it and the system keeps it unpublished.
Non functional requirements	Listing approval workflows should ensure accurate validation and secure publication processes.
Postconditions	The property listing is either approved and published or rejected and kept unpublished.

UC Name	UC_PRS_30 - Monitor Rental Requests
Summary	Review rental requests and approval activity at an operational level.
Dependency	UC_PRS_17 - Accept or Reject Rental Request
Actors	Primary: Manager Secondary: Tenant, Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The manager logs into the system. 2. The manager views the rental requests. 3. The manager reviews requests. 4. The manager approves or rejects the request
Description of alternative sequence	1. If there are no requests pending, the manager waits for new requests.
Non functional requirements	The system should support efficient monitoring of large numbers of rental requests.
Postconditions	The manager can monitor rental requests and approvals.

Done by:Ornelt Lemellari (Responsible for UC_PRS_31 until UC_PRS_36)

UC Name	UC_PRS_31 - Generate Management Reports
Summary	Produce reports for operational oversight.
Dependency	UC_PRS_24 - View System Reports and property listings
Actors	Primary: Manager
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The manager logs in. 2. The manager opens the reports section. 3. The manager selects report criteria or period. 4. The manager generates the report. 5. The manager displays the report.
Description of alternative sequence	1. If no data exists for the selected criteria, a no data message is shown.
Non functional requirements	Reports should be generated accurately and within acceptable response time.
Postconditions	A management report is generated and displayed.

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UC Name	UC_PRS_32 - Approve Large Discounts and Promotions
Summary	Authorize high-impact discounts or promotions.
Dependency	None
Actors	Primary: Manager Secondary: Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The manager logs in. 2. The manager opens the pending discounts or promotions list. 3. The manager reviews the proposal details. 4. The manager approves or rejects the request and records his answer.
Description of alternative sequence	1. If required information is missing, the manager may return the proposal without approval.
Non functional requirements	Promotion approval actions should be stored securely and support audit tracking.
Postconditions	The discount or promotion request is approved or rejected.

UC Name	UC_PRS_33 - Review Complaints and Forward Unresolved Issues
Summary	Allow the manager to review user complaints and forward unresolved issues to the administrator.
Dependency	UC_PRS_38 - Manage Complaints
Actors	Primary: Manager Secondary: Administrator
Preconditions	The manager is logged into the system and there are complaints submitted by users in the system.
Description of main sequence	1. The manager logs in. 2. The manager navigates to the complaints management section. 3. The manager views submitted complaints. 4. The manager reviews the details of a complaint. 5. The manager attempts to resolve the issue. 6. If the issue cannot be resolved, the manager forwards the complaint to the administrator.
Description of alternative sequence	1. If the complaint can be resolved by the manager, the manager updates the complaint status as resolved and no escalation is needed.
Non functional requirements	Complaint handling should maintain confidentiality and accurate issue tracking.
Postconditions	The complaint is either resolved by the manager or forwarded to the administrator for further action.

UC Name	UC_PRS_34 - View Transaction History
Summary	Review past payment and transaction records.
Dependency	UC_PRS_8 - Submit Rental Payment
Actors	Primary: Financier Secondary: Tenant, Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The financier logs in. 2. The financier opens transaction history. 3. The system retrieves transaction records. 4. The system displays the transaction history.
Description of alternative sequence	1. If no transactions are found, the system shows an empty-history message.
Non functional requirements	Transaction records should be retrieved securely and displayed accurately.
Postconditions	The financier can review transaction history.

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UC Name	UC_PRS_35 - Monitor Payment Process and Security
Summary	Track payment processing activity and payment-related security status.
Dependency	UC_PRS_8 - Submit Rental Payment
Actors	Primary: Financier
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The financier logs in. 2. The financier opens payment monitoring tools or records. 3. The system displays payment statuses, failures, or suspicious events. 4. The system reviews normal payment processing information.
Description of alternative sequence	1. If a payment issue or security alert is detected, the system highlights it for follow-up.
Non functional requirements	Payment monitoring should support secure transaction tracking and real-time alerts.
Postconditions	The financier has reviewed payment processing and security information.

UC Name	UC_PRS_36 - Communicate Payment Issues with Tenants and Owners
Summary	Contact users about payment-related problems.
Dependency	UC_PRS_35 - Monitor Payment Process and Security
Actors	Primary: Financier Secondary: Tenant, Property Owner
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The financier logs in. 2. The financier identifies a payment issue. 3. The financier opens the communication interface. 4. The financier writes a message to the relevant tenant or owner. 5. The financier sends the message. 6. The financier stores and delivers the communication.
Description of alternative sequence	1. If the message is empty, content is asked before sending.
Non functional requirements	Communication messages should be delivered reliably and securely stored.
Postconditions	Users are informed about the payment issue and the communication is stored.

Done by: Kevin Muho (Responsible for UC_PRS_37 until UC_PRS_43)

UC Name	UC_PRS_37 - Approve Refunds for cancelled Rental Transactions
Summary	Allow the financier to approve refunds for cancelled rental transactions.
Dependency	UC_PRS_8 - Submit Rental Payment
Actors	Primary: Financier
Preconditions	The financier is logged into the system and there is at least one cancelled rental transaction with a pending refund request.
Description of main sequence	1. The financier logs in. 2. The financier accesses the transaction management section. 3. The system displays cancelled rental transactions with pending refund requests. 4. The financier selects a refund request for review. 5. The financier reviews the transaction details and cancellation information. 6. The financier approves the refund request. 7. The system updates the refund status and records the approval.

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Description of alternative sequence	1. If the refund request does not meet the refund conditions, the request is rejected and the system records the rejection.
Non functional requirements	Refund processing should ensure transaction security and accurate financial record updates.
Postconditions	The refund request is either approved and recorded in the system or rejected and marked accordingly.

UC Name	UC_PRS_38 - Manage Complaints
Summary	Review and process user complaints.
Dependency	None
Actors	Primary: HR
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The HR user logs in. 2. The HR user opens the complaints section. 3. The system retrieves submitted complaints. 4. The HR user reviews a complaint and records a handling action or update. 5. The system saves the complaint status update.
Description of alternative sequence	1. If required complaint details are missing, the system requests more information before progressing the case.
Non functional requirements	Complaint records should be securely maintained and accessible only to authorized staff.
Postconditions	The complaint record is updated.

UC Name	UC_PRS_39 - Send Notifications and Reminders
Summary	Notify users about deadlines, updates, or follow-up actions.
Dependency	UC_PRS_38 - Manage Complaints
Actors	Primary: HR
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The HR user logs in. 2. The HR user opens the notifications interface. 3. The HR user selects recipients and writes the message. 4. The HR user sends the notification or reminder. 5. The HR delivers and records the notification.
Description of alternative sequence	1. If recipient or message data is missing, HR asks for completion before sending.
Non functional requirements	Notifications should be delivered promptly and recorded for tracking purposes.
Postconditions	The notification or reminder is sent and recorded.

UC Name	UC_PRS_40 - Maintain Records of User Issues and Resolutions
Summary	Keep a record of issues reported by users and how they were resolved.
Dependency	UC_PRS_38 - Manage Complaints
Actors	Primary: HR
Preconditions	The actor is authenticated when the function requires account access, and the relevant page or module is available to the actor.
Description of main sequence	1. The HR user logs in. 2. The HR user opens the issues and resolutions records area. 3. The HR user creates or updates a record with issue details and status. 4. The HR user saves the record.

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	5. The HR stores the issue and resolution data.
Description of alternative sequence	1. If required information is missing, HR asks for correction before saving.
Non functional requirements	Issue records should be stored securely and support future retrieval and auditing.
Postconditions	The user issue record is created or updated.

UC Name	UC_PRS_41 - Register Account
Summary	Allow a user to create a new account in the system.
Dependency	None
Actors	Primary: User
Preconditions	The user does not already have an account and has access to the registration page.
Description of main sequence	1. The user opens the registration page. 2. The user enters the required personal and account information. 3. The user submits the registration form. 4. The user validates the entered information. 5. If information is valid, if account doesn't exist already, then new account is created
Description of alternative sequence	1. If required fields are missing or invalid, user sees error message. 2. If an account with the same email or username already exists, user is informed
Non functional requirements	User registration should validate credentials securely and protect personal information.
Postconditions	A new user account is created and stored in the system.

UC Name	UC_PRS_42 - Log In
Summary	Allow a user to log into the system using valid credentials.
Dependency	UC_PRS_41 - Register Account
Actors	Primary: User
Preconditions	The user already has a registered account in the system.
Description of main sequence	1. The user opens the login page. 2. The user enters username or email and password. 3. The user submits the login request. 4. The system validates the entered credentials. 5. The system grants access to the user.
Description of alternative sequence	1. If the entered credentials are incorrect, the system displays an error message.
Non functional requirements	Authentication processes should use encrypted credentials and secure session management.
Postconditions	The user is logged into the system successfully.

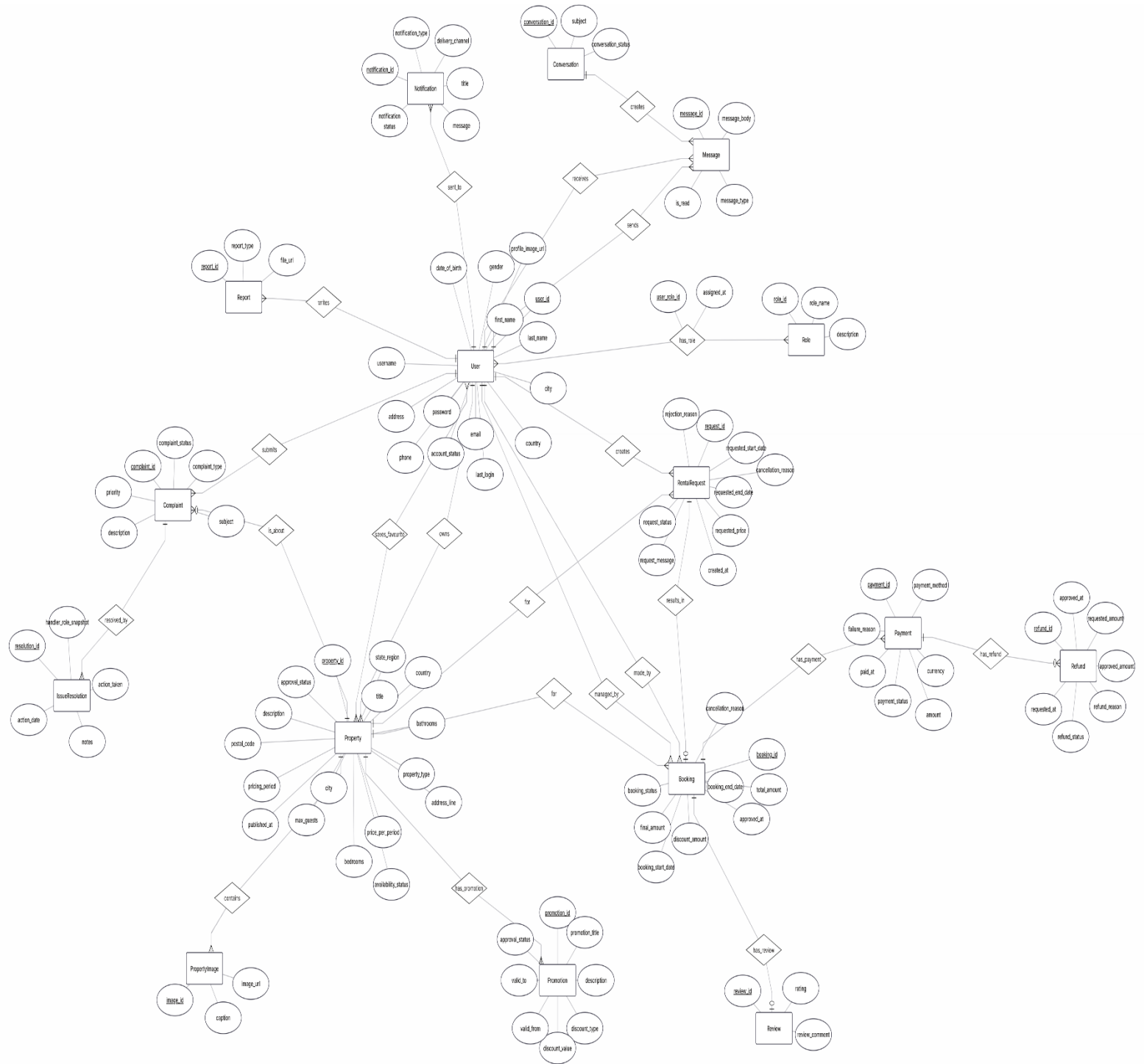
UC Name	UC_PRS_43 - Log Out
Summary	Allow a user to securely log out of the system and terminate their active session.
Dependency	UC_PRS_42 - Log In
Actors	Primary: User
Preconditions	The user is logged into the system. The system session is active.
Description of main sequence	1. The user selects the "Log Out" option from the system interface. 2. The system processes the logout request. 3. The system terminates the user session. 4. The system clears any session-related data.

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	5. The system redirects the user to the login page or home page.
Description of alternative sequence	1. If the session has already expired, the system redirects the user to the login page automatically. 2. If a system error occurs during logout, the system shows an error message but ensures the session is terminated.
Non functional requirements	Logout functionality should terminate sessions securely and prevent unauthorized reuse of sessions.
Postconditions	The user is logged out successfully. The session is terminated and no longer accessible without re-authentication.

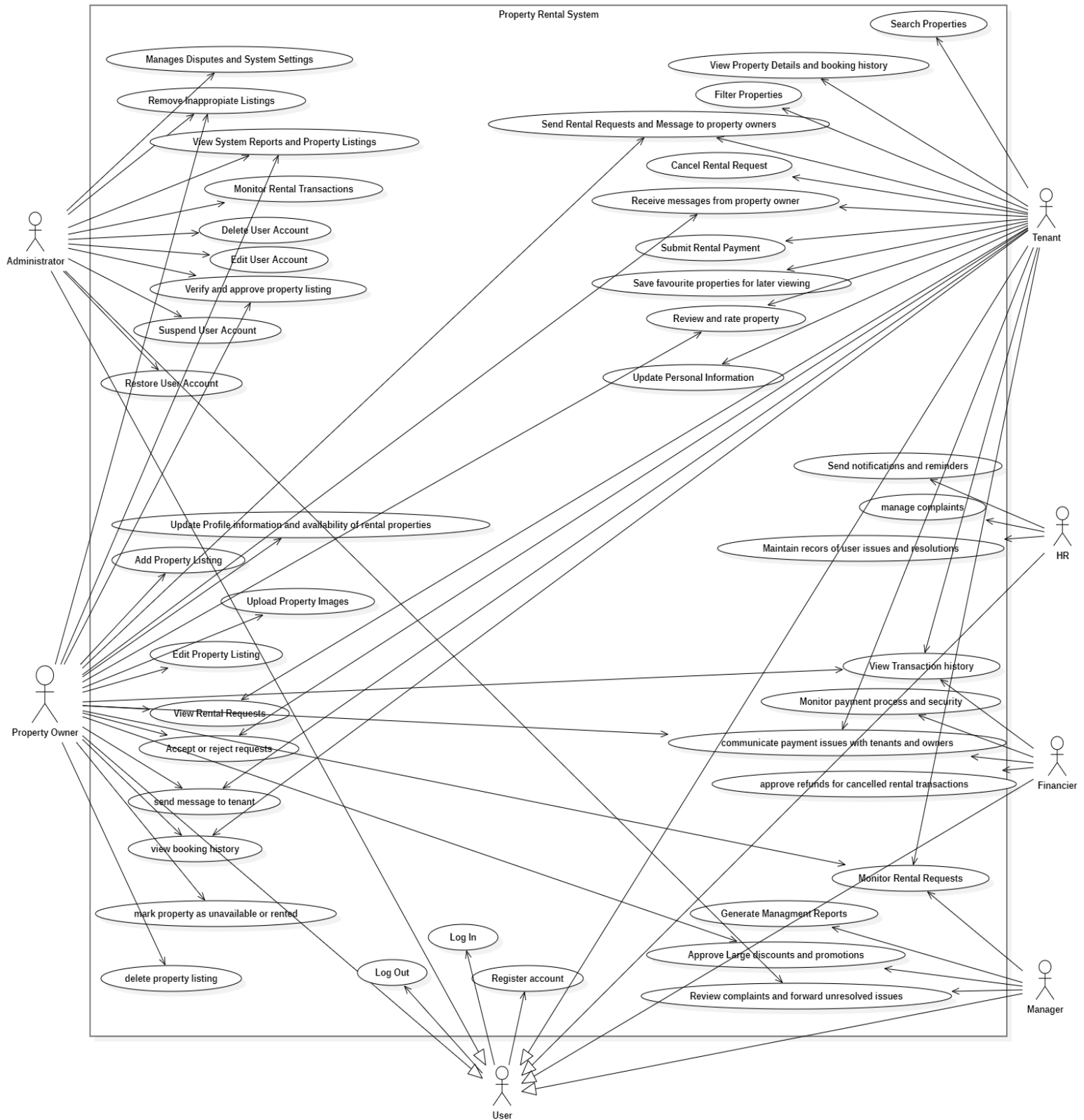
5. Diagrams

5.1 ER Diagram



Property Rental System Requirement Specification

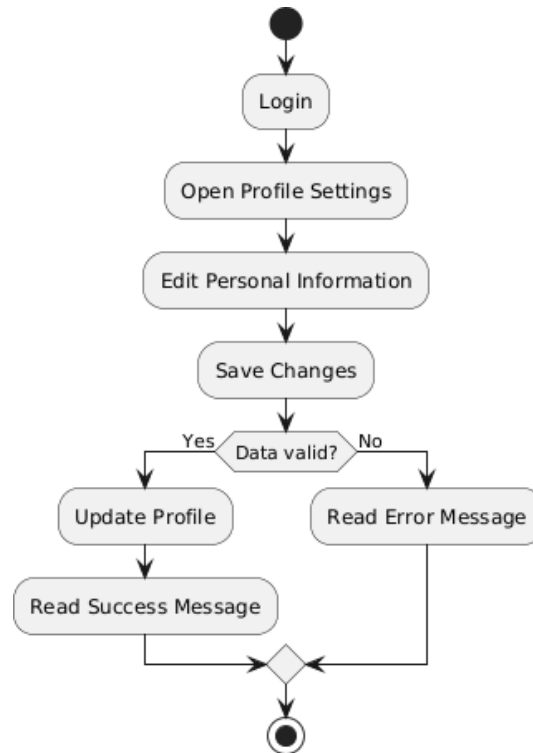
5.2 Use Case Diagram



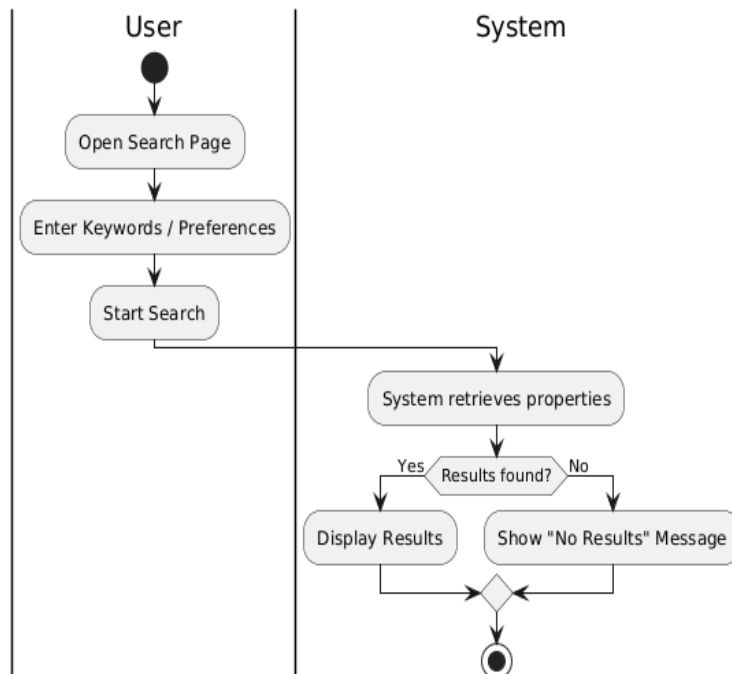
Property Rental System Requirement Specification

5.3 Activity Diagram

AC_PRS_1- Update personal information (Done by: Anisa Domi)

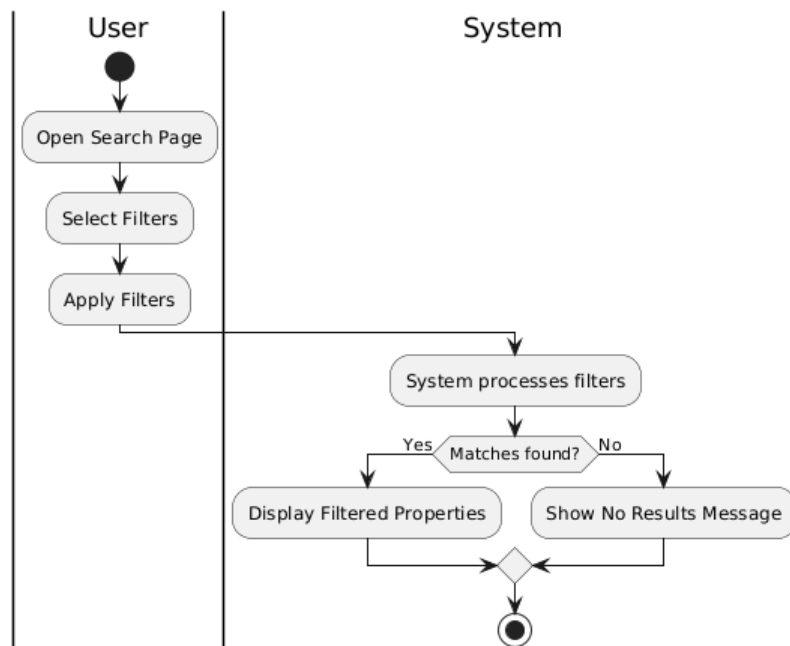


AC_PRS_2 - Search properties (Done by: Anisa Domi)

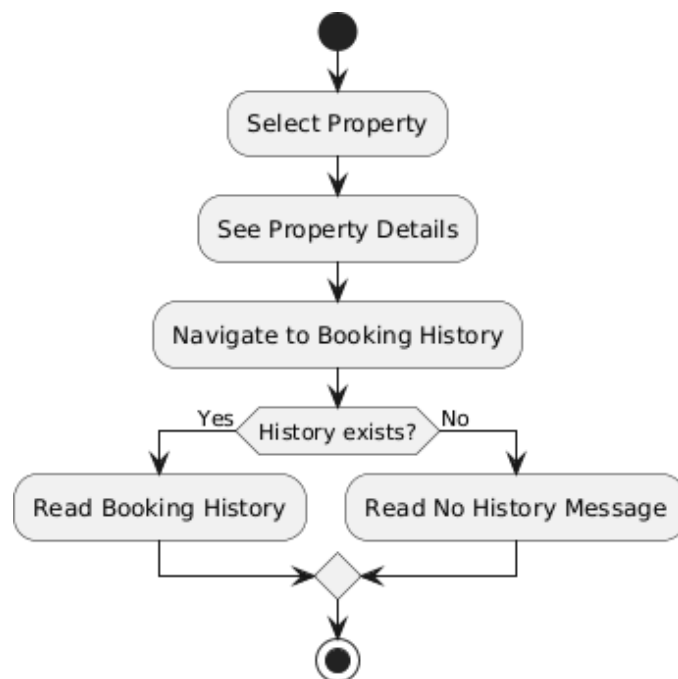


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AC_PRS_3 - Filter properties (Done by: Anisa Domi)

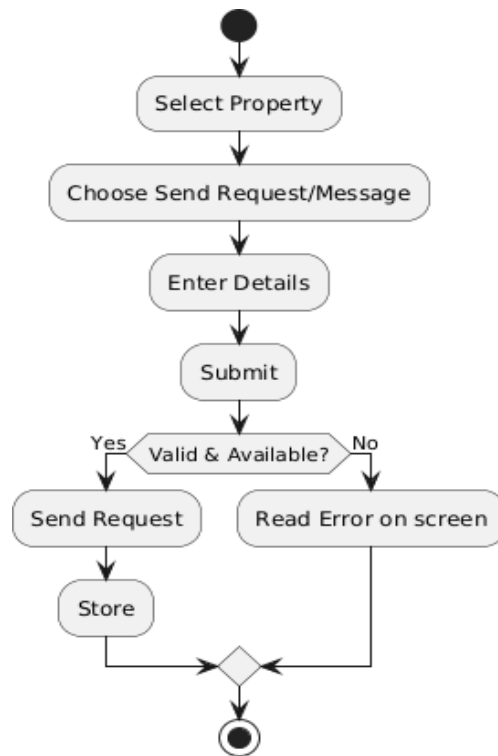


AC_PRS_4 - View property details and booking history. (Done by: Anisa Domi)

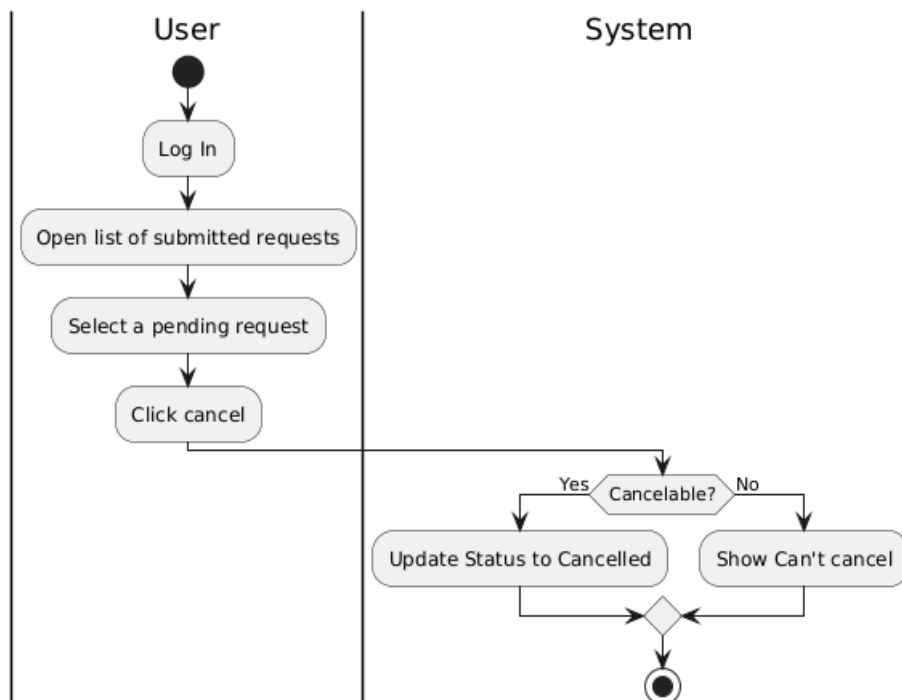


Property Rental System Requirement Specification

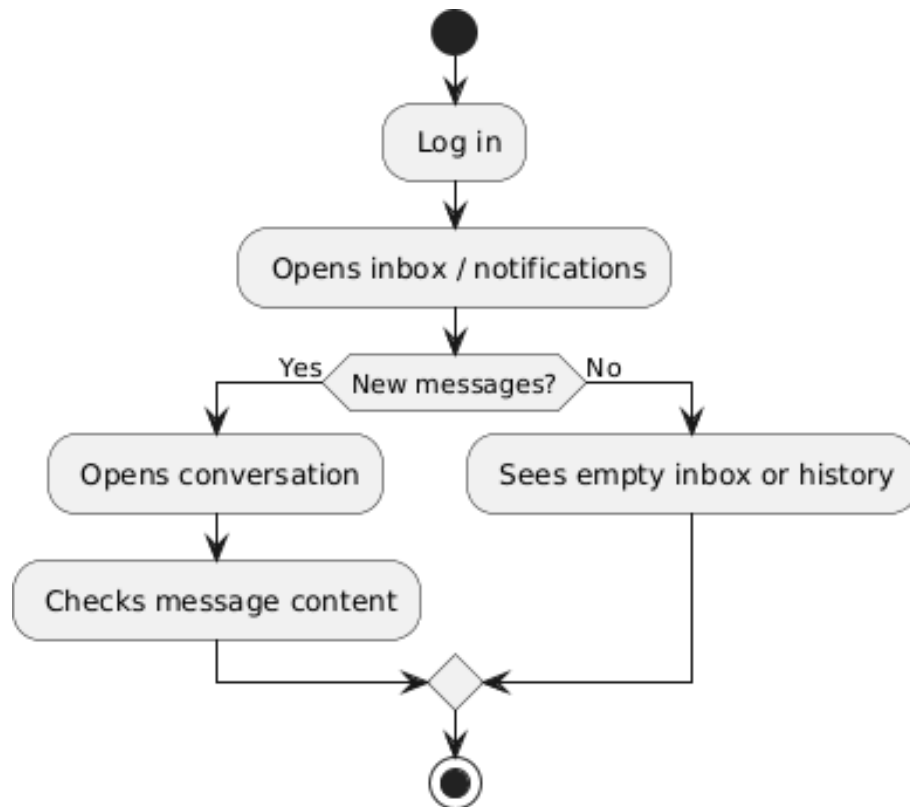
AC_PRS_5 - Send rental requests and message to property owners. (Done by: Anisa Domi)



AC_PRS_6 - Cancel rental request (Done by: Anisa Domi)

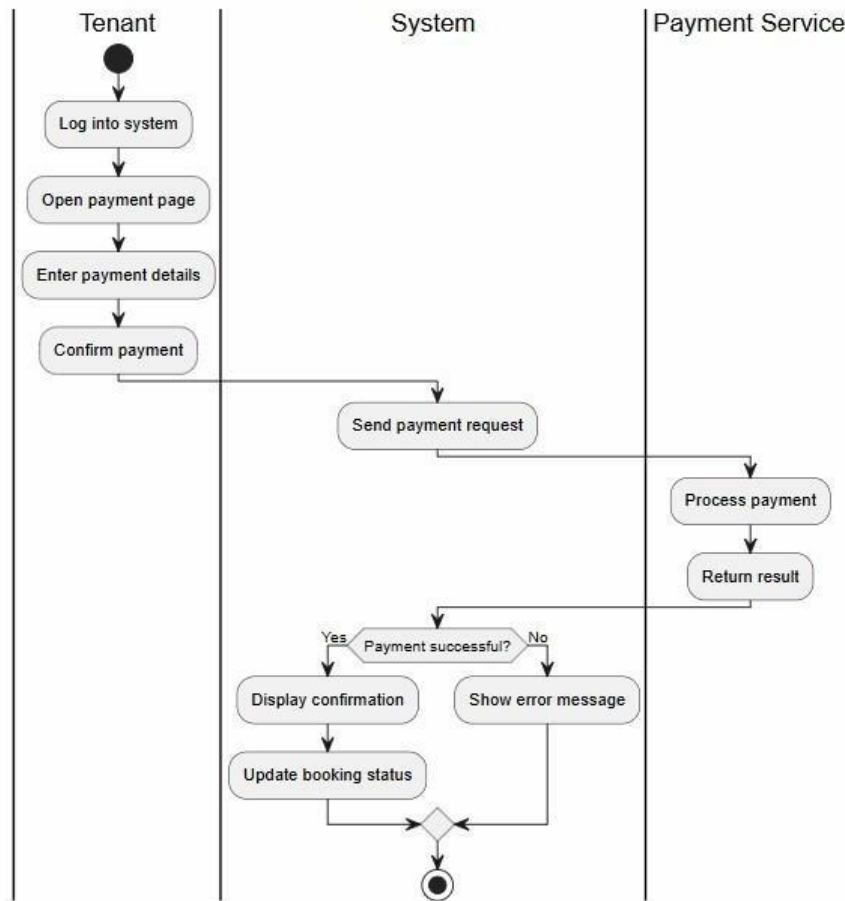


AC_PRS_7 - Receive messages from property owners. (Done by: Haidi Toto)

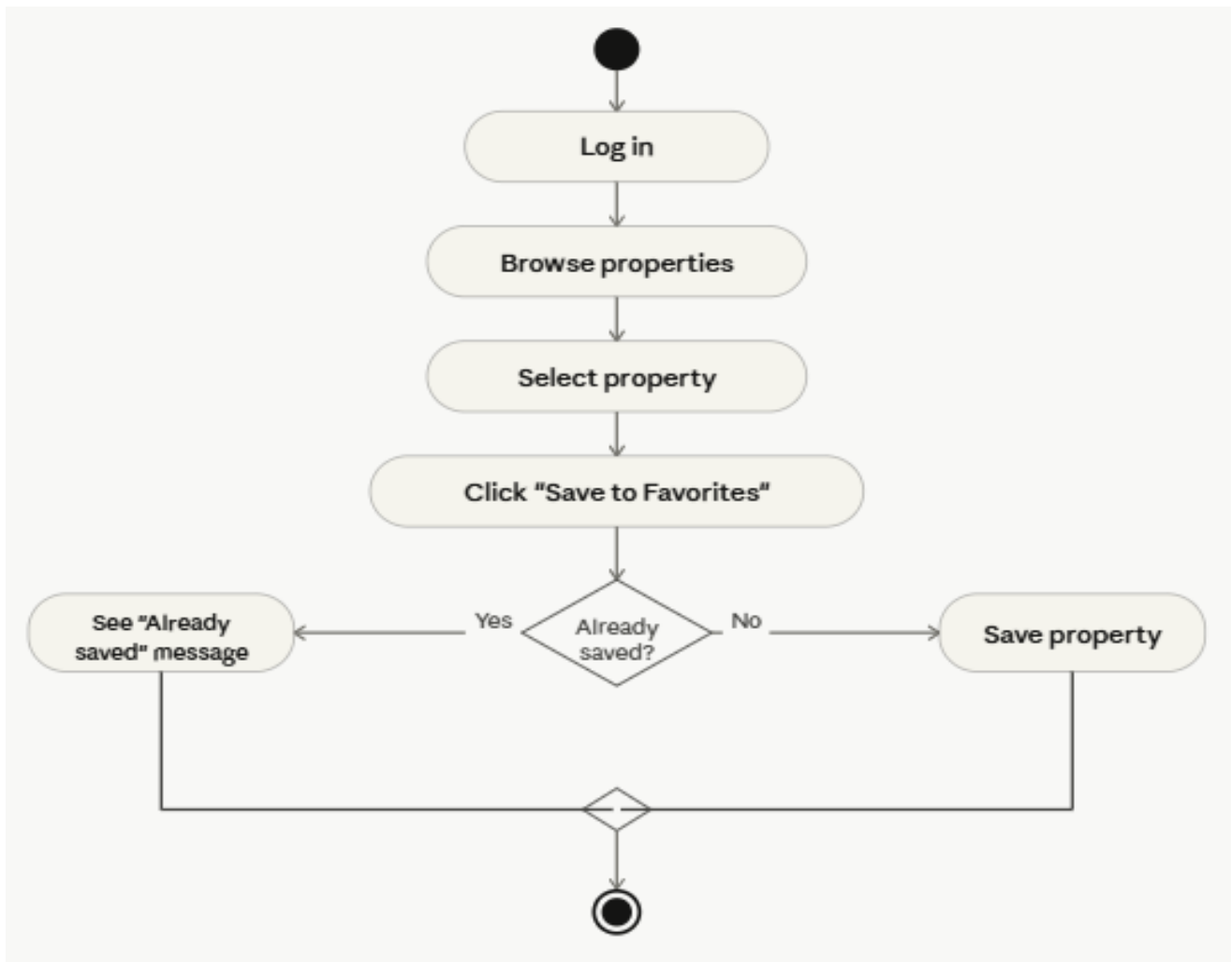


Property Rental System Requirement Specification

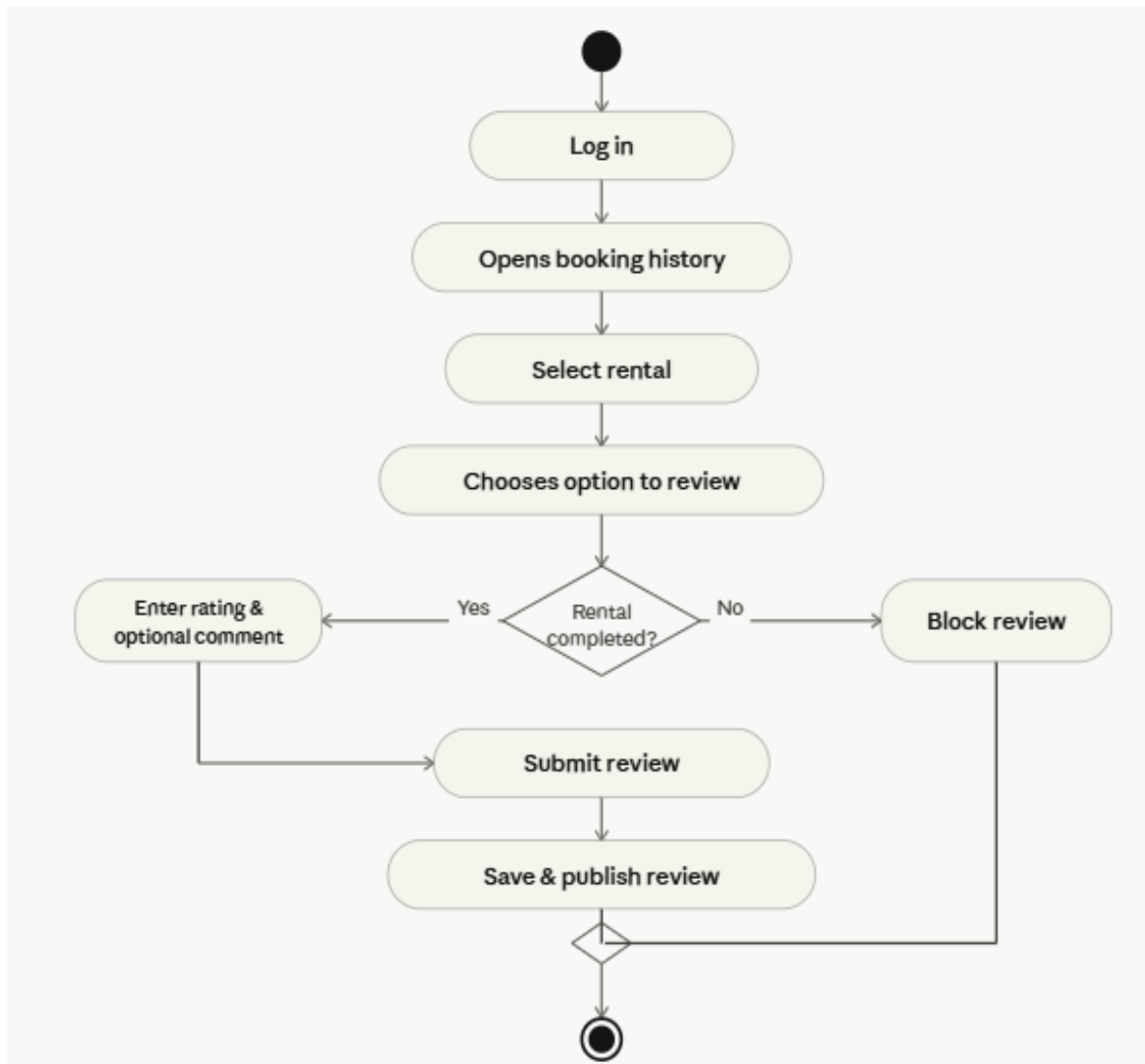
AC_PRS_8 – Submit rental payments. (Done by: Haidi Toto)



AC_PRS_9 - Save favourite properties for later viewing. (Done by: Haidi Toto)

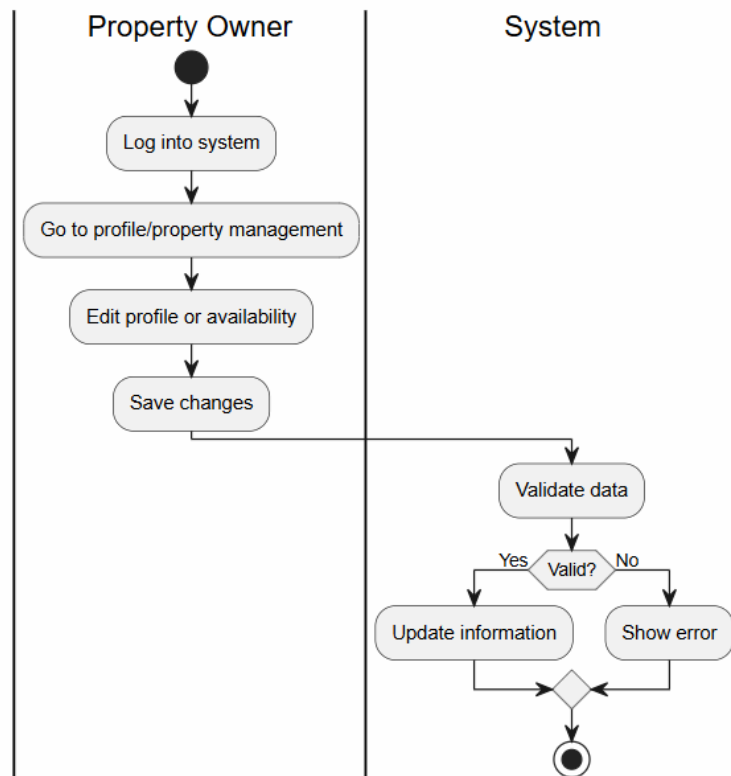


AC_PRS_10 – Review and Rate property (Done by: Haidi Toto)

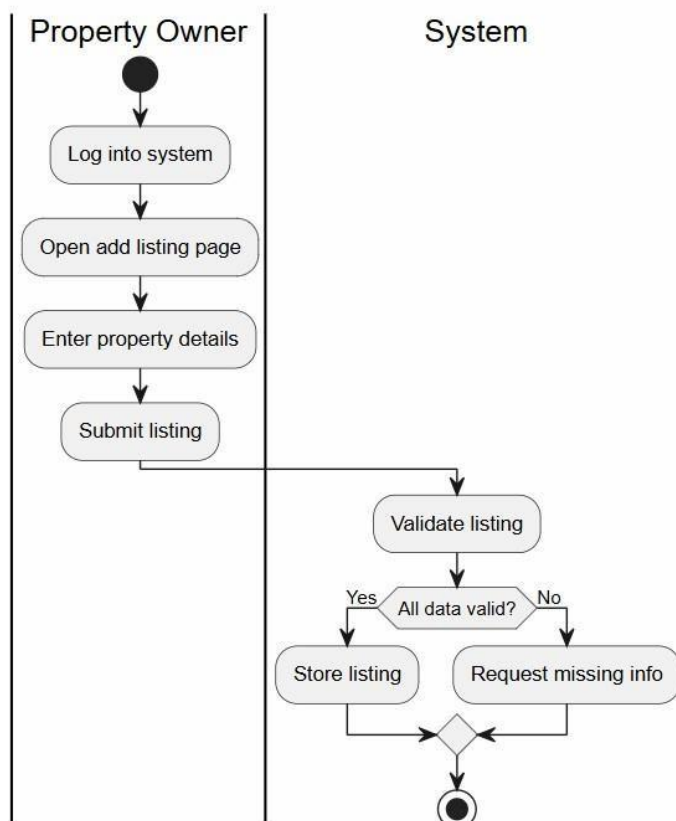


Property Rental System Requirement Specification

AC_PRS_11 - Update Profile Information and availability of rental properties (Done by: Haidi Toto)

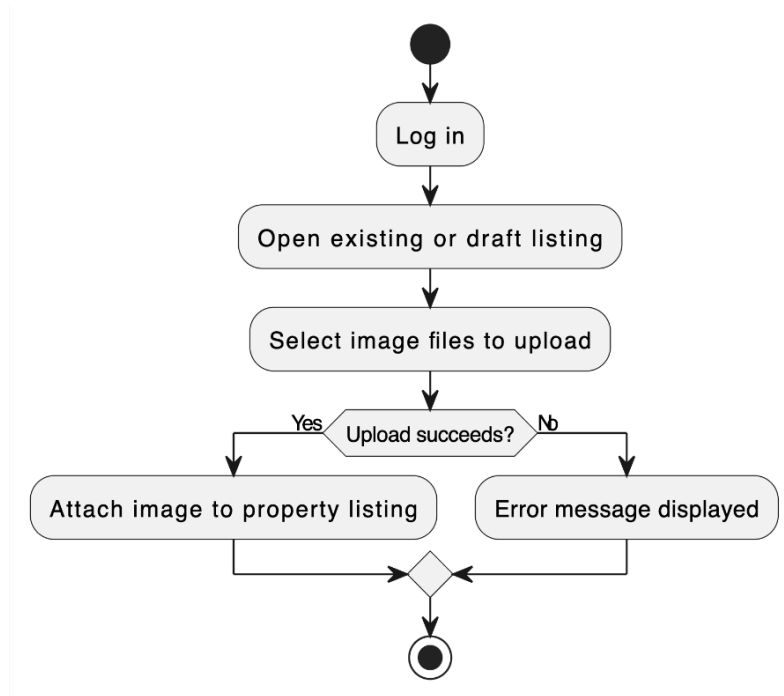


AC_PRS_12 - Add Property Listing (Done by: Haidi Toto)

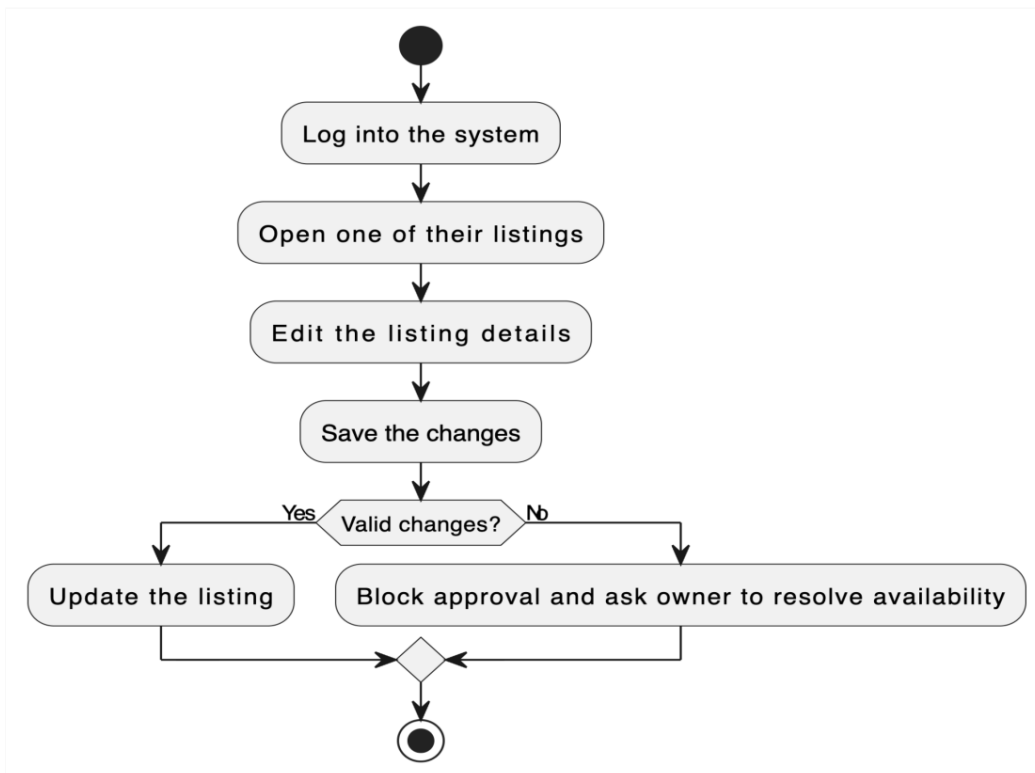


Property Rental System Requirement Specification

AC_PRS_13 - Upload Property Images (Done by: Ledio Tafciu)

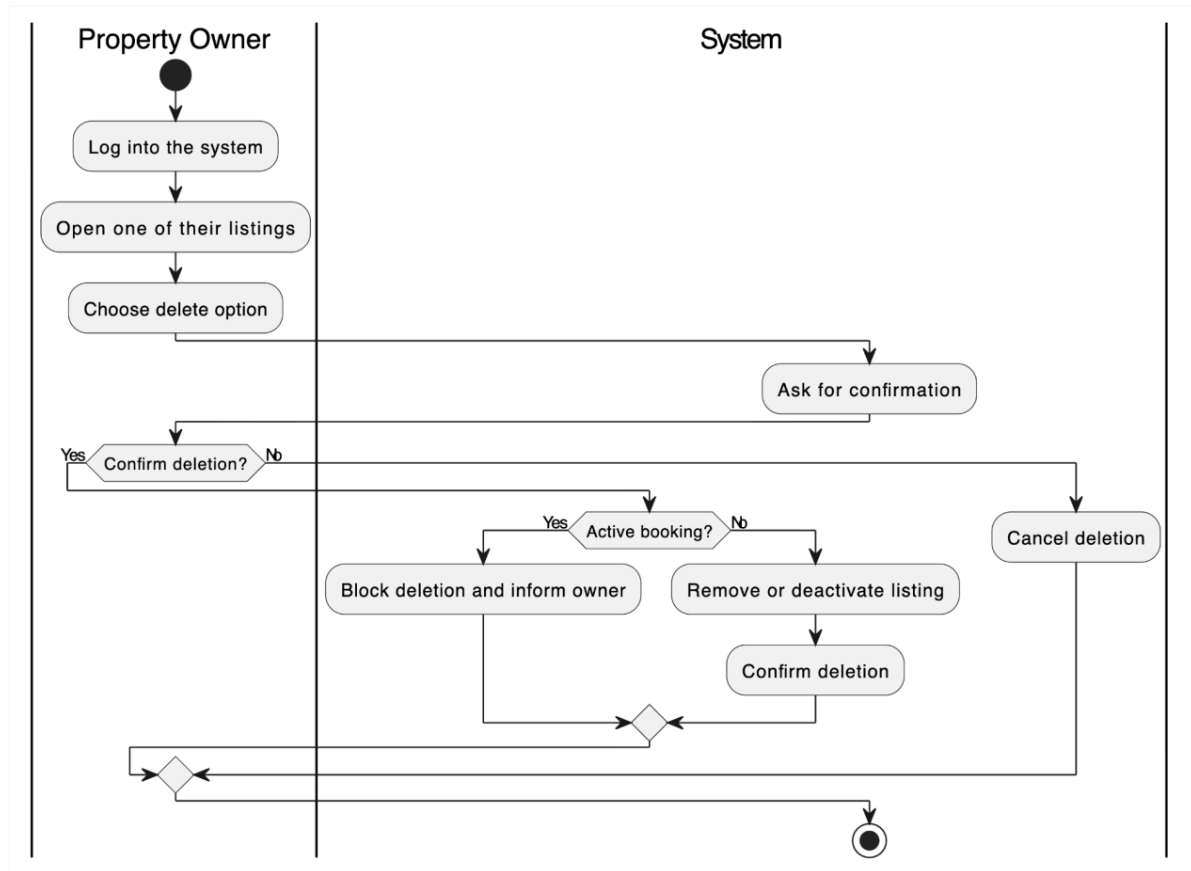


AC_PRS_14 - Edit Property Listing (Done by: Ledio Tafciu)

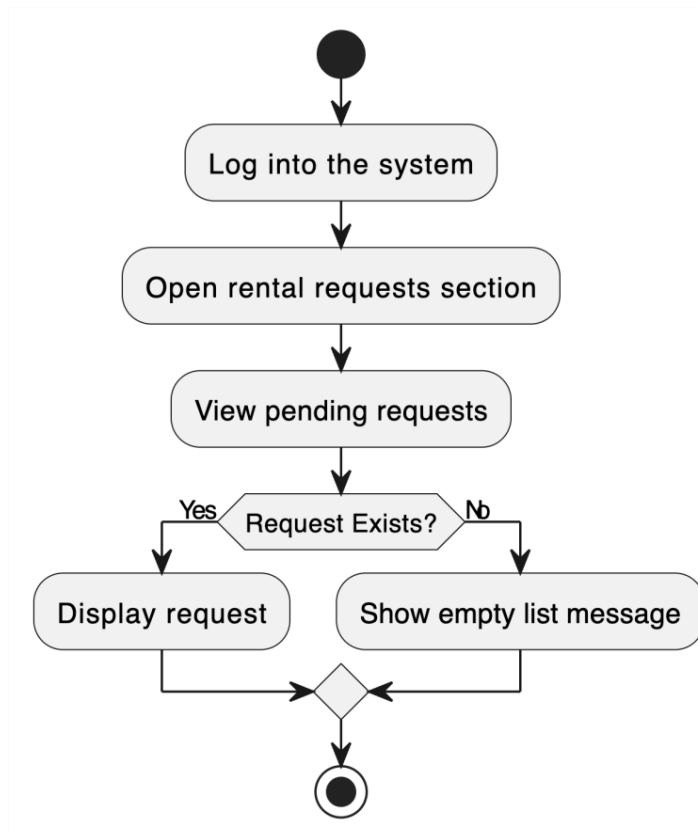


Property Rental System Requirement Specification

AC_PRS_15 - Delete Property Listing (Done by: Ledio Tafciu)

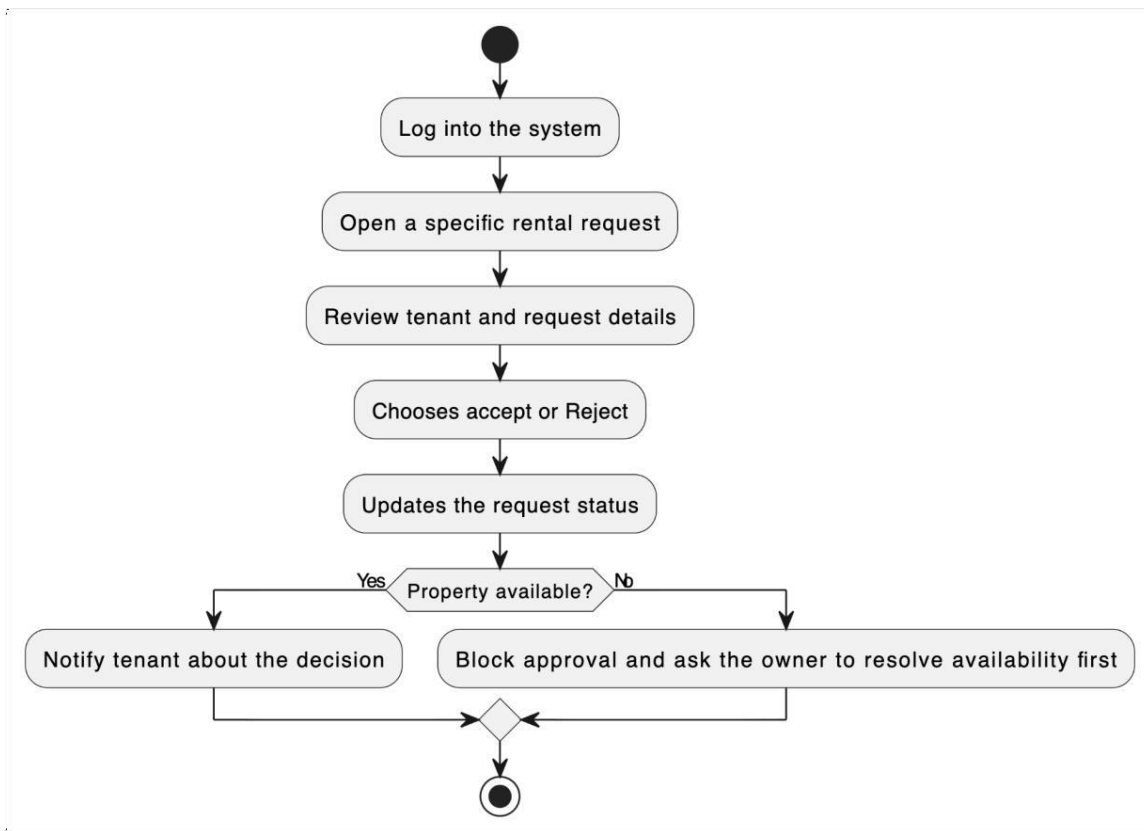


AC_PRS_16 - View Rental Requests (Done by: Ledio Tafciu)

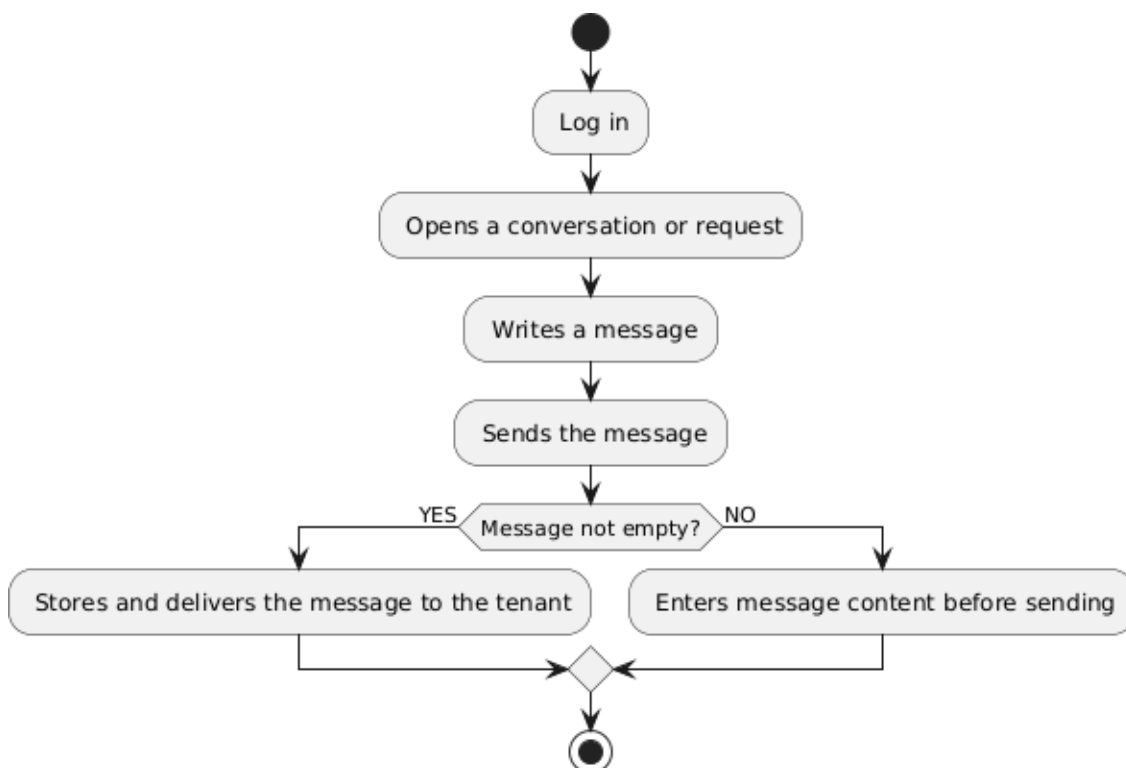


Property Rental System Requirement Specification

AC_PRS_17 - Accept or Reject Rental Request (Done by: Ledio Tafciu)

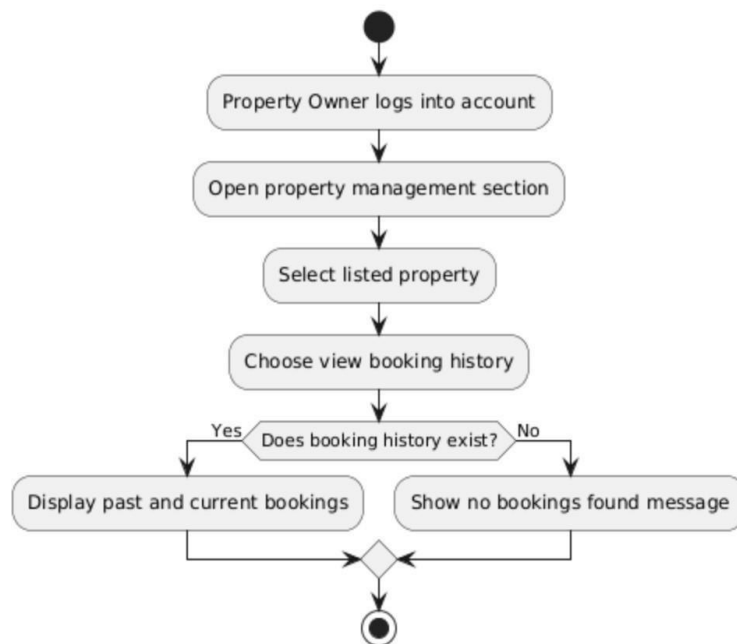


AC_PRS_18 - Send Message to Tenant (Done by: Ledio Tafciu)

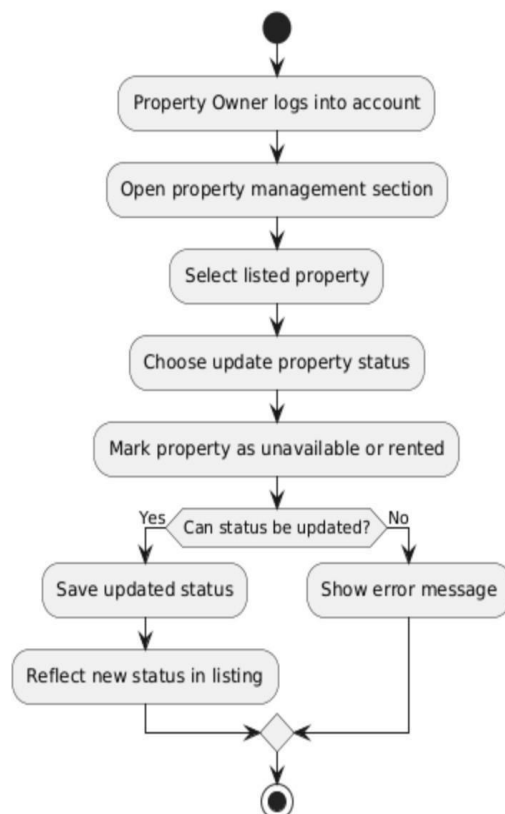


Property Rental System Requirement Specification

AC_PRS_19 - View Booking History (Done by: Desard Lybeshari)

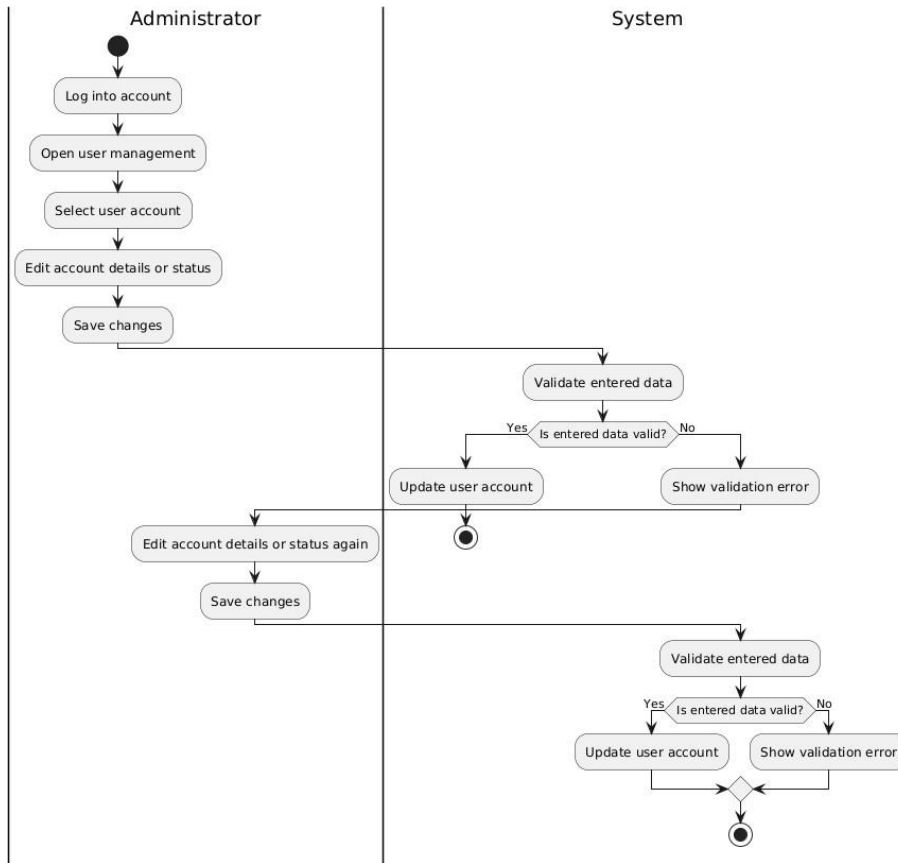


AC_PRS_20 - Mark Property as Unavailable or Rented (Done by: Desard Lybeshari)

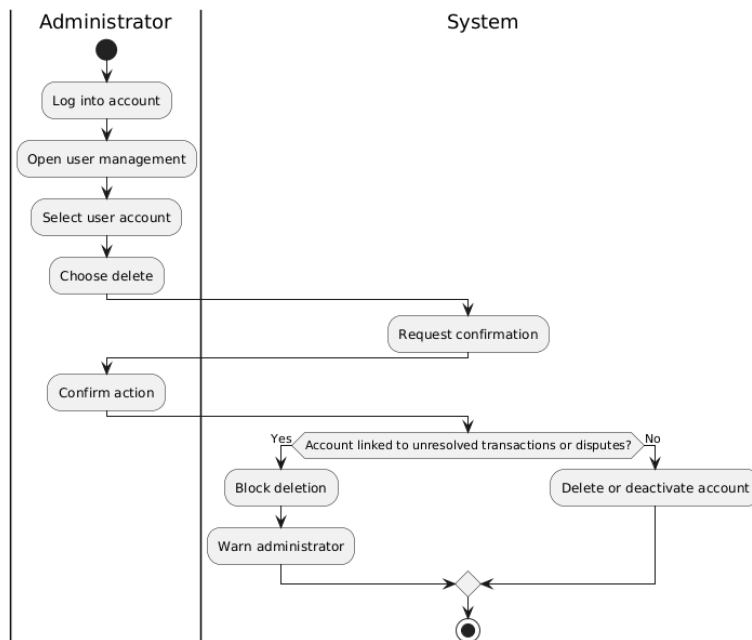


Property Rental System Requirement Specification

AC_PRS_21 - Edit User Accounts (Done by: Desard Lybeshari)

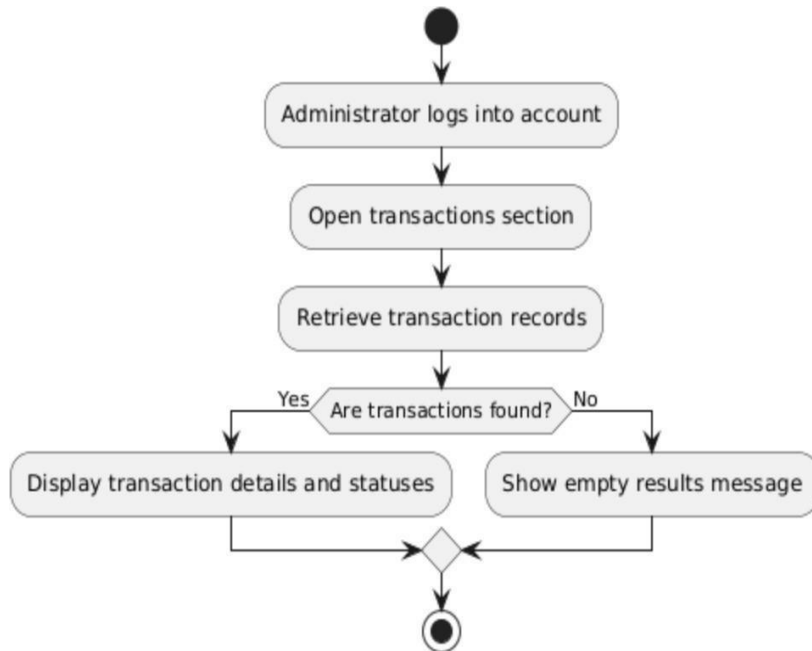


AC_PRS_22 - Delete User Accounts (Done by: Desard Lybeshari)

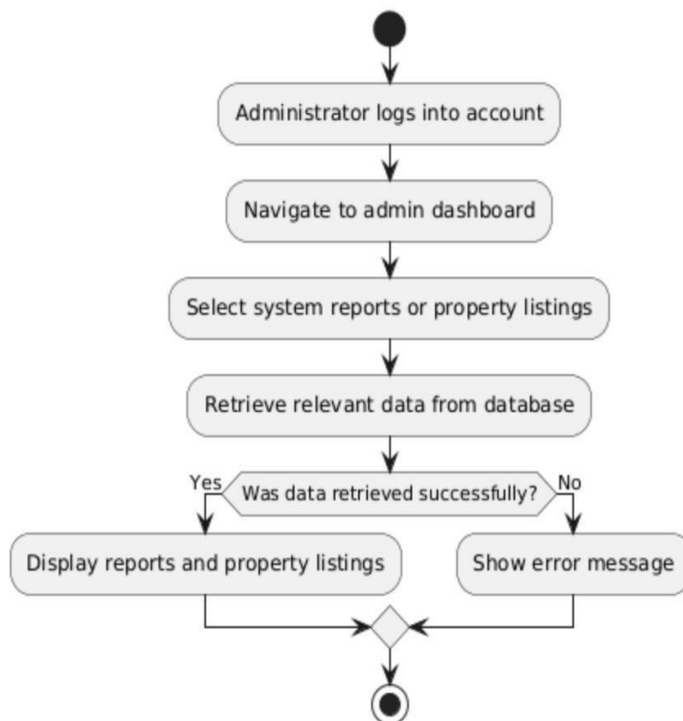


Property Rental System Requirement Specification

AC_PRS_23 - Monitor Rental Transactions (Done by: Desard Lybeshari)



AC_PRS_24 - View System Reports and Property Listings (Done by: Desard Lybeshari)

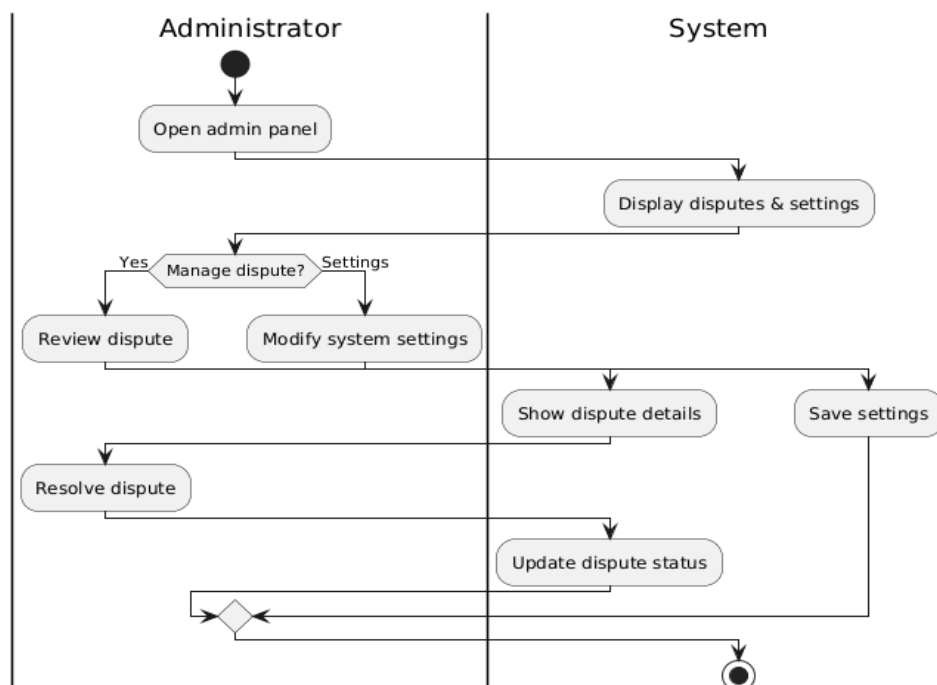


Property Rental System Requirement Specification

AC_PRS_25 - Remove Inappropriate Listings (Done by: Matias Bali)

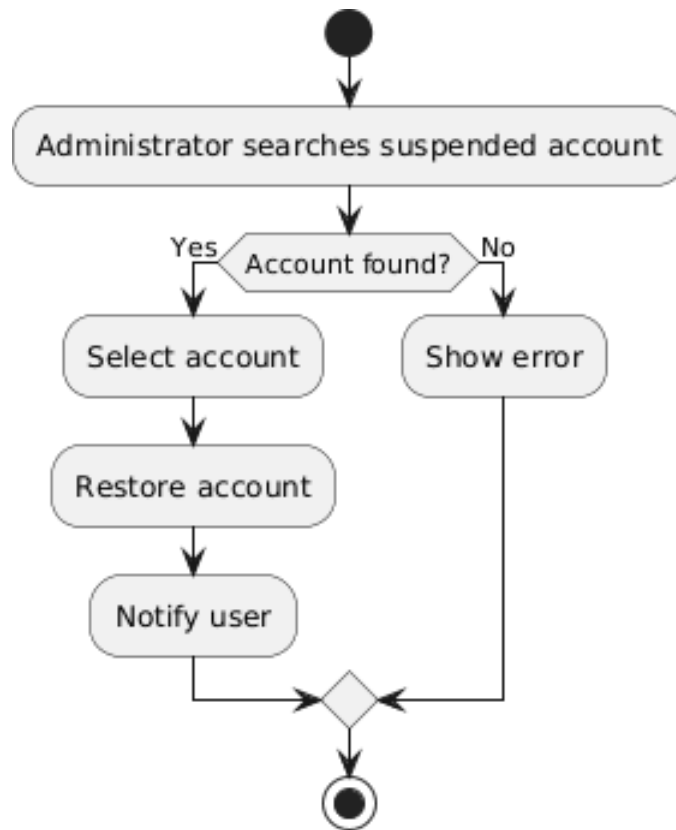


AC_PRS_26 - Manage Disputes and System Settings (Done by: Matias Bali)

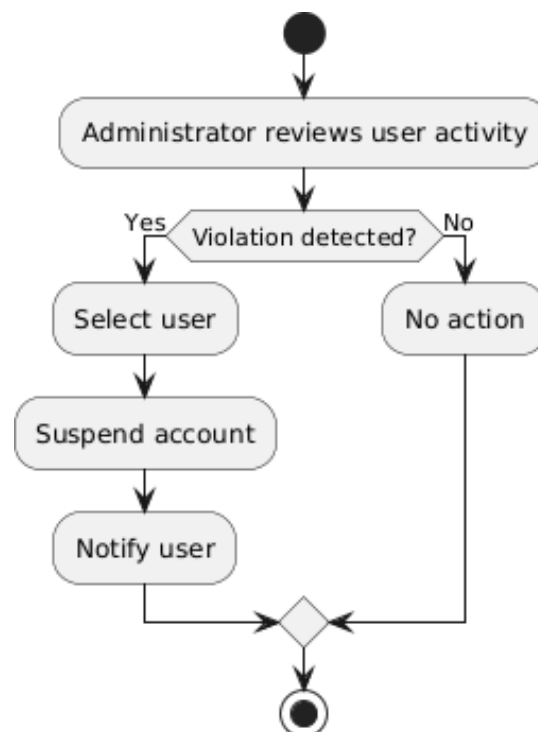


Property Rental System Requirement Specification

AC_PRS_27 - Restore User Accounts (Done by: Matias Bali)

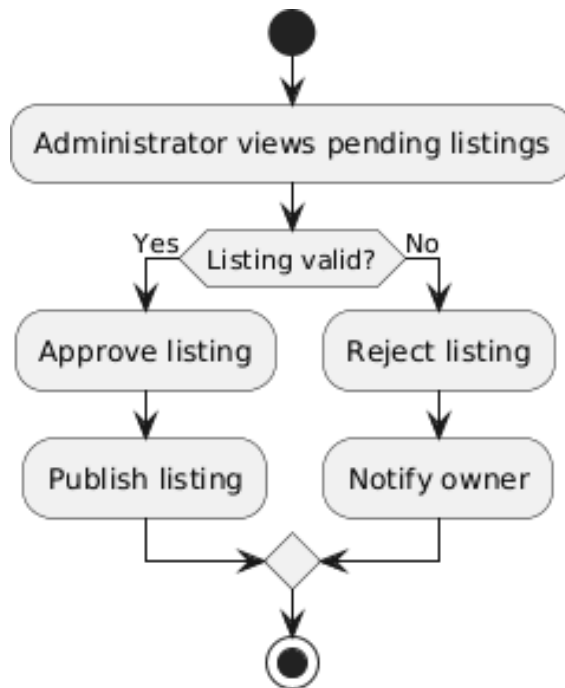


AC_PRS_28 - Suspend User Accounts (Done by: Matias Bali)

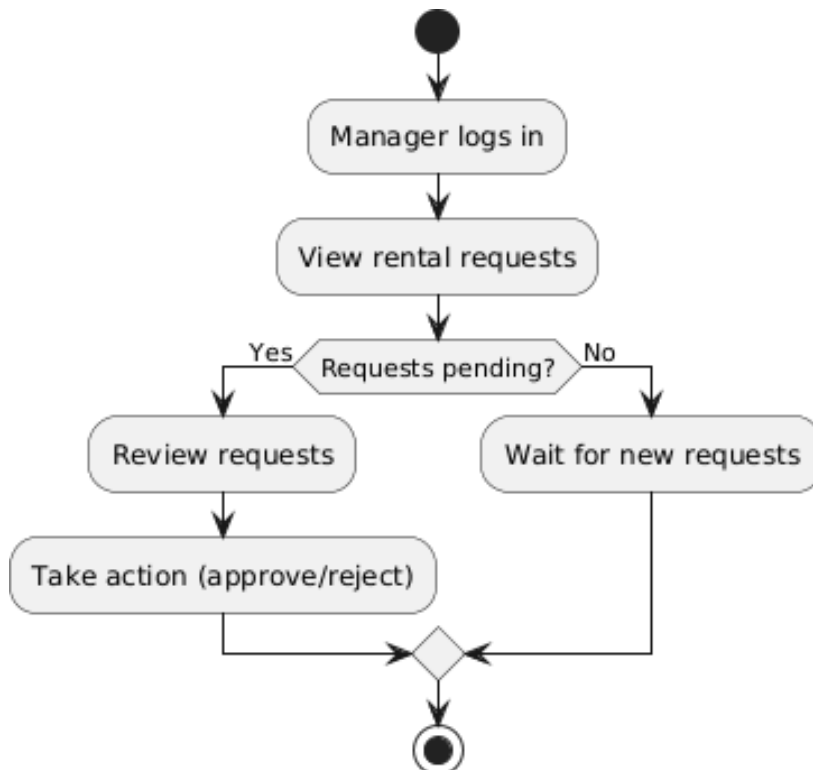


Property Rental System Requirement Specification

AC_PRS_29 - Verify and Approve property listings (Done by: Matias Bali)

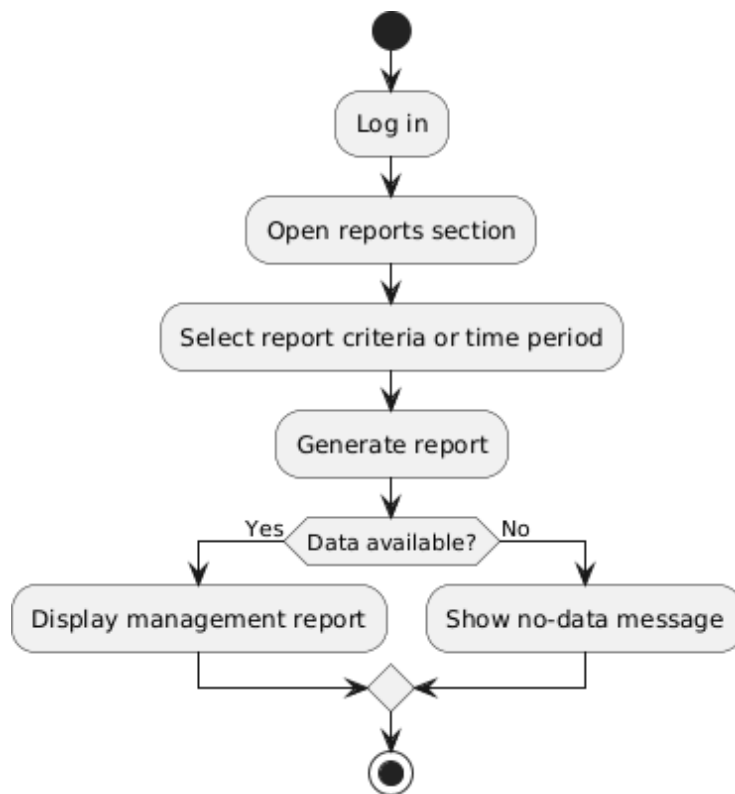


AC_PRS_30 - Monitor Rental Requests (Done by: Matias Bali)



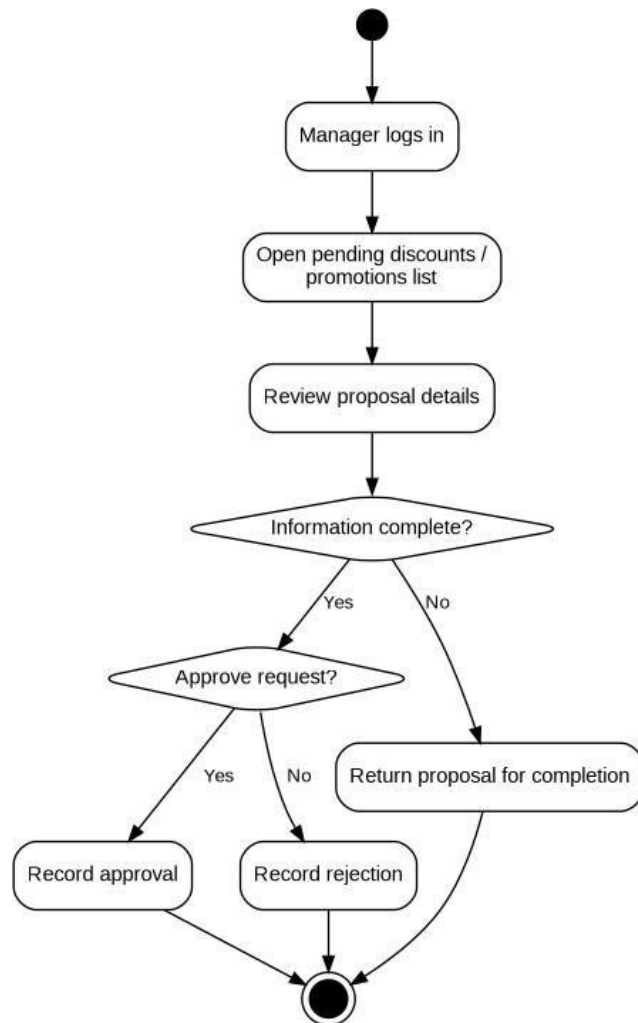
Property Rental System Requirement Specification

AC_PRS_31 - Generate Management Reports (Done by: Ornelt Lemellari)



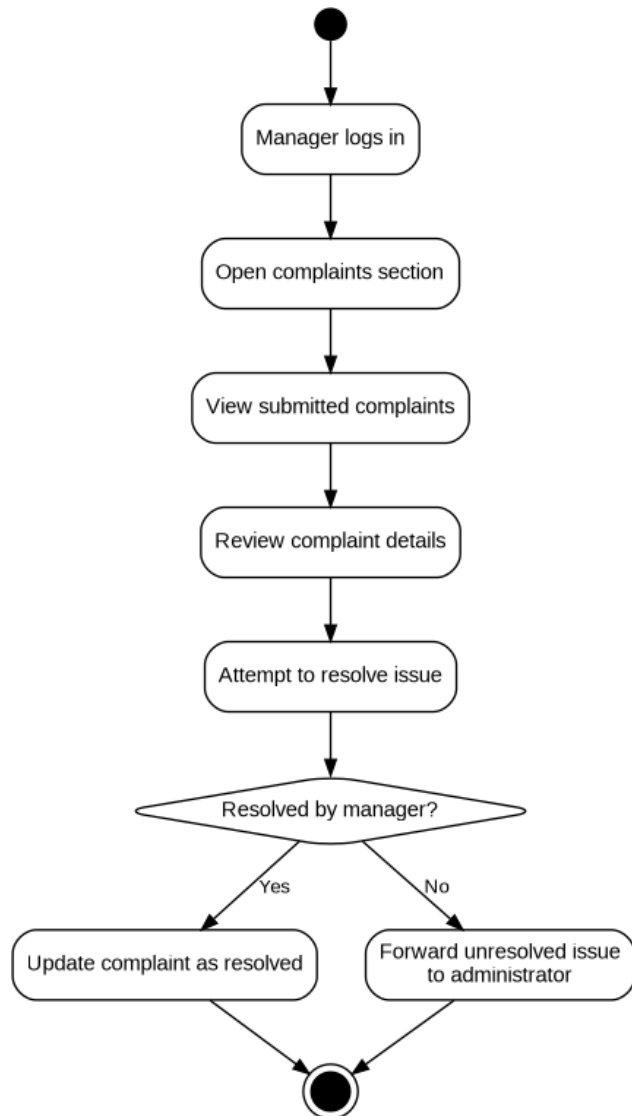
Property Rental System Requirement Specification

AC_PRS_32 - Approve Large Discounts and Promotions (Done by: Ornelt Lemellari)



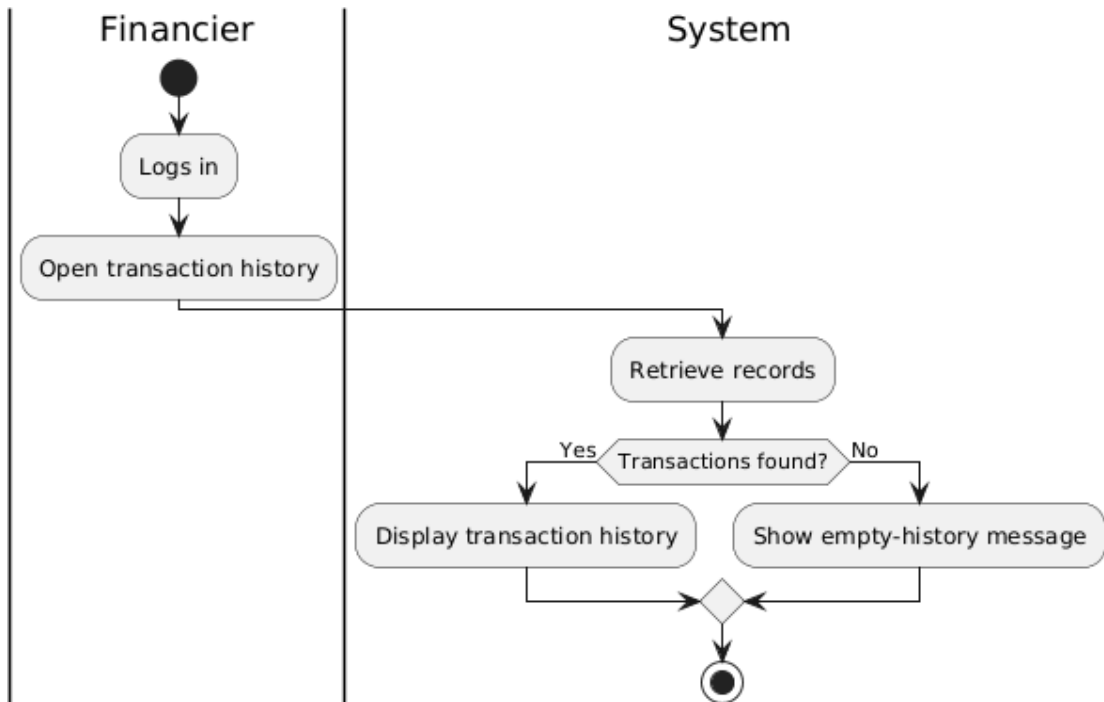
Property Rental System Requirement Specification

AC_PRS_33 - Review Complaints and Forward Unresolved Issues (Done by: Ornelt Lemellari)

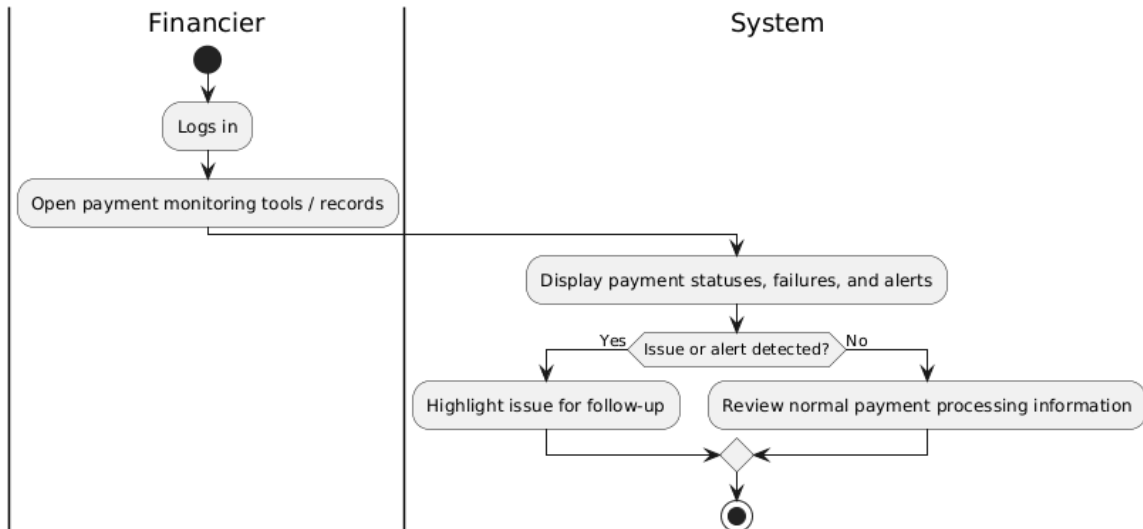


Property Rental System Requirement Specification

AC_PRS_34 - View Transaction History (Done by: Ornel Lemellari)

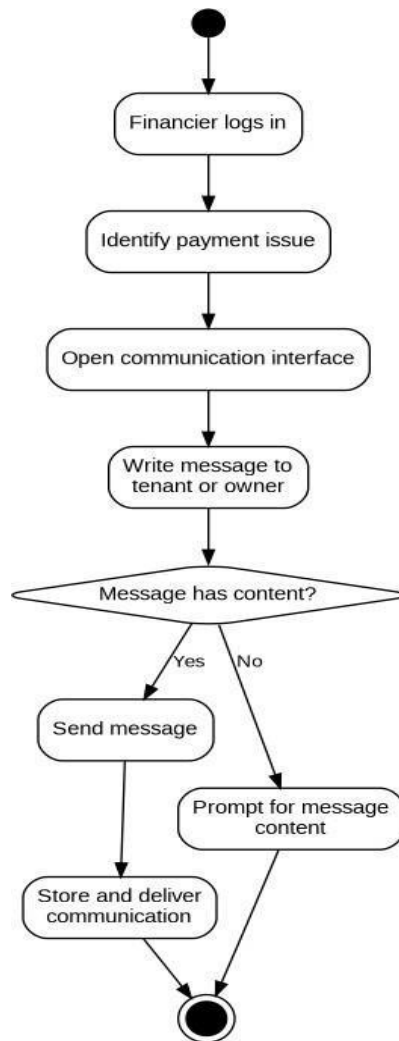


AC_PRS_35 - Monitor Payment Process and Security (Done by: Ornel Lemellari)



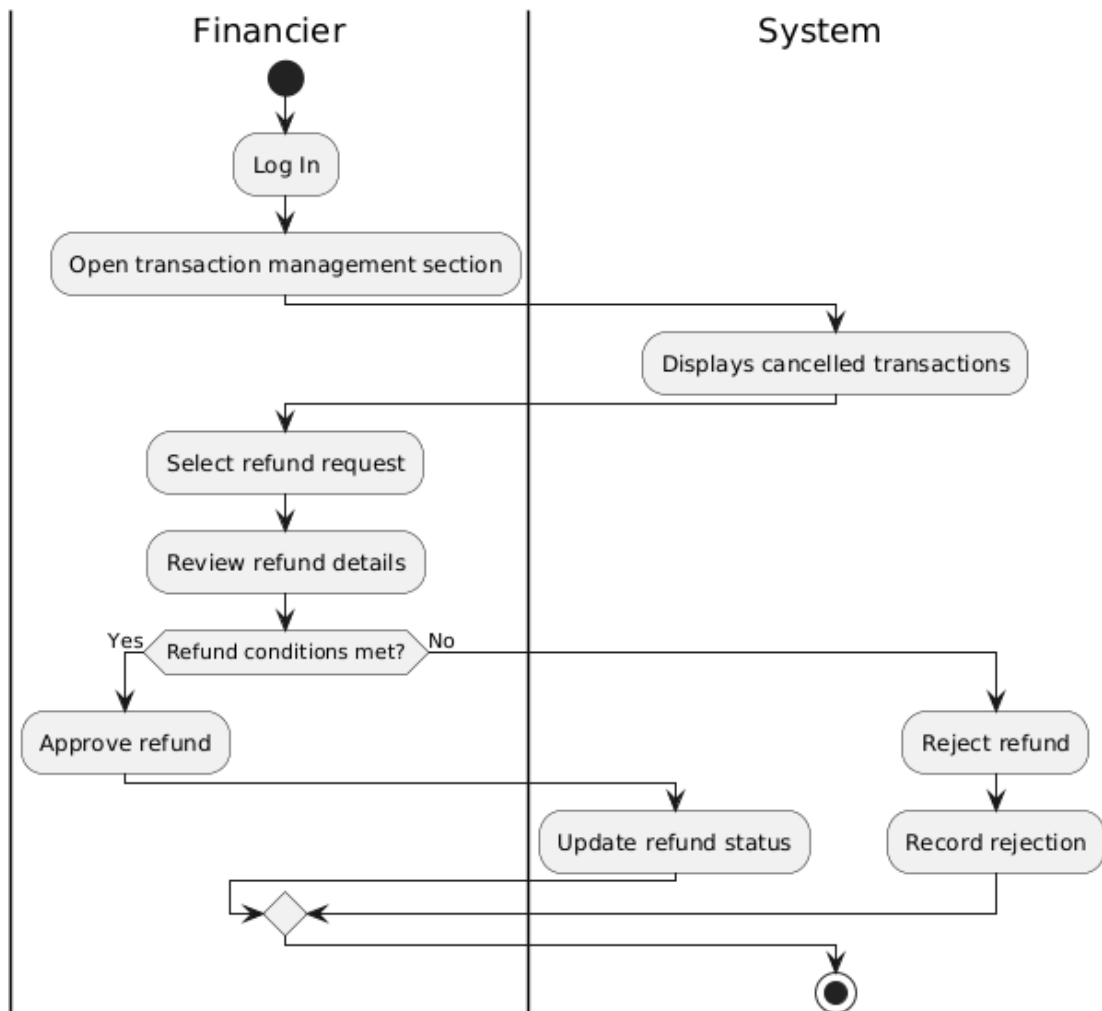
Property Rental System Requirement Specification

AC_PRS_36 - Communicate Payment Issues with Tenants and Owners (Done by: Ornelt Lemellari)



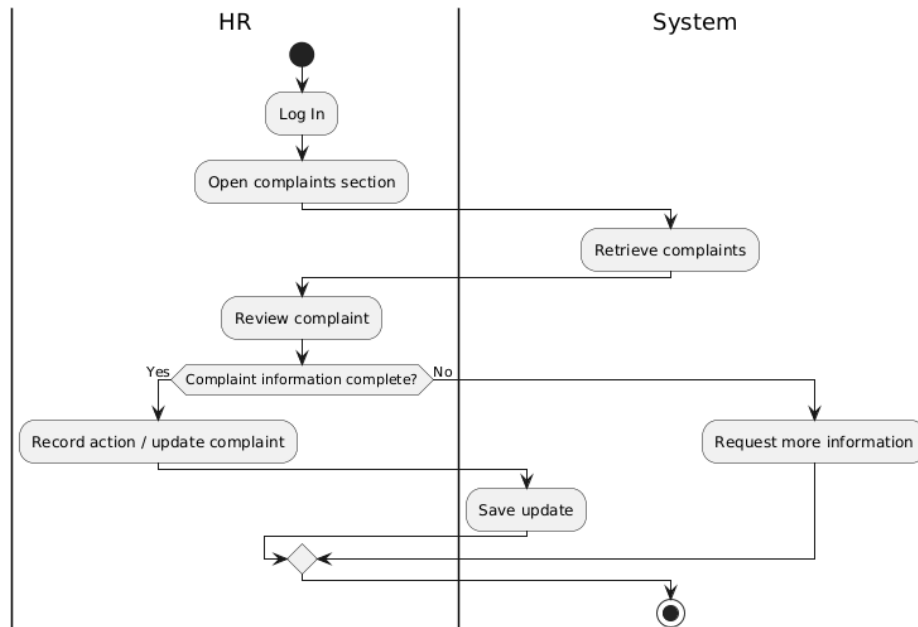
Property Rental System Requirement Specification

AC_PRS_37 - Approve Refunds for cancelled Rental Transactions (Done by: Kevin Muho)

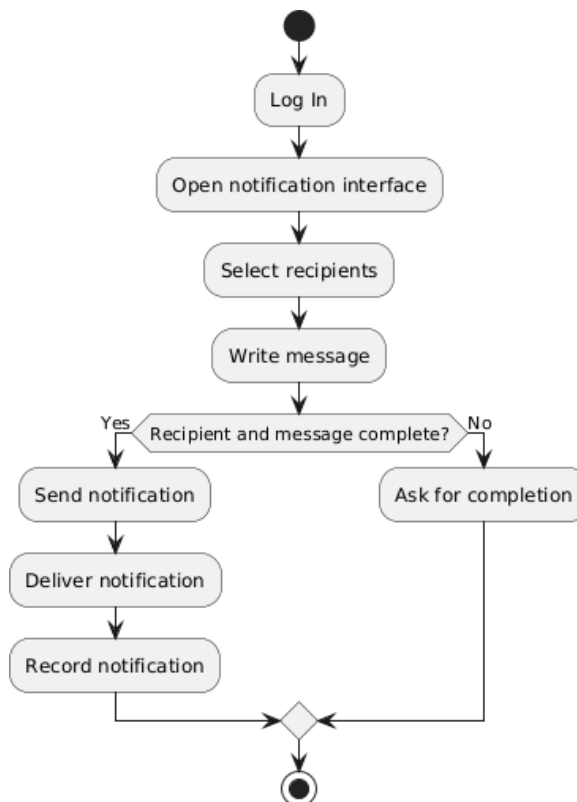


Property Rental System Requirement Specification

AC_PRS_38 - Manage Complaints (Done by: Kevin Muho)

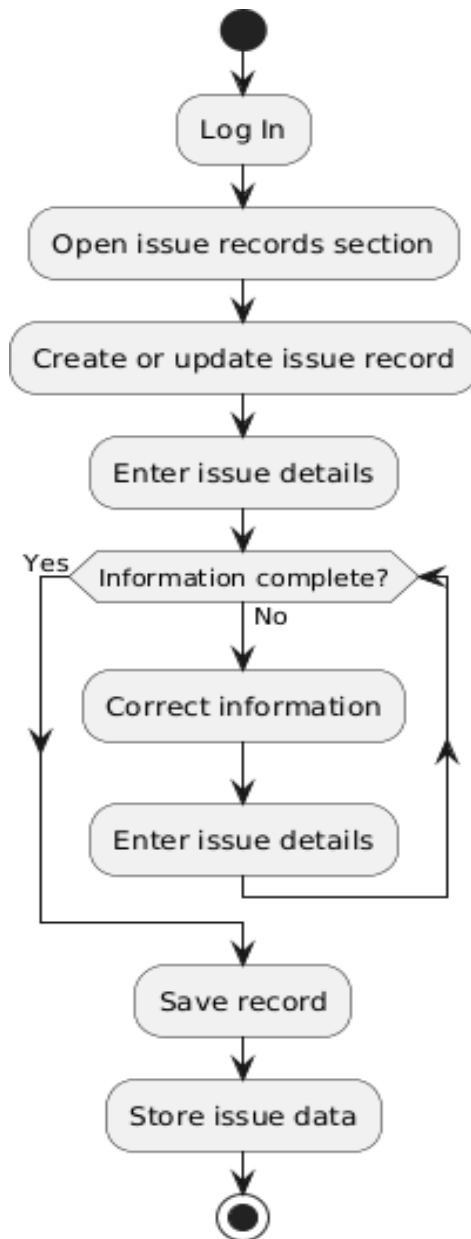


AC_PRS_39 - Send Notifications and Reminders (Done by: Kevin Muho)

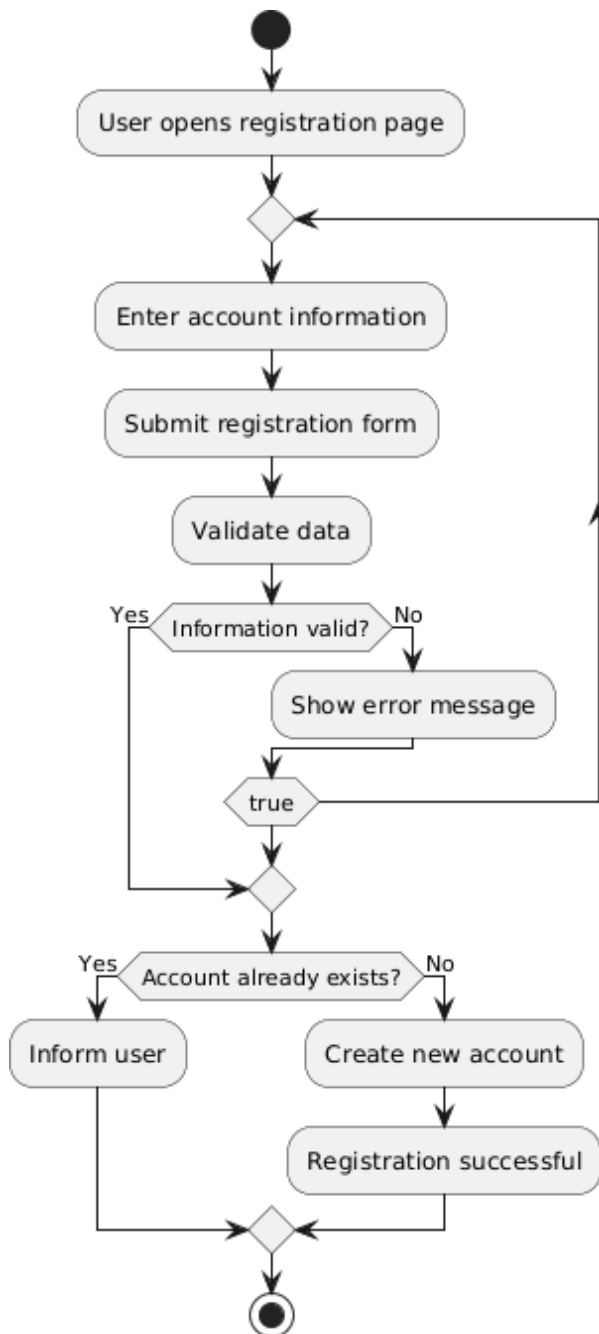


Property Rental System Requirement Specification

AC_PRS_40 - Maintain Records of User Issues and Resolutions (Done by: Kevin Muho)

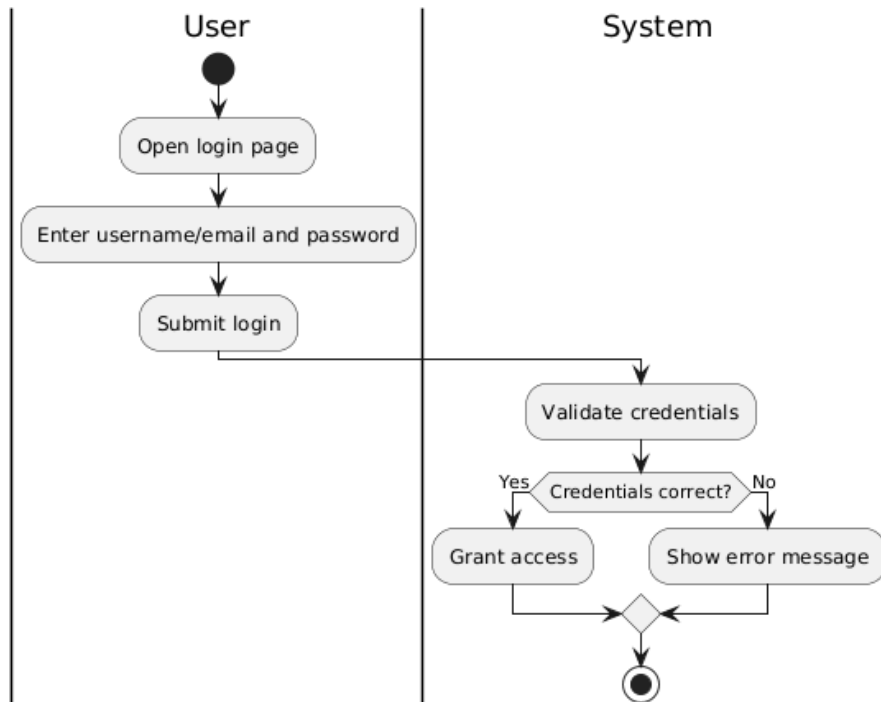


AC_PRS_41 - Register Account (Done by: Kevin Muho)

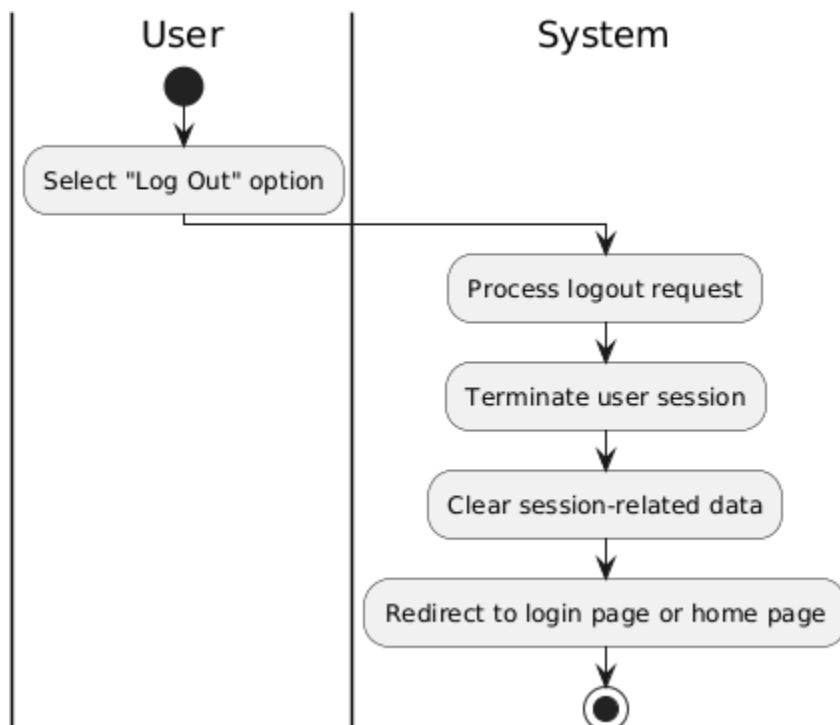


Property Rental System Requirement Specification

AC_PRS_42 - Log In (Done by: Kevin Muho)

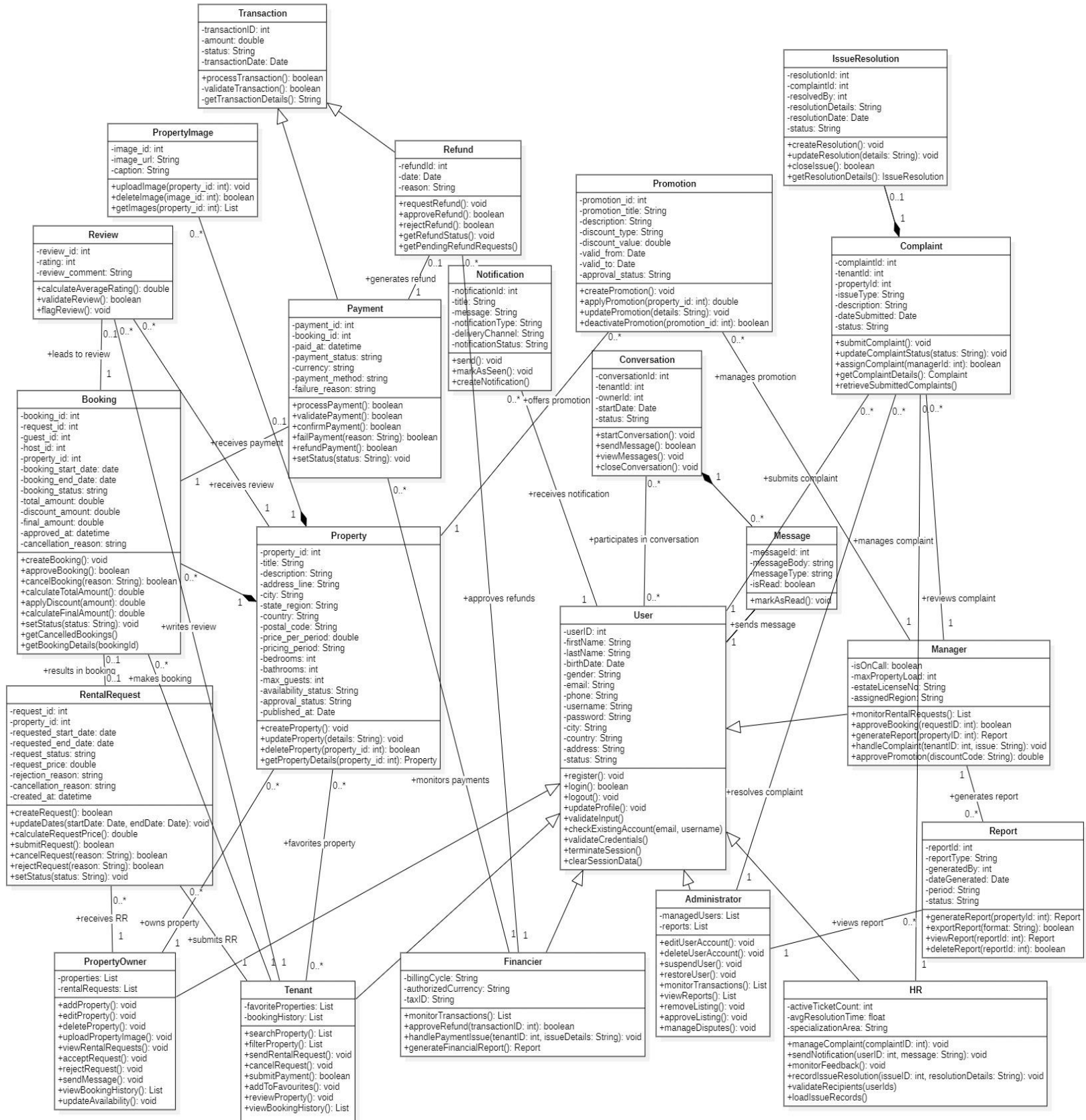


AC_PRS_43 – Log Out (Done by: Kevin Muho)



Property Rental System Requirement Specification

5.4. Class diagram.

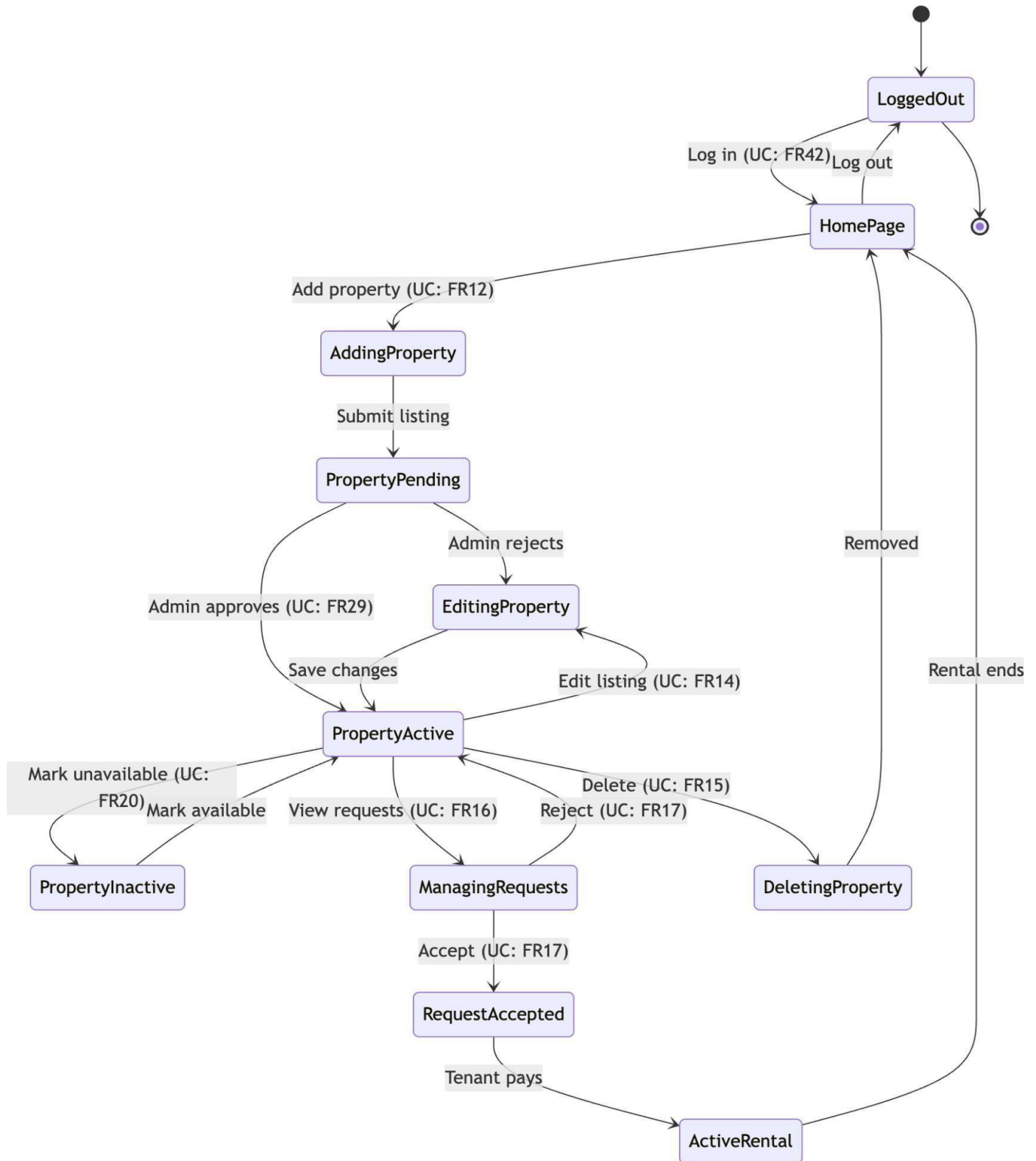


5.5 State diagram

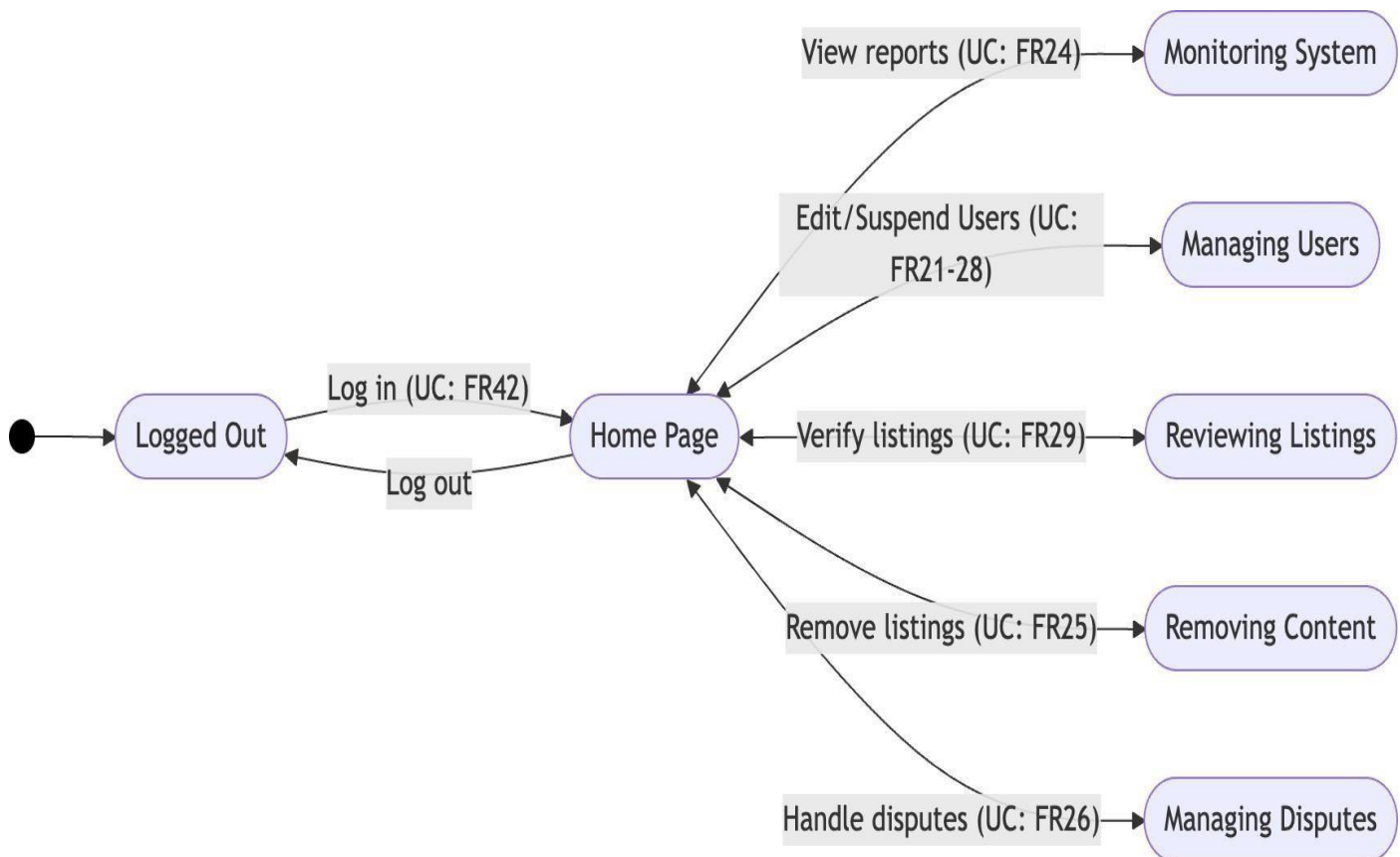
State Diagram – Tenant (Done by: Anisa Domi)



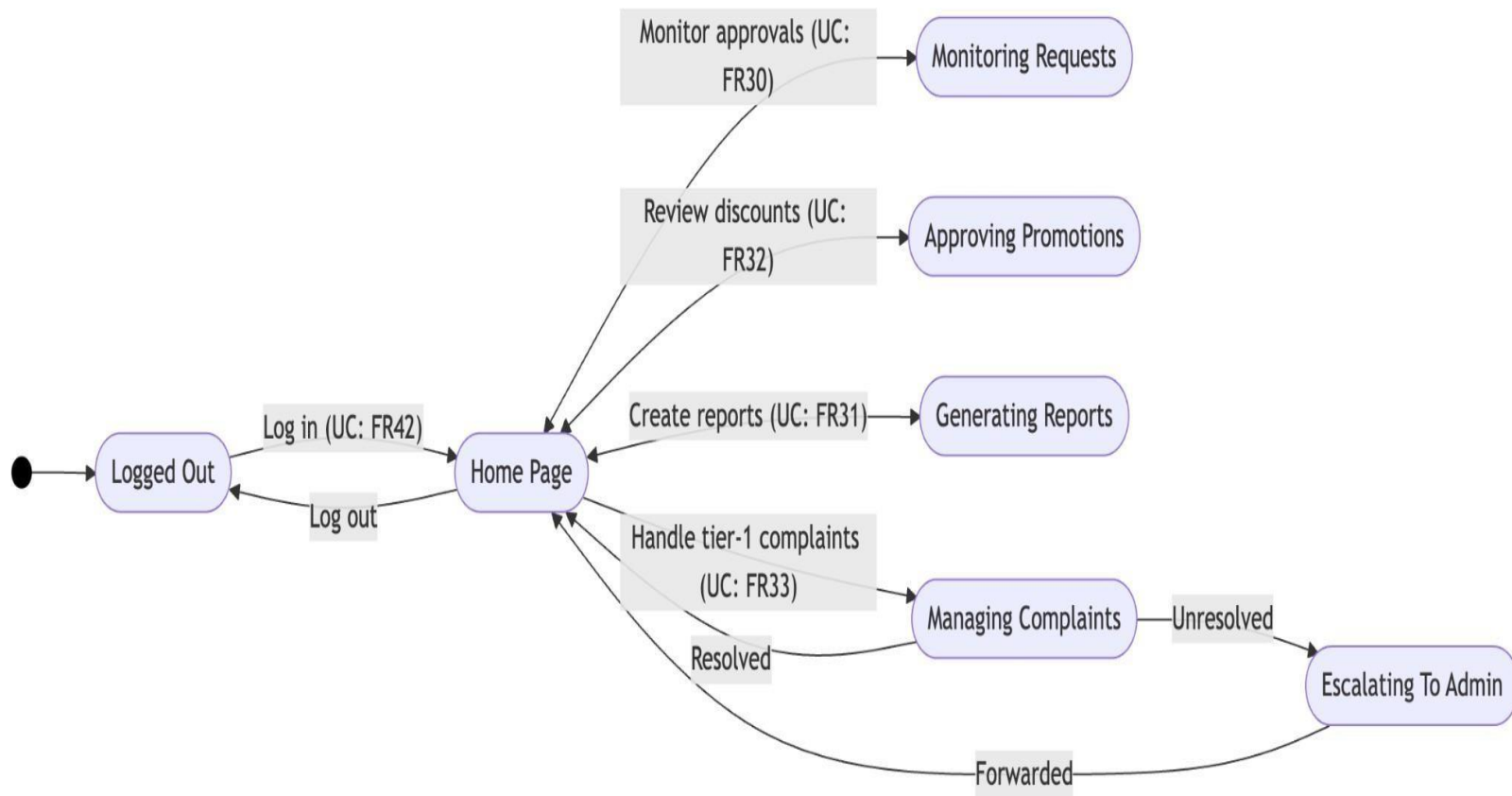
State Diagram - Property Owner (Done by:Haidi Toto)



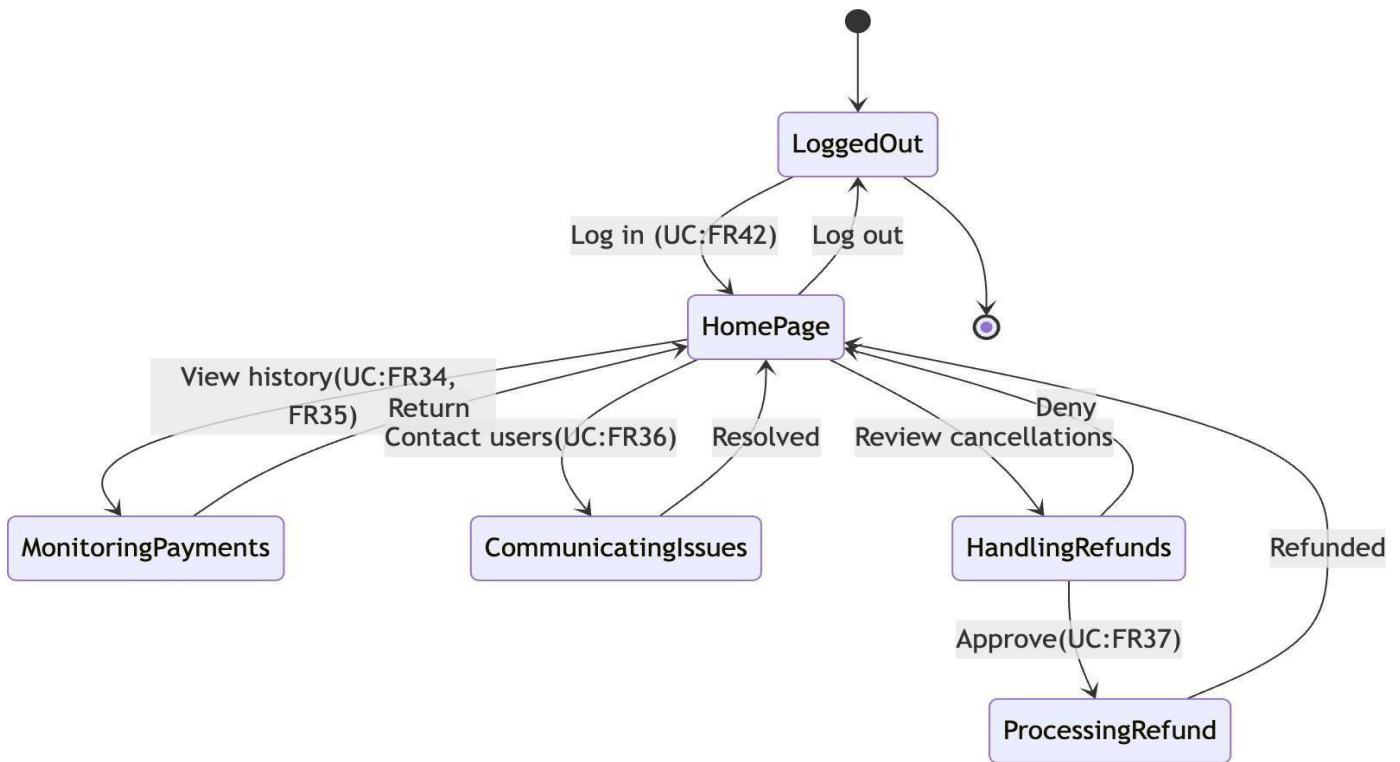
State Diagram – Administrator (Done by:Ledio Tafciu)



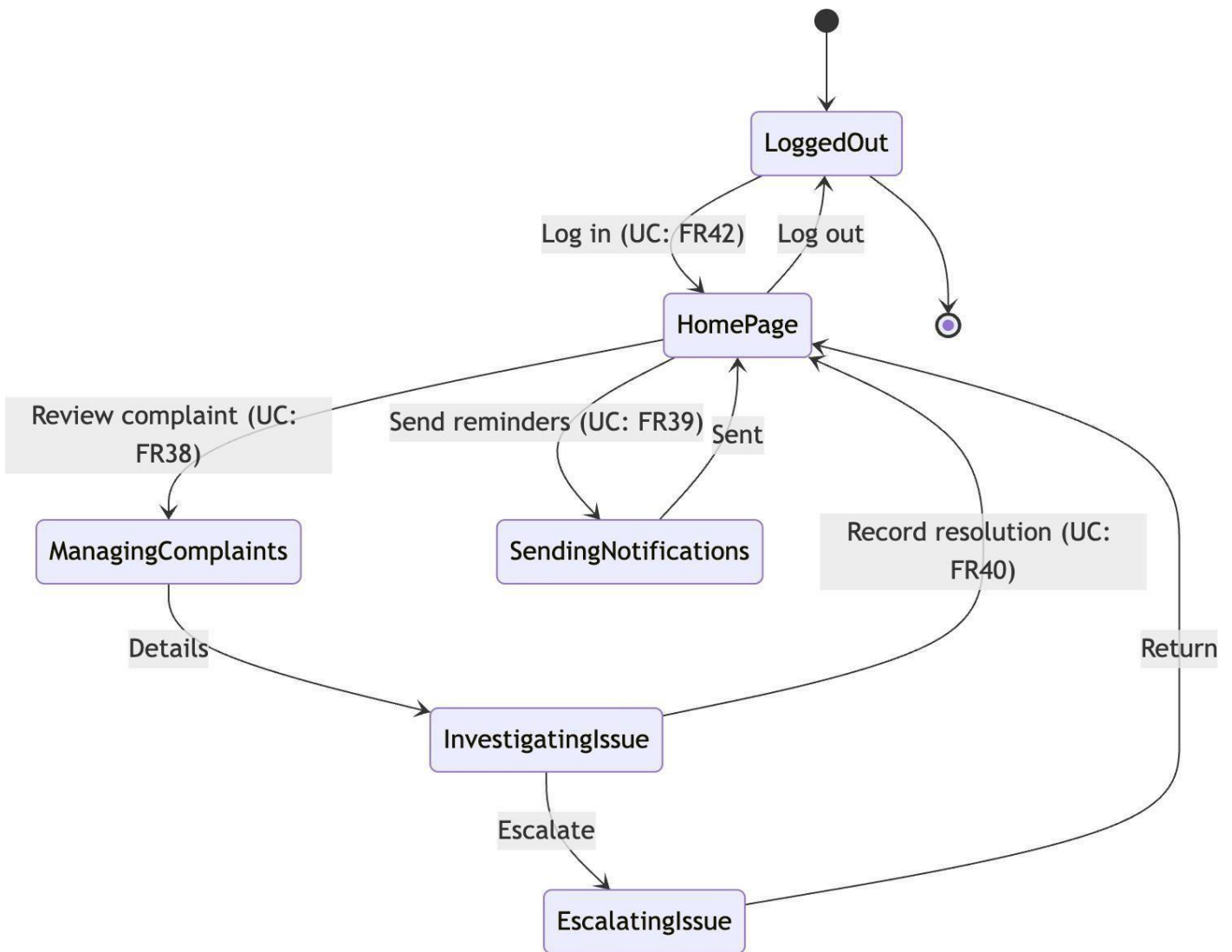
State Diagram – Manager (Done by:Matias Bali)



State Diagram – Financier (Done by:Ornelt Lemellari)



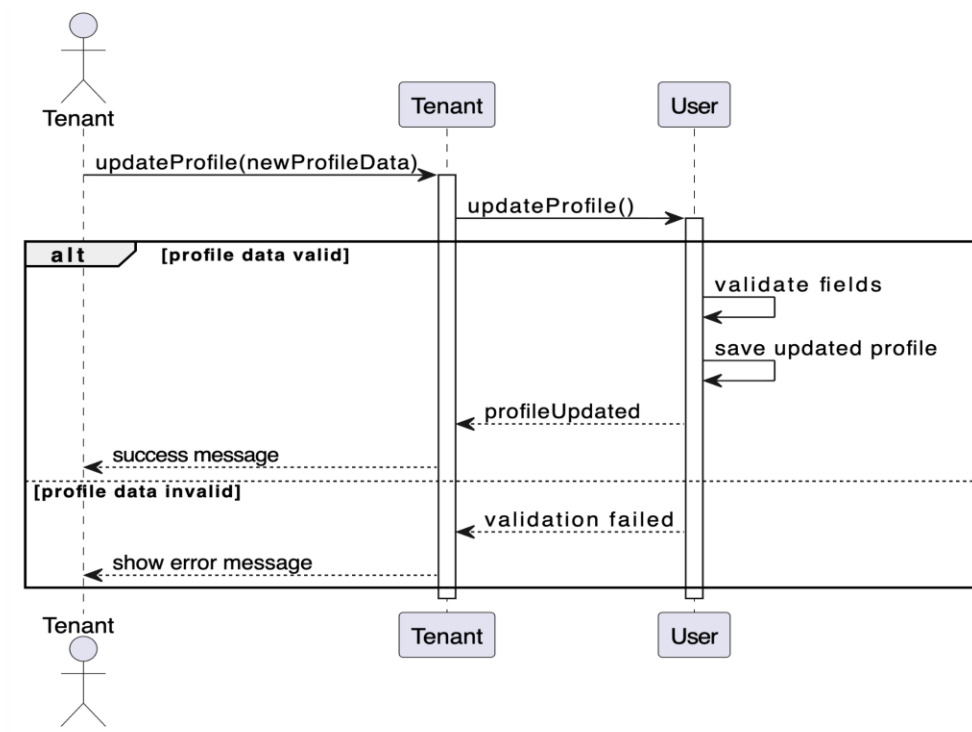
Property Rental System Requirement Specification
State Diagram – HR (Done by:Desard Lybeshari)



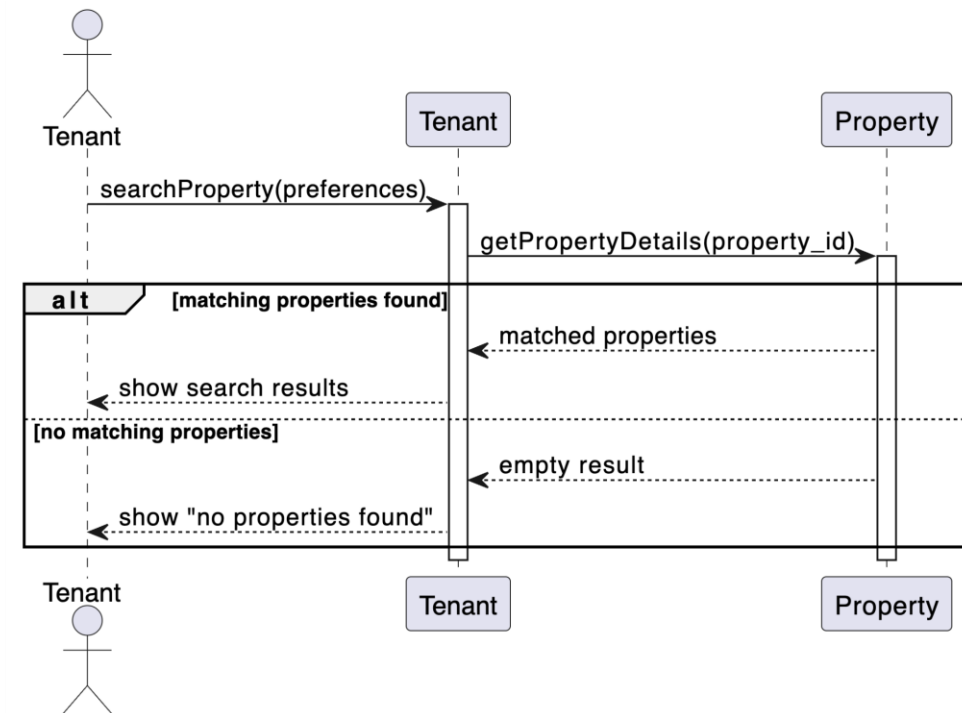
Property Rental System Requirement Specification

5.6 Sequence diagram.

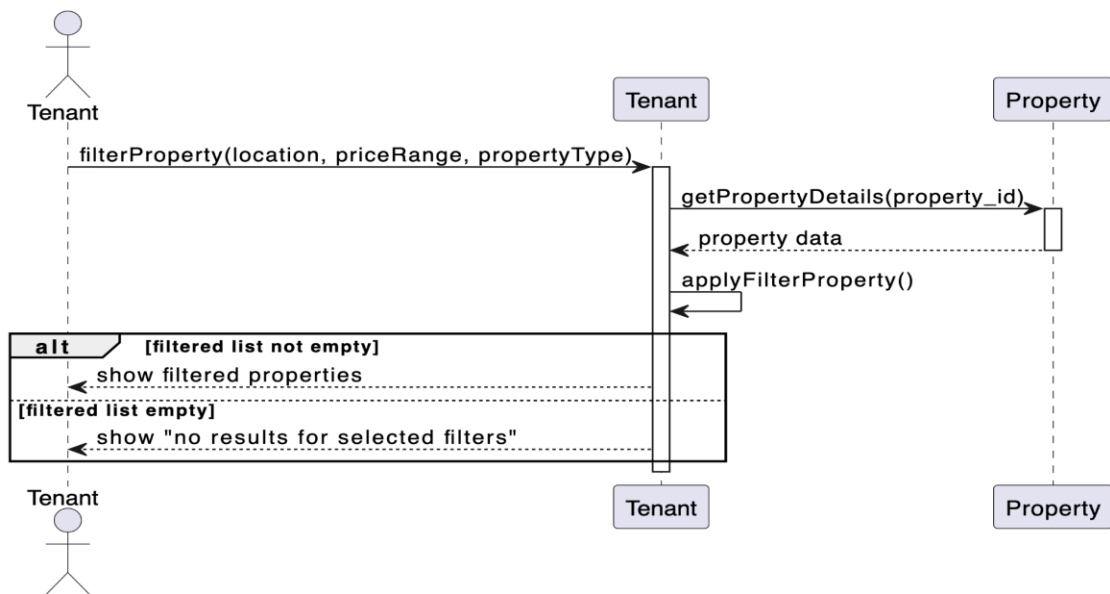
SC_PRS_1 – Update Personal Information (Done by: Anisa Domi)



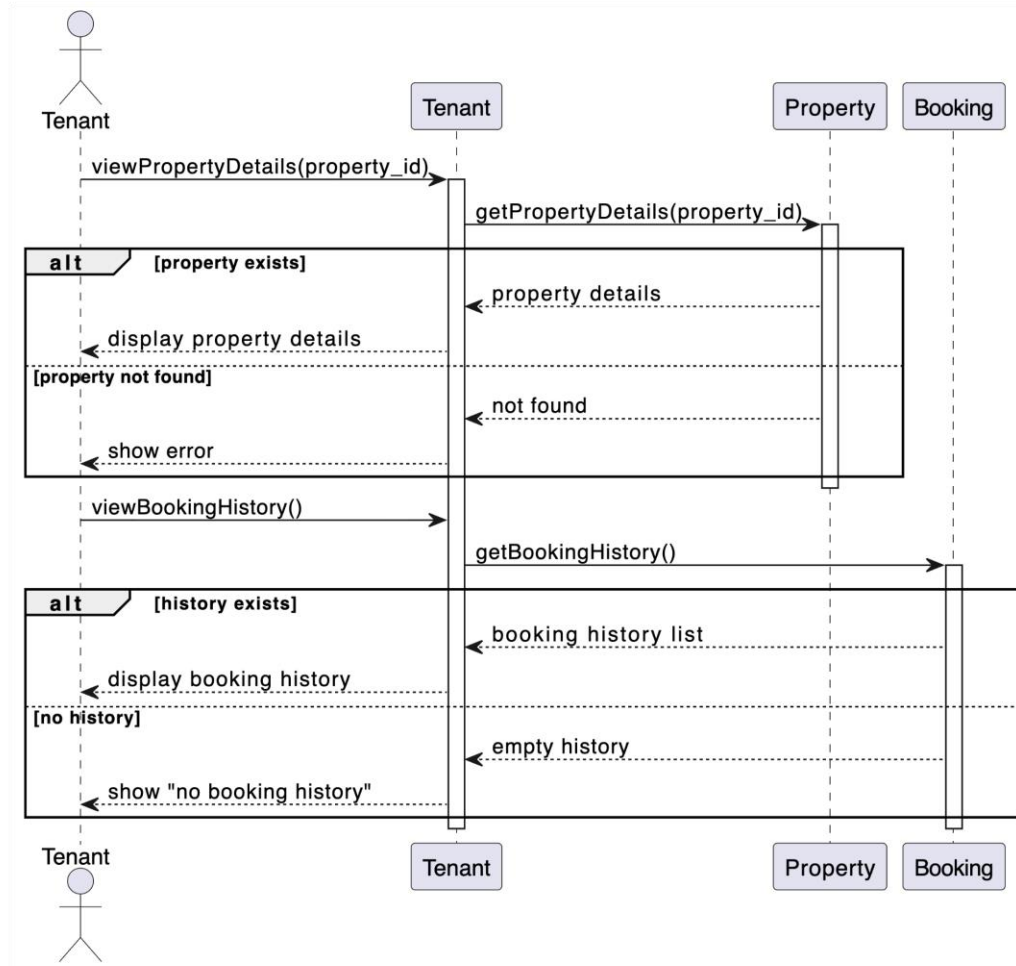
SC_PRS_2 – Search Properties (Done by: Anisa Domi)



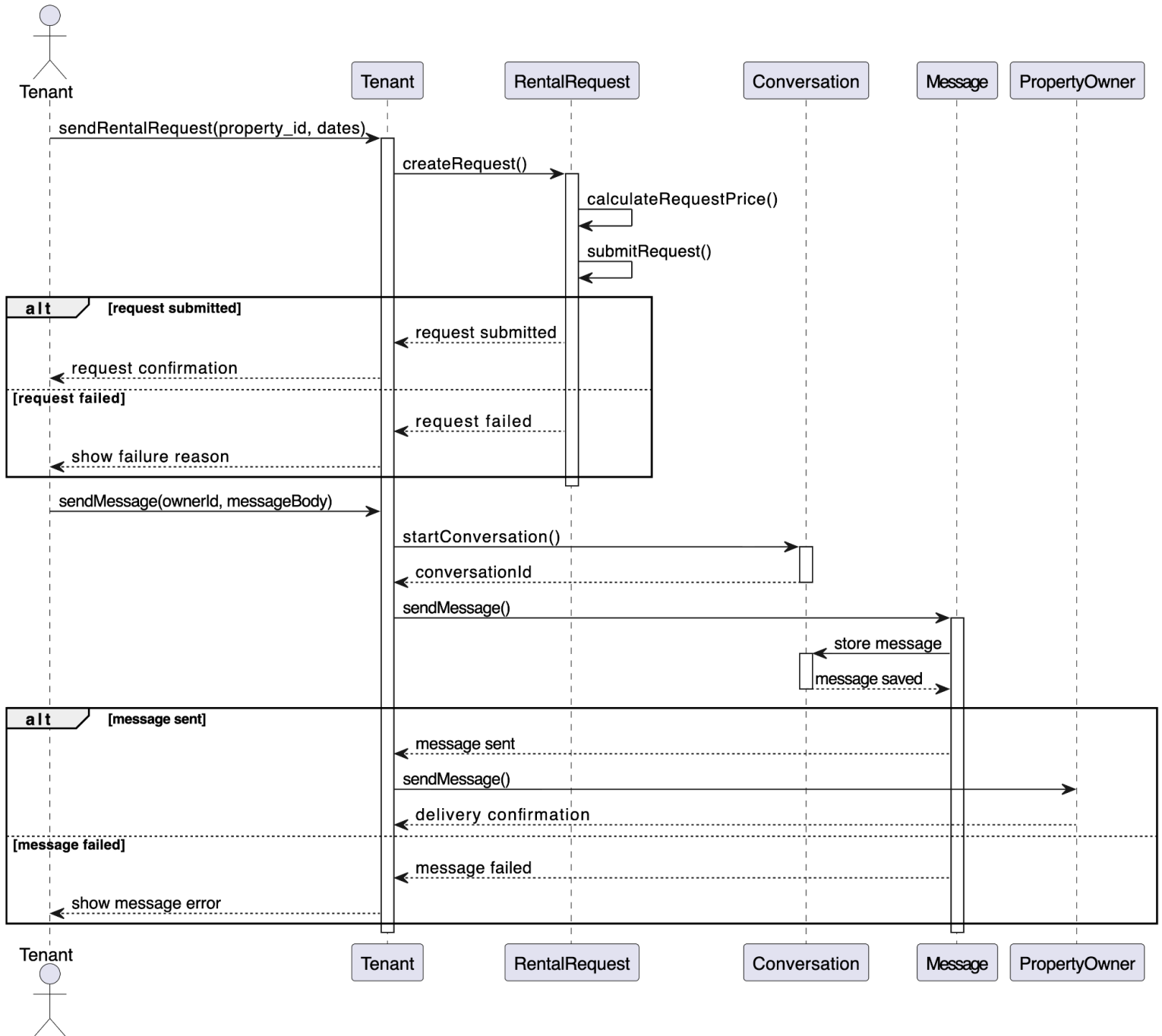
SC_PRS_3 – Filter Properties (Done by: Anisa Domi)



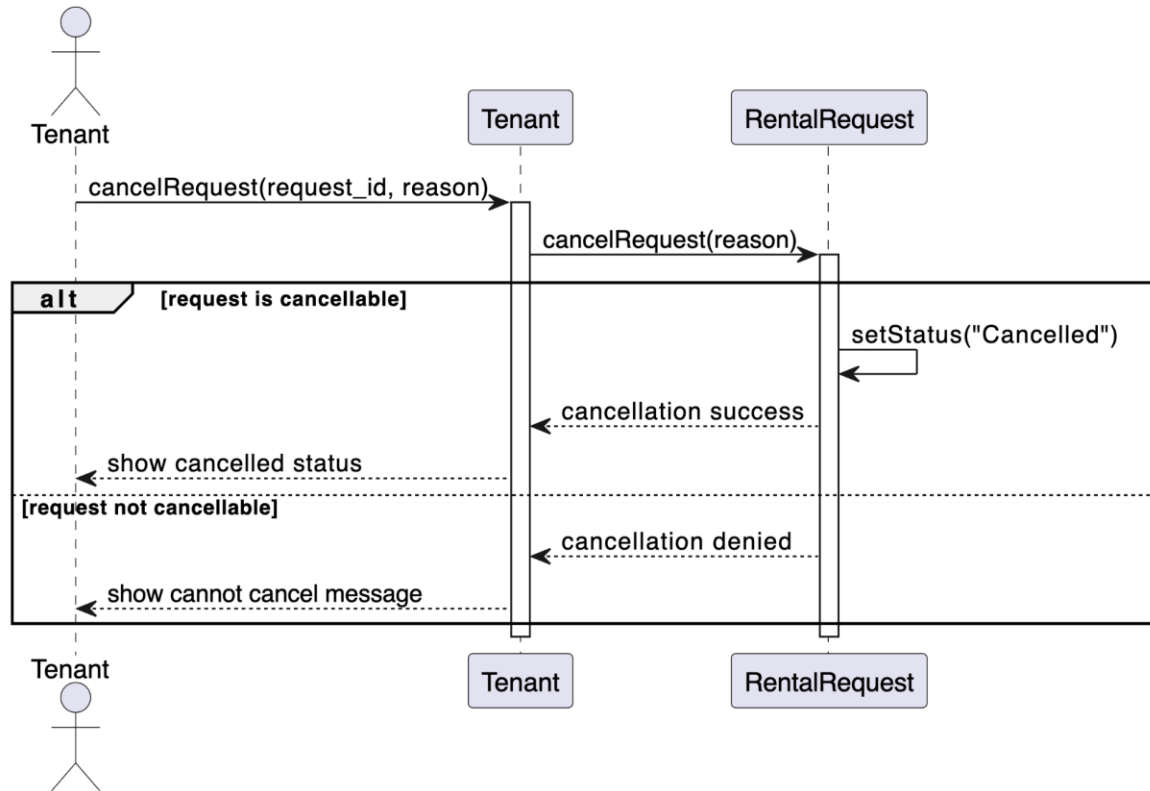
SC_PRS_4 – View Property Details and Booking History (Done by: Anisa Domi)



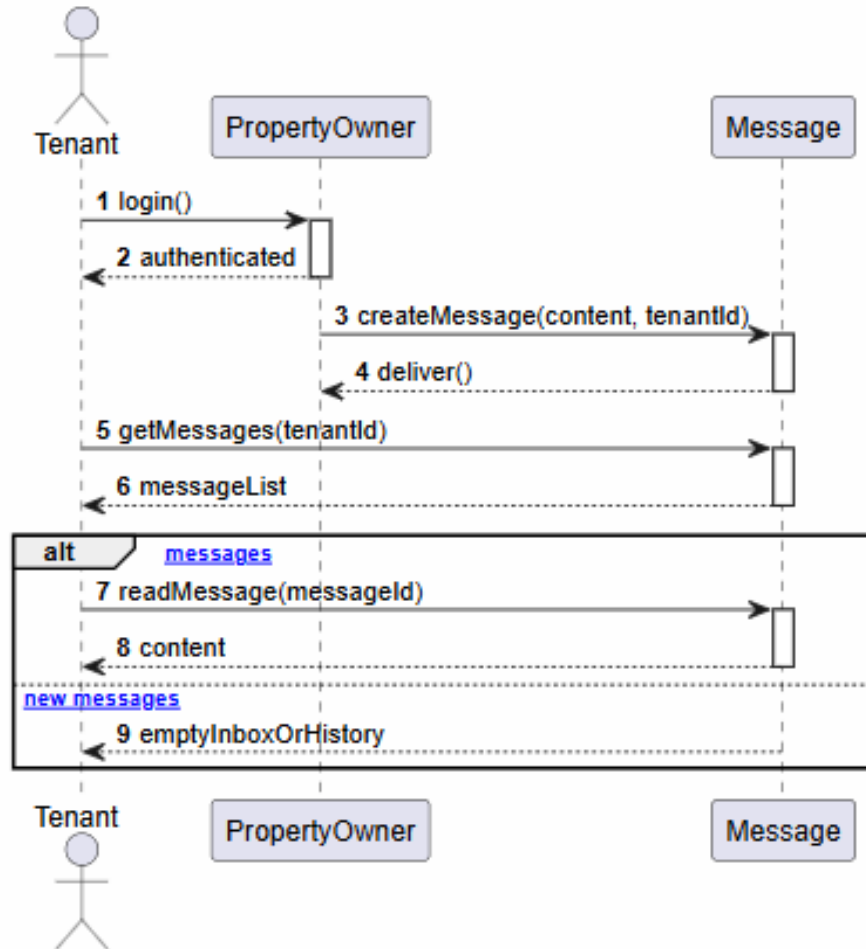
SC_PRS_5 – Send Rental Requests and Message to Property Owners (Done by: Anisa Domi)



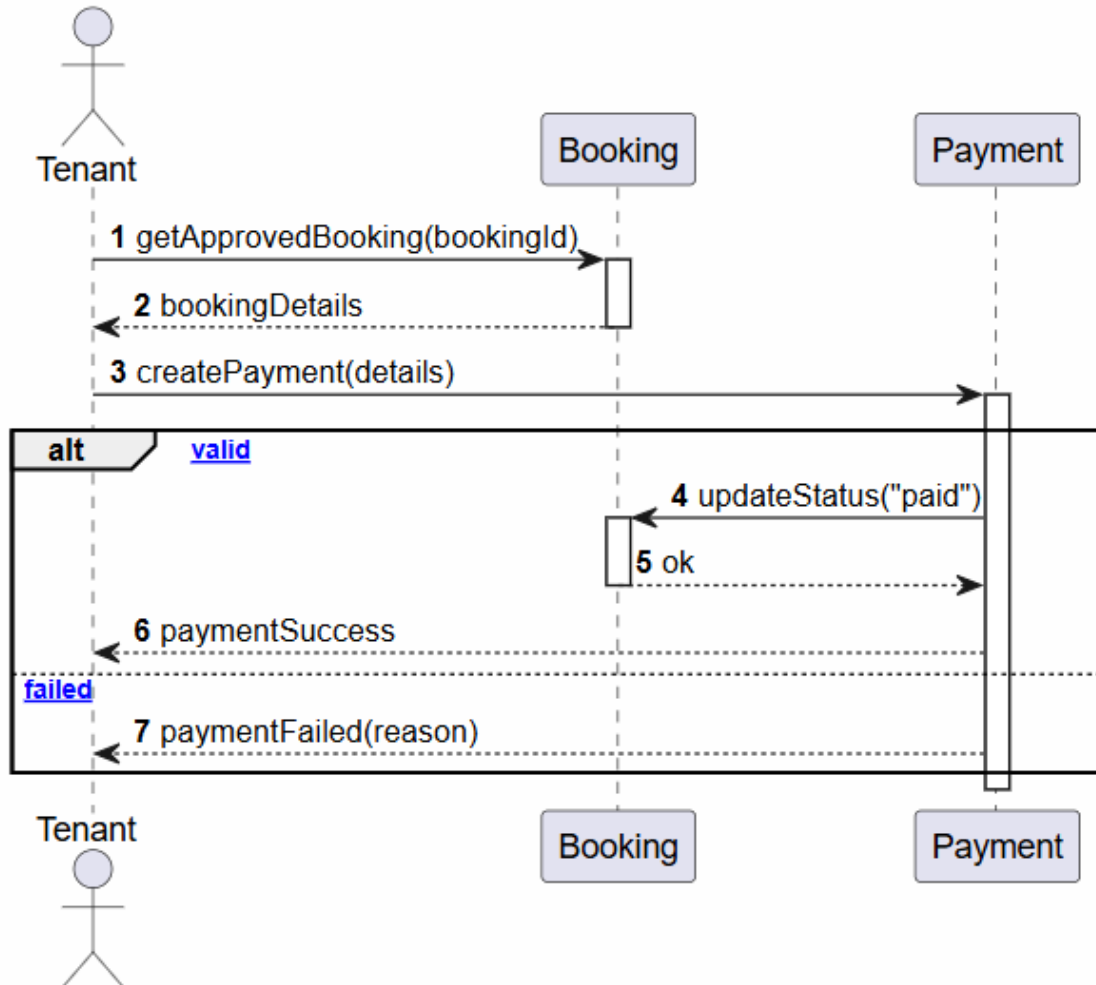
SC_PRS_6 – Cancel Rental Request (Done by: Anisa Domi)



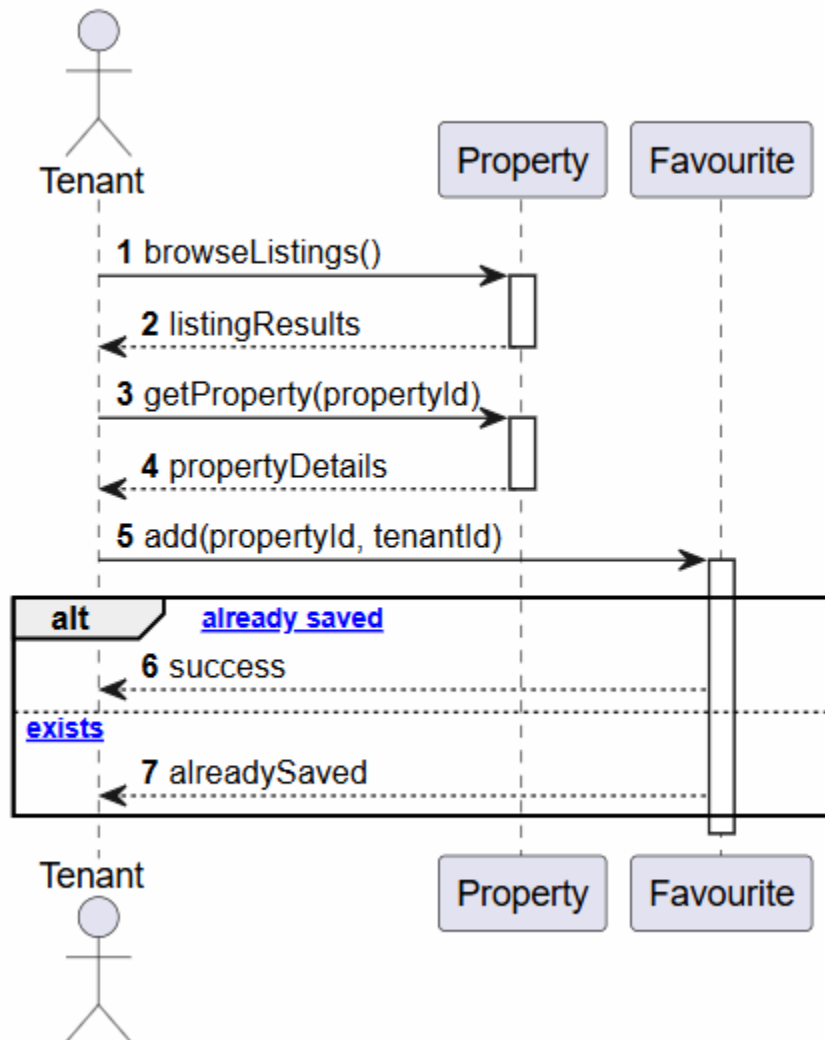
SC_PRS_7 – Receive Messages from Property Owners (Done by: Haidi Toto)



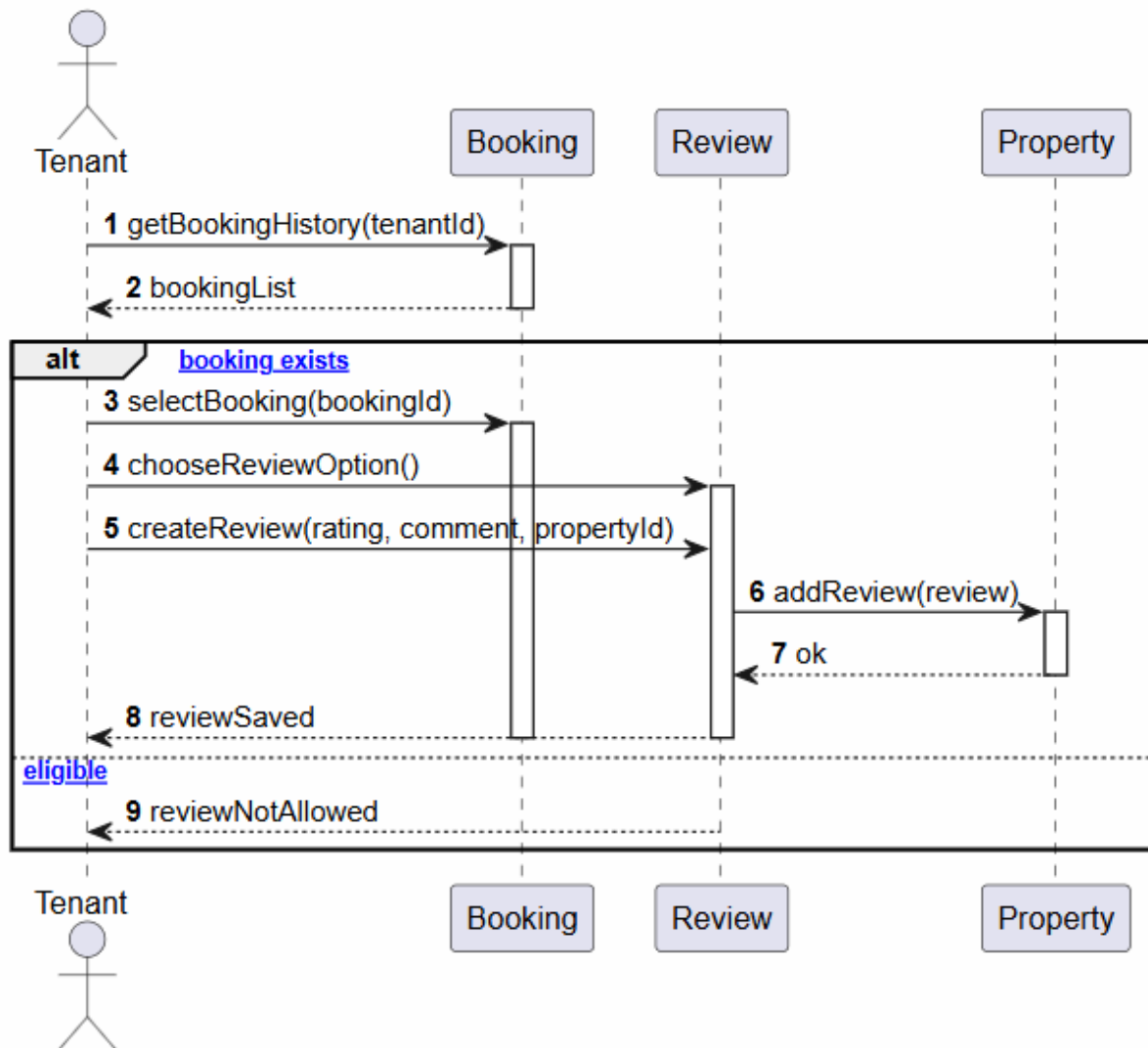
SC_PRS_8 – Submit Rental Payments (Done by: Haidi Toto)



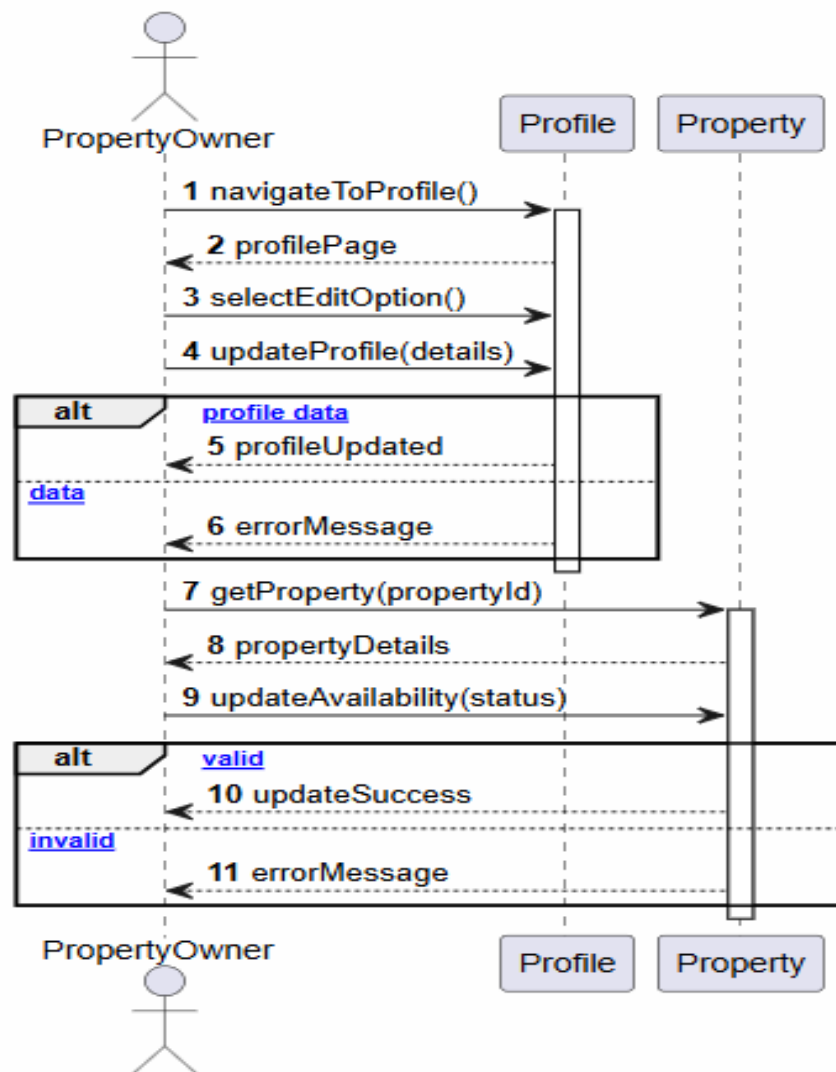
SC_PRS_9 – Save Favourite Properties for Later Viewing (Done by: Haidi Toto)



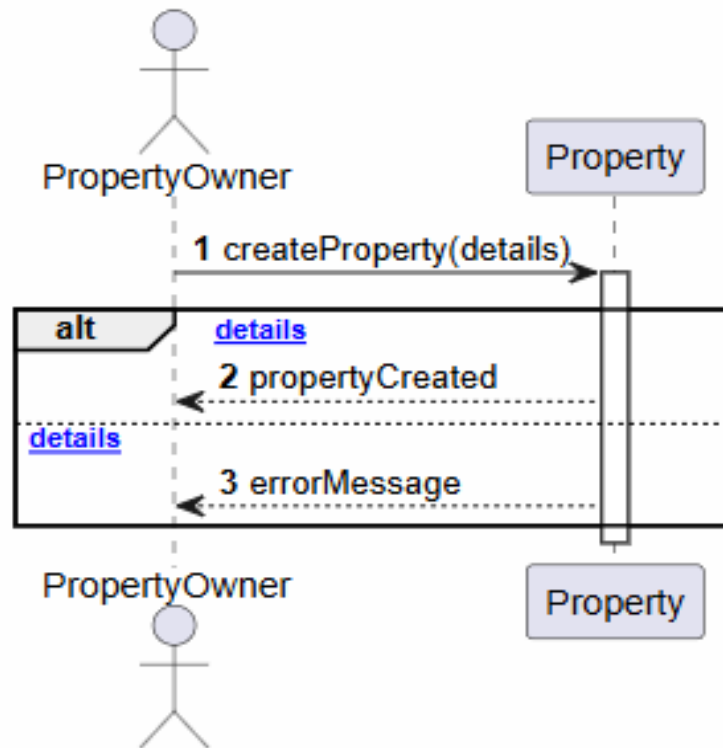
SC_PRS_10 – Review and Rate Property (Done by: Haidi Toto)



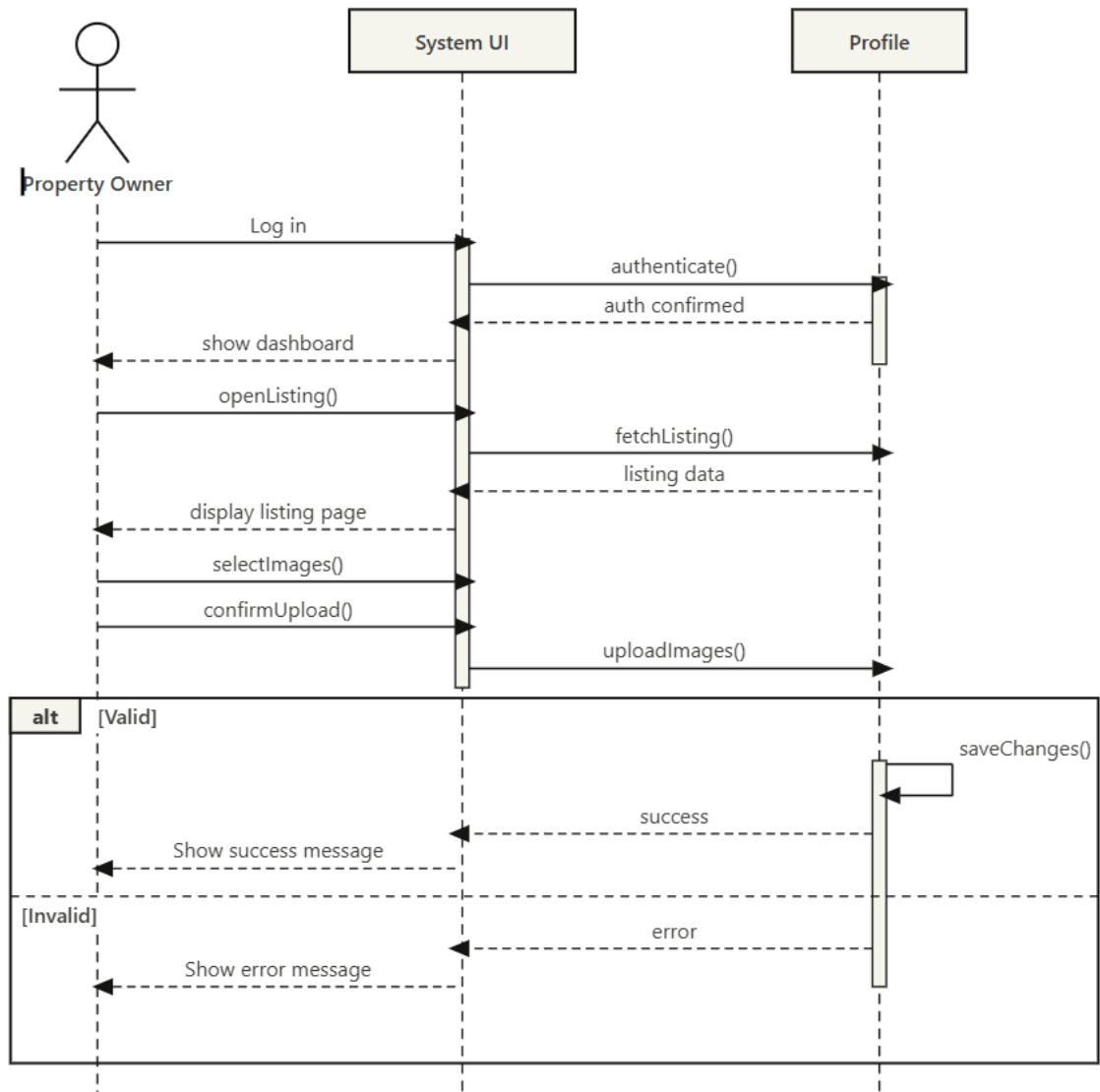
SC_PRS_11 – Update Profile Information and Availability of Rental Properties
(Done by: Haidi Toto)



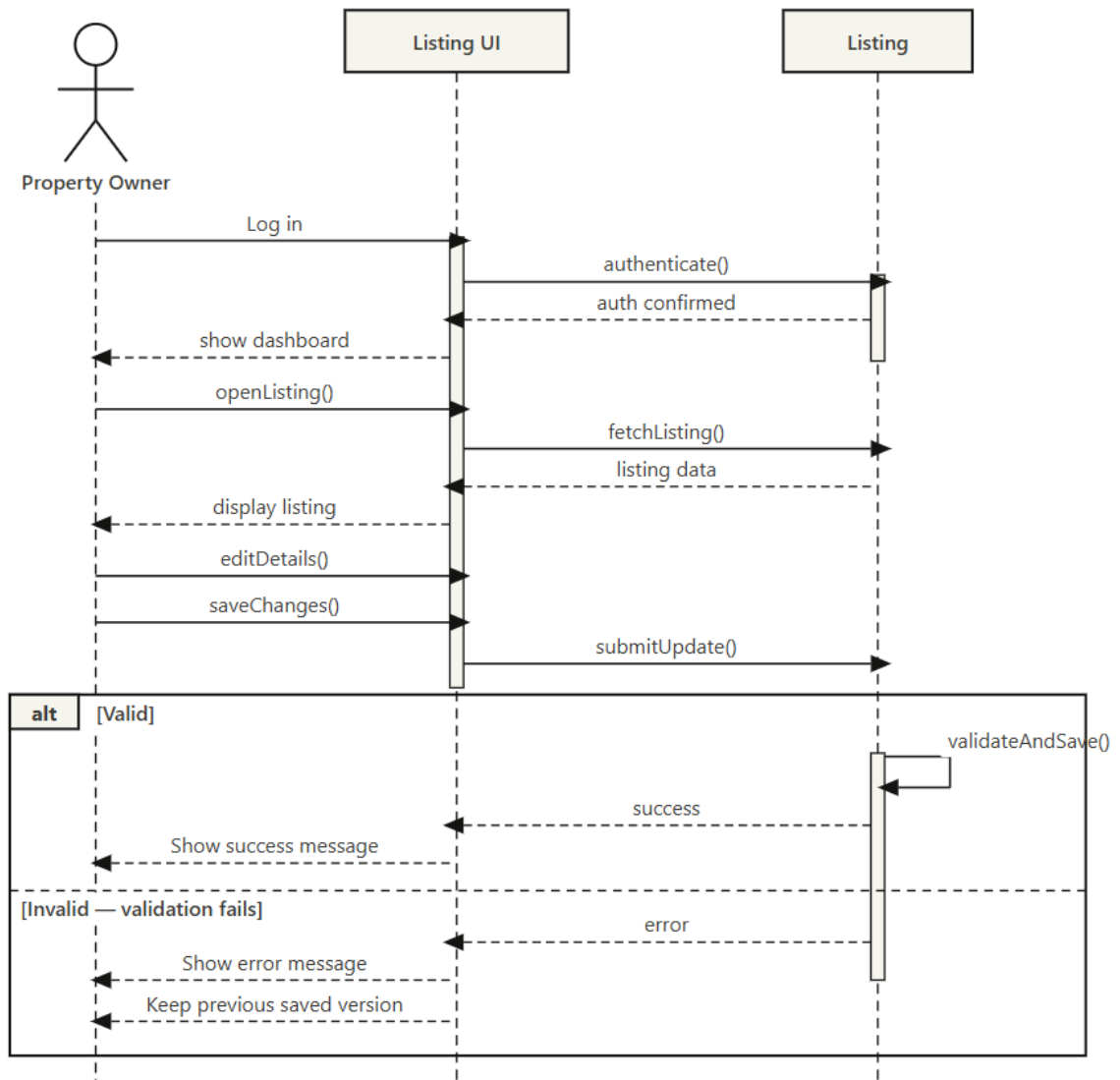
SC_PRS_12 – Add Property Listing (Done by: Haidi Toto)



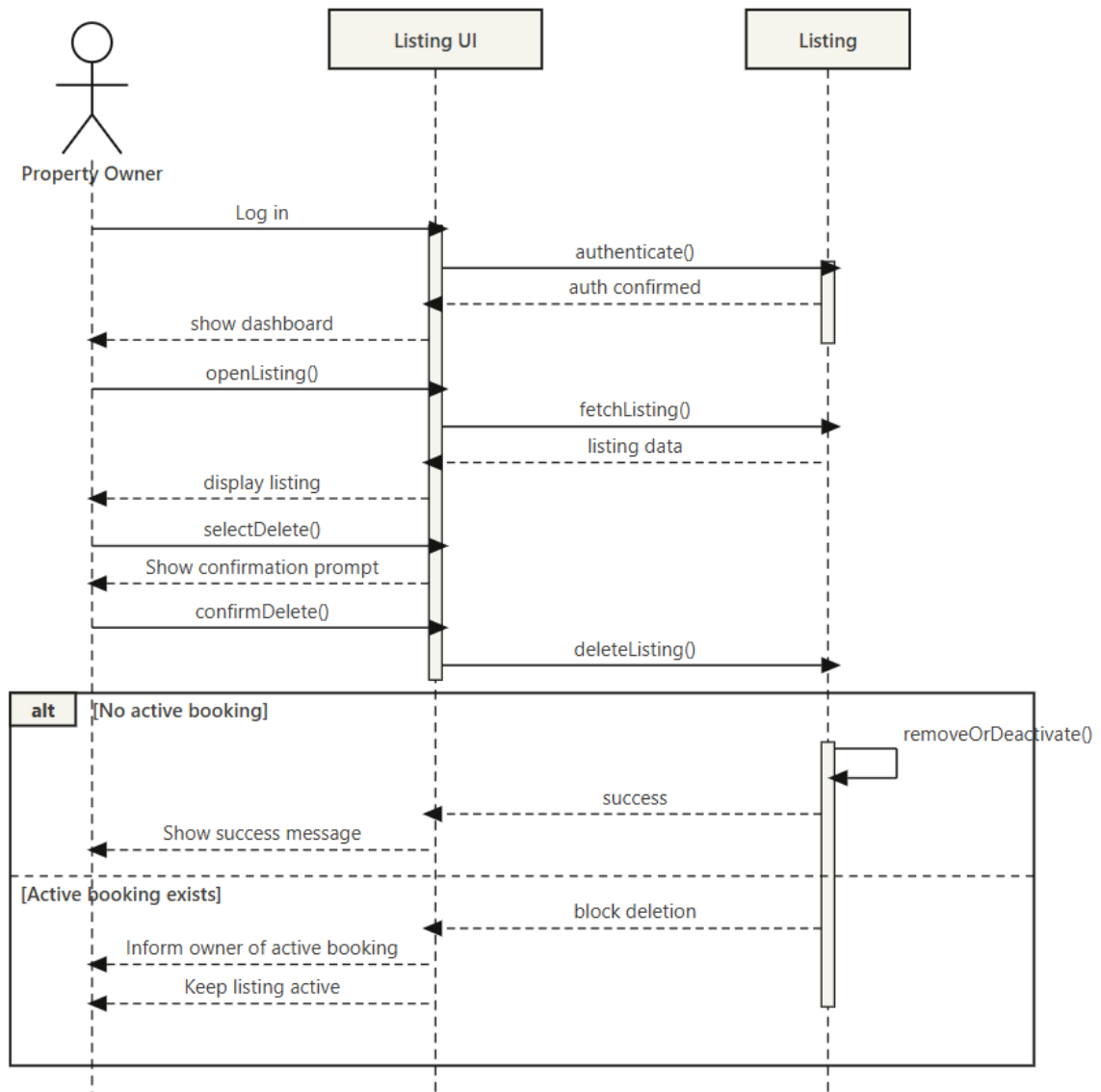
SC_PRS_13 – Upload Property Images (Done by: Ledio Tafciu)



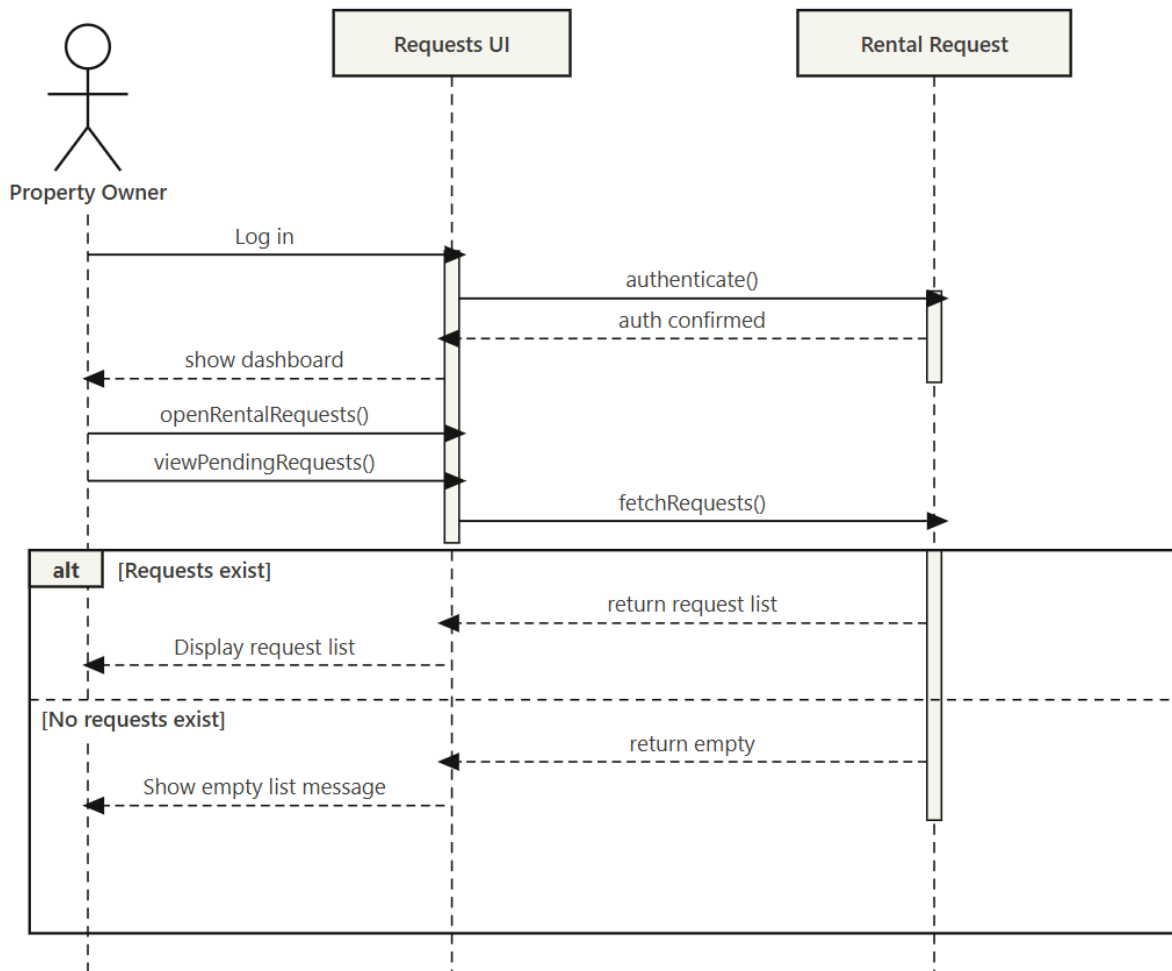
SC_PRS_14 – Edit Property Listing (Done by: Ledio Tafciu)



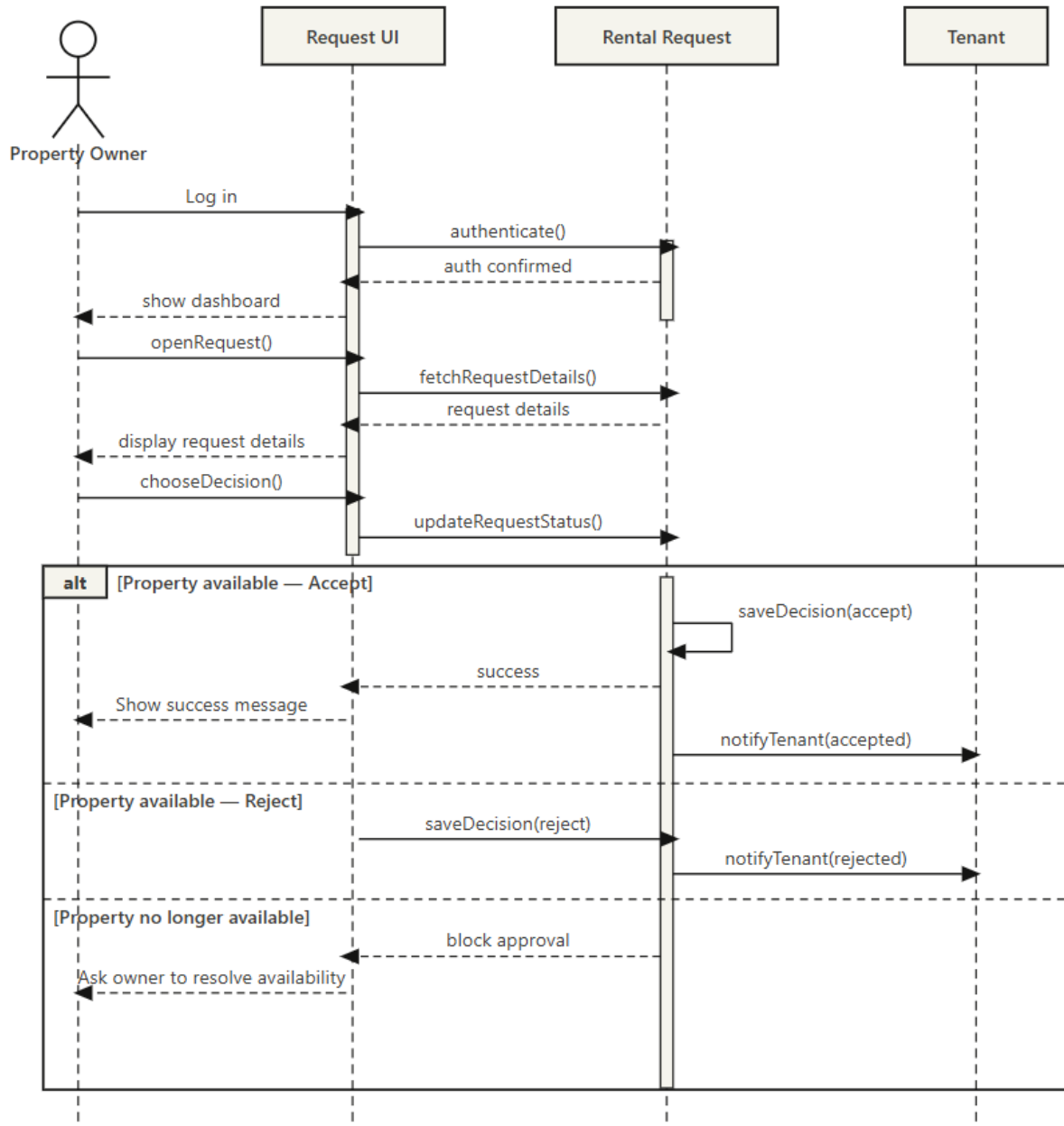
SC_PRS_15 – Delete Property Listing (Done by: Ledio Tafciu)



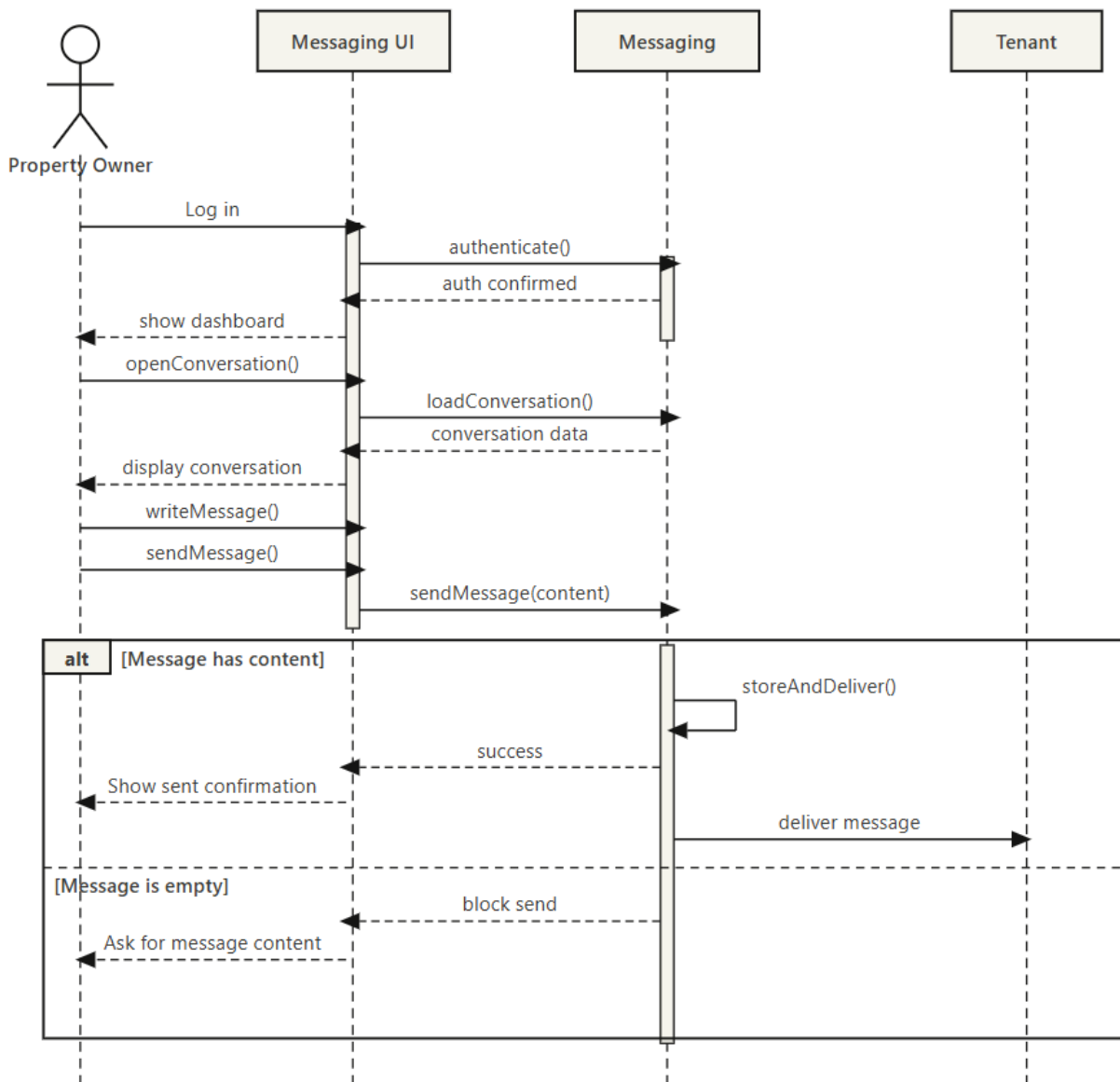
SC_PRS_16 – View Rental Requests (Done by: Ledio Tafciu)



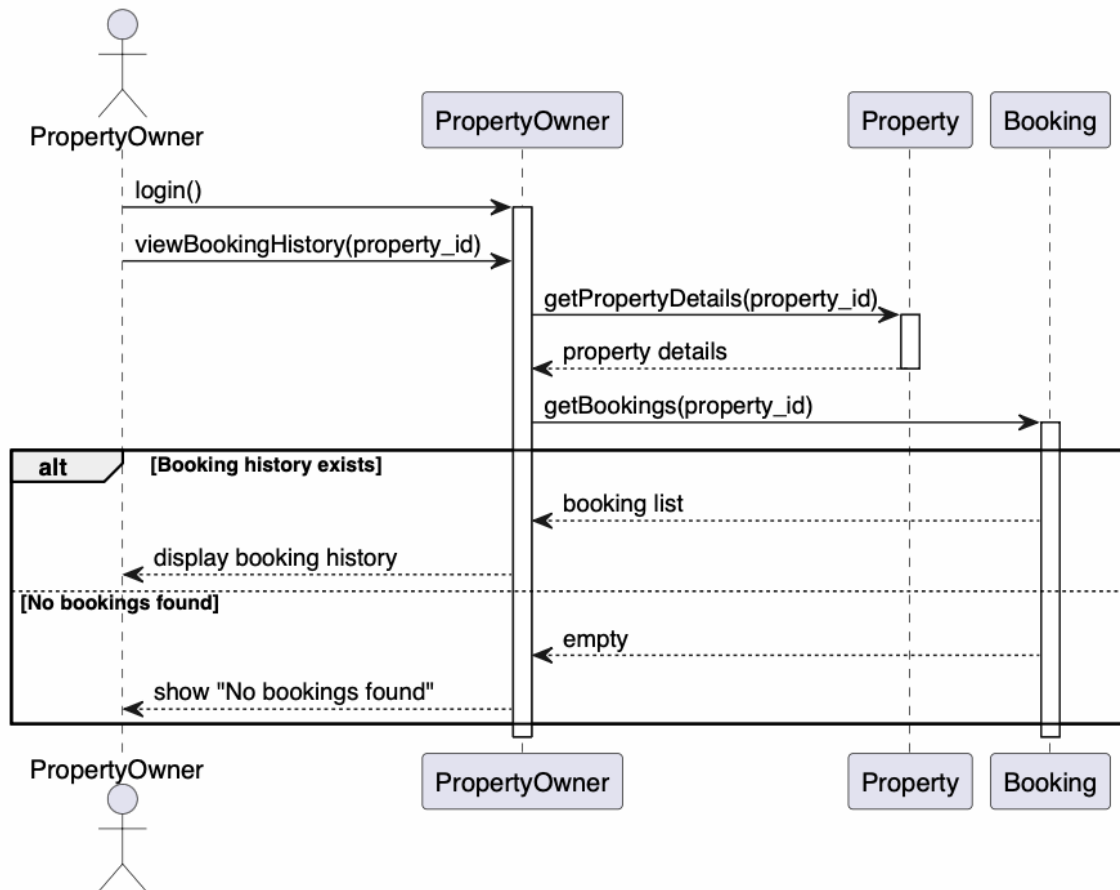
SC_PRS_17 – Accept or Reject Rental Request (Done by: Ledio Tafciu)



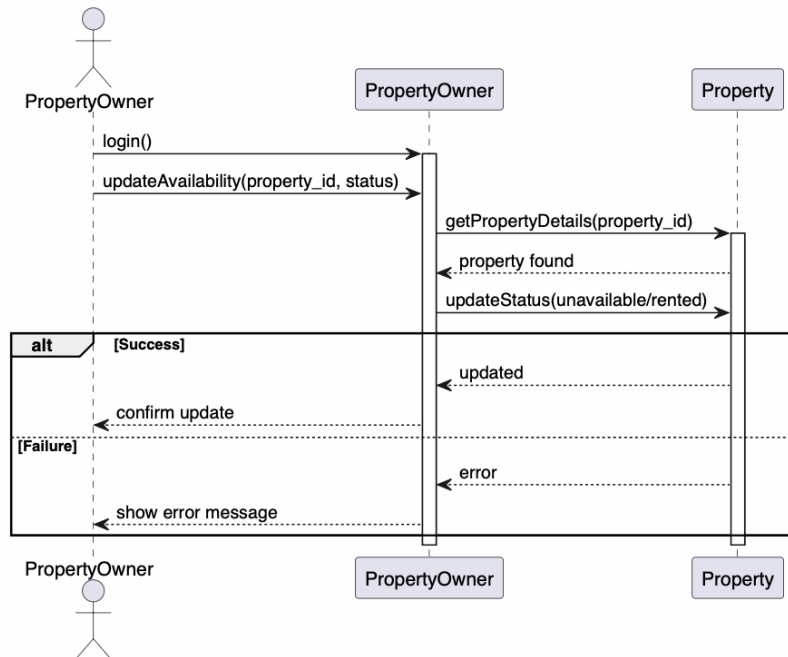
SC_PRS_18 – Send Message to Tenant (Done by: Ledio Tafciu)



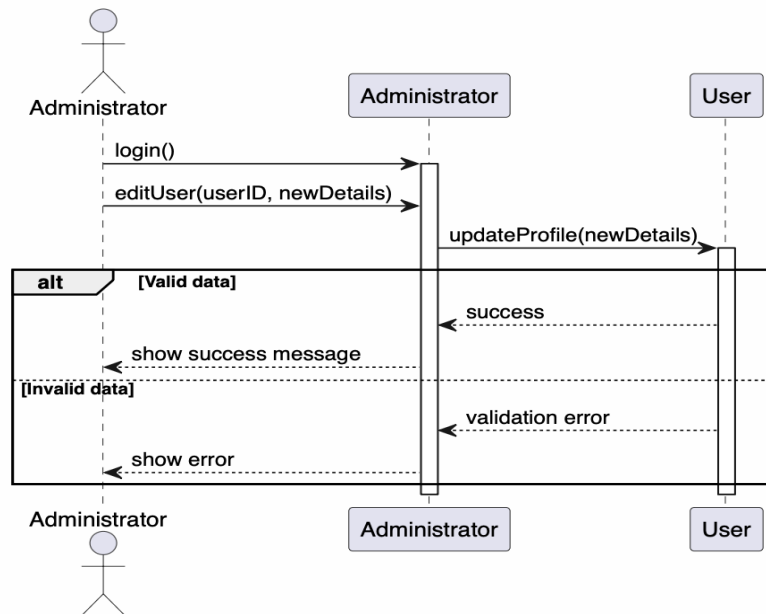
SC_PRS_19 – View Booking History (Done by: Desard Lybeshari)



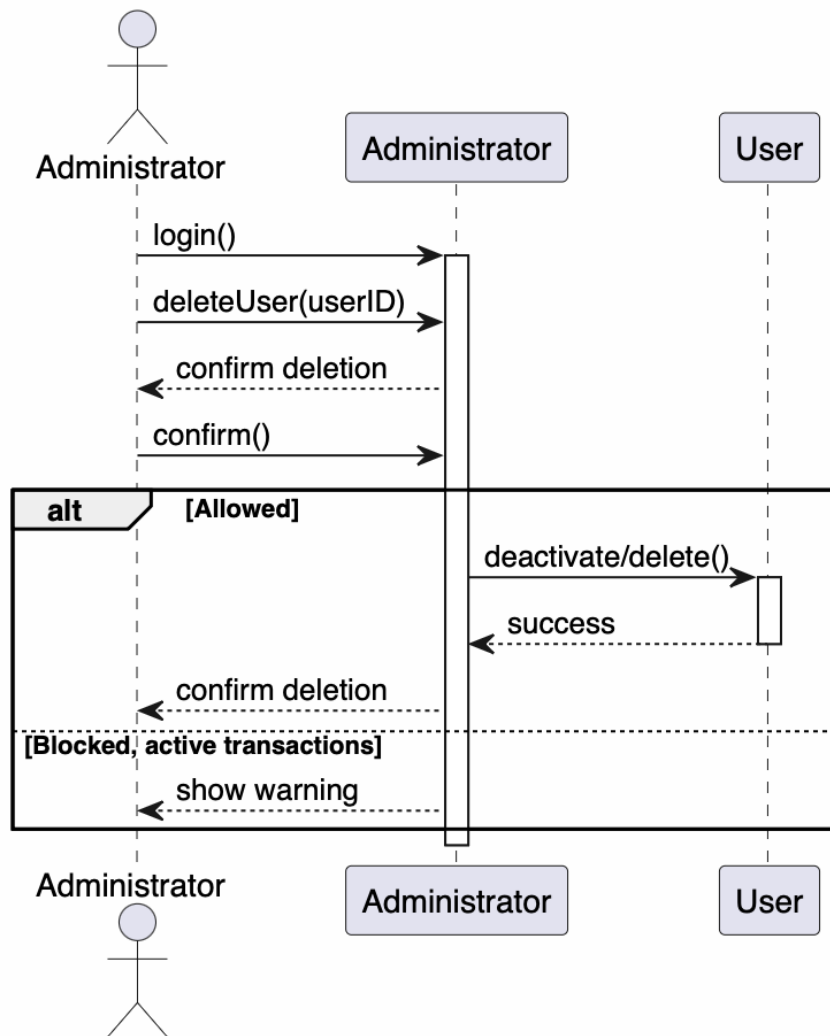
SC_PRS_20 – Mark Property as Unavailable or Rented (Done by: Desard Lybeshari)



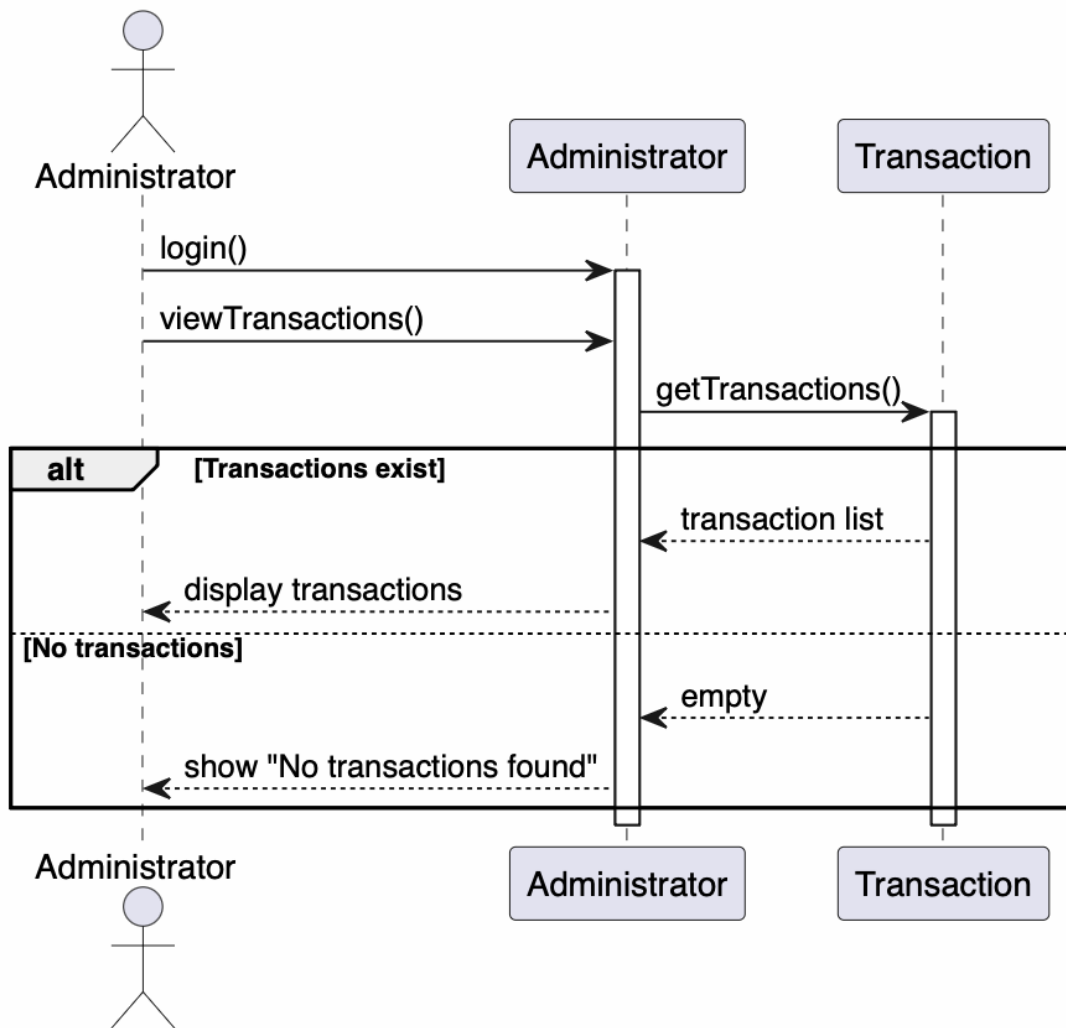
SC_PRS_21 – Edit User Accounts (Done by: Desard Lybeshari)



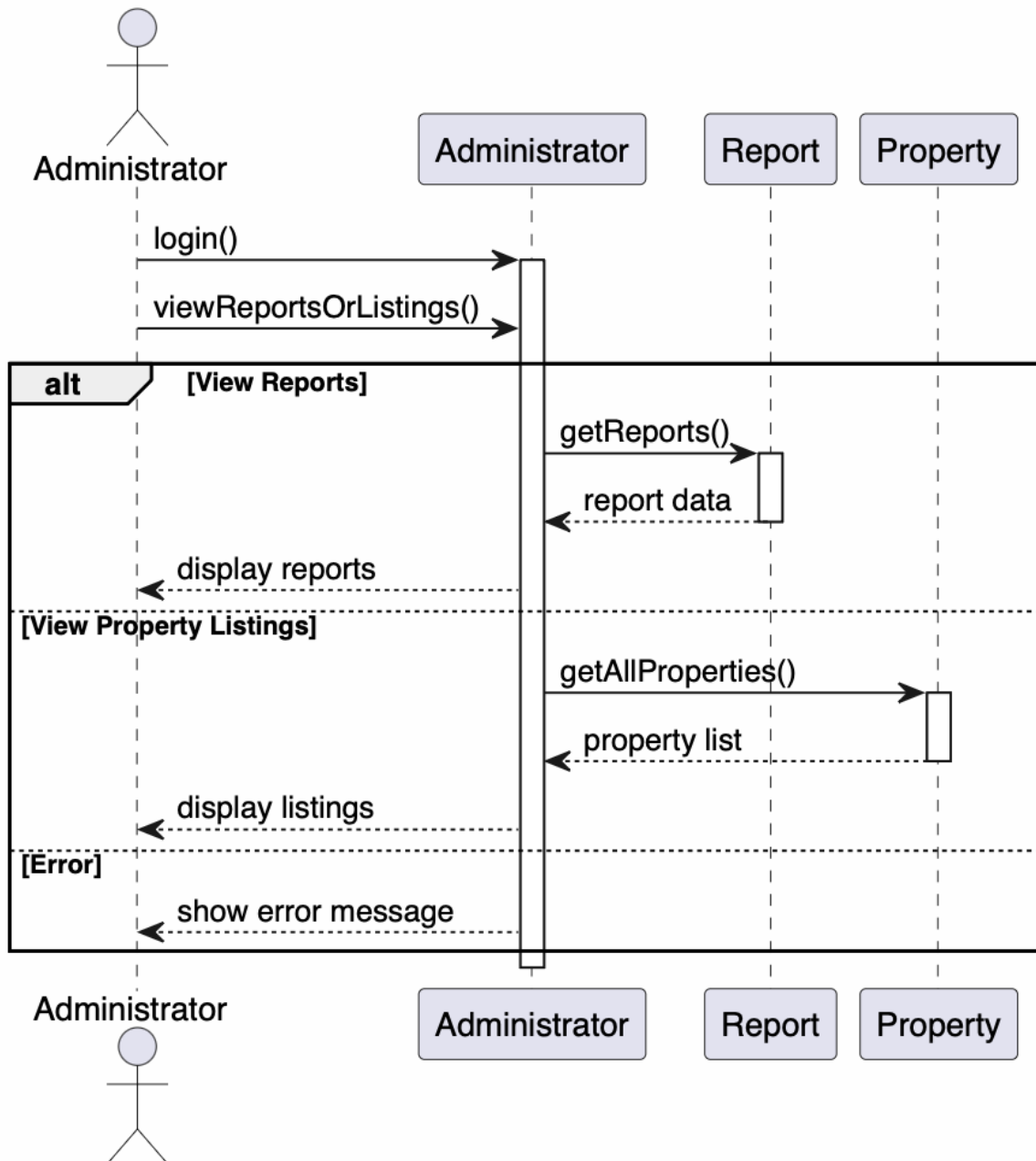
SC_PRS_22 – Delete User Accounts (Done by: Desard Lybeshari)



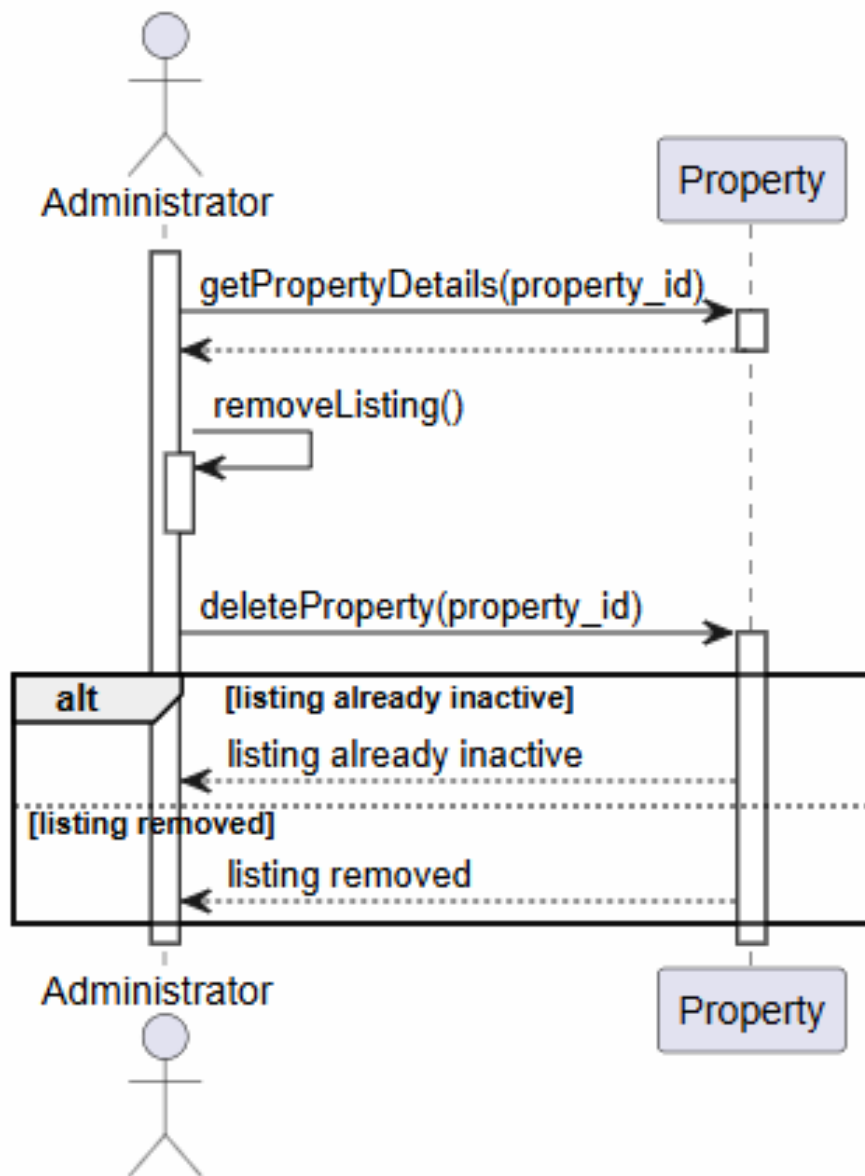
SC_PRS_23 – Monitor Rental Transactions (Done by: Desard Lybeshari)



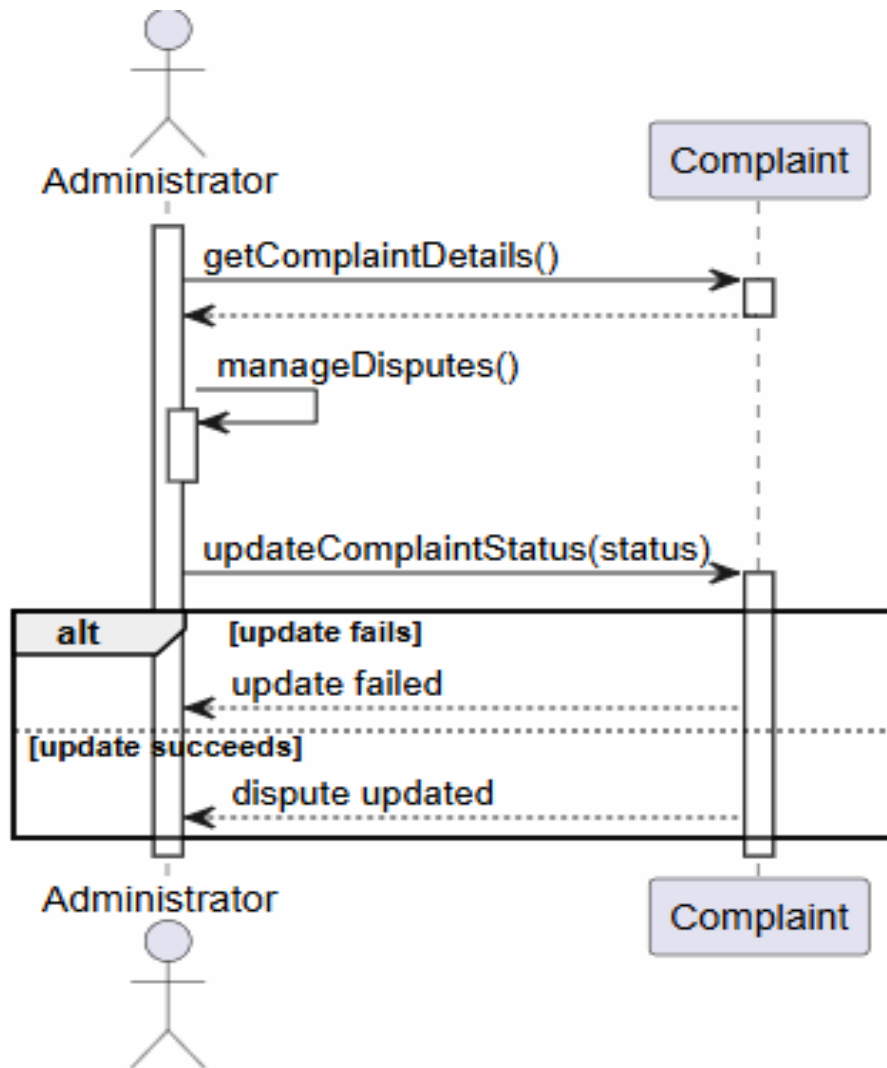
SC_PRS_24 – View System Reports and Property Listings (Done by: Desard Lybeshari)



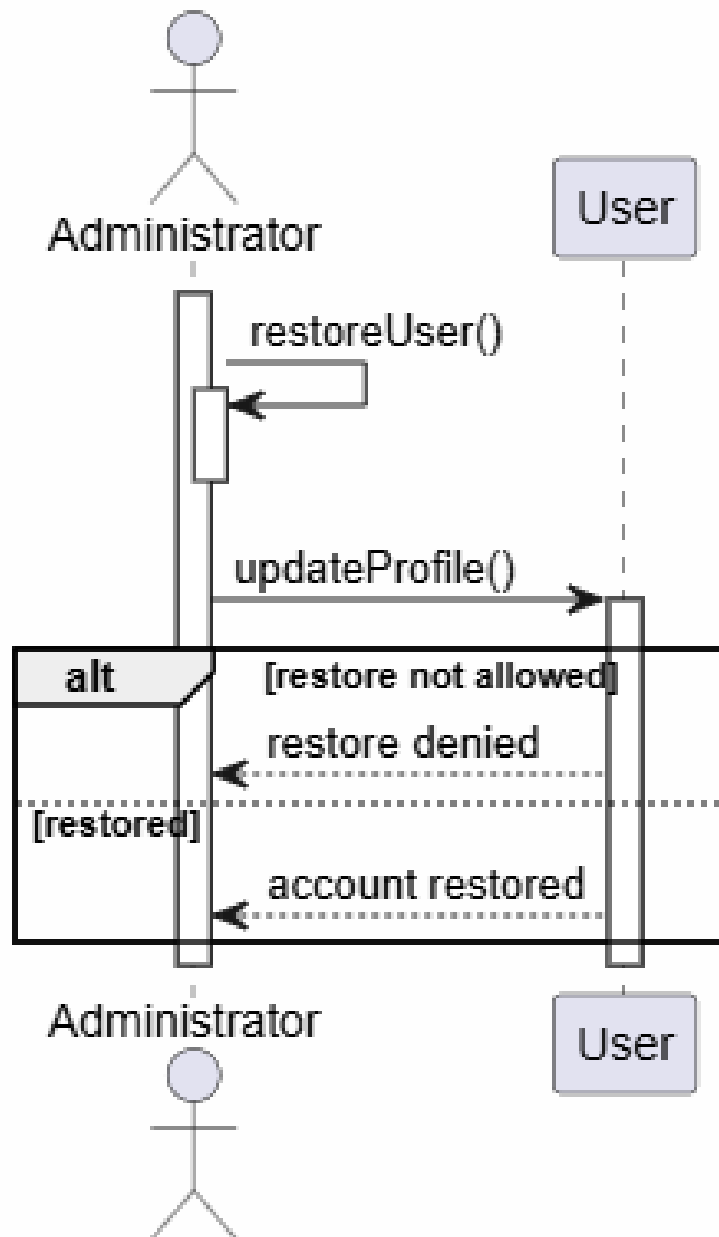
SC_PRS_25 – Remove Inappropriate Listings (Done by: Matias Bali)



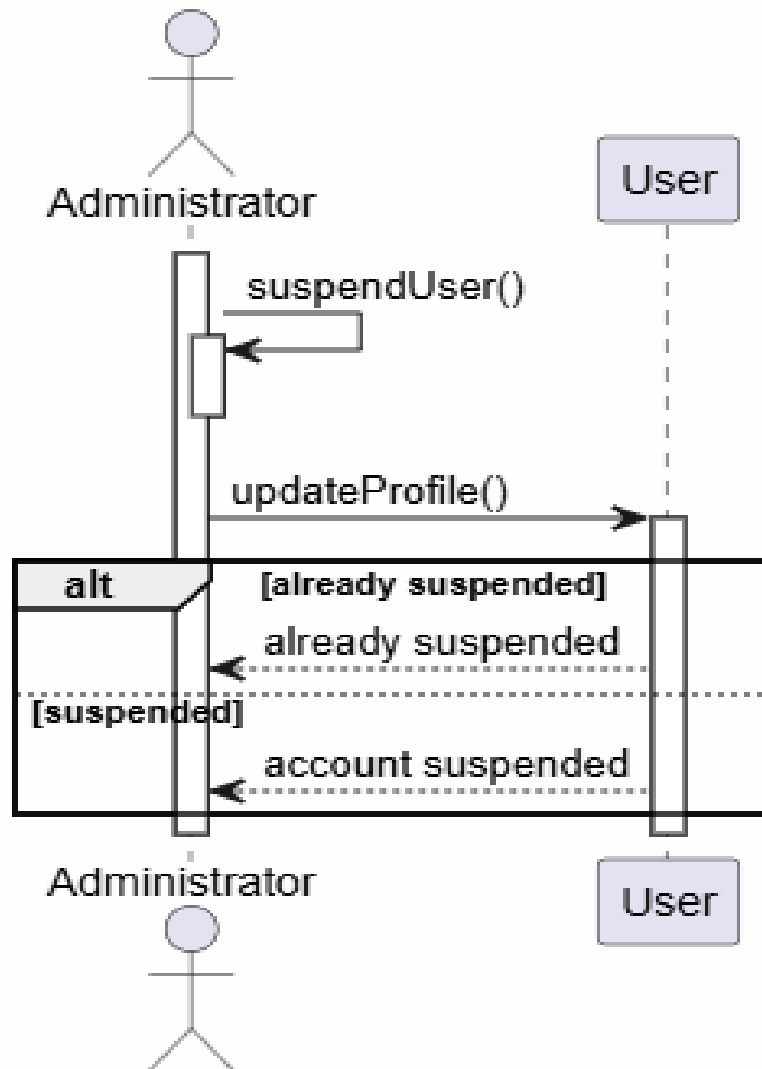
SC_PRS_26 – Manage Disputes and System Settings (Done by: Matias Bali)



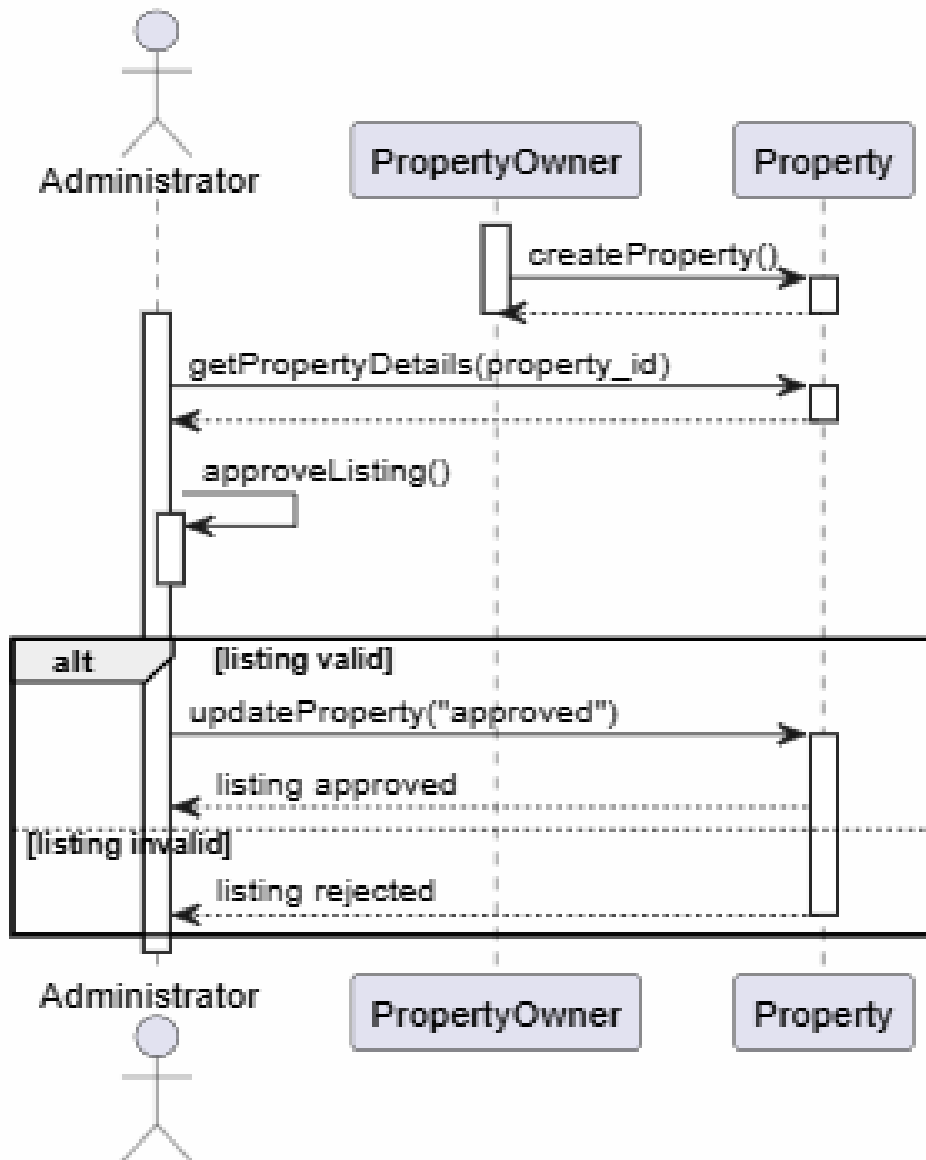
SC_PRS_27 – Restore User Account (Done by: Matias Bali)



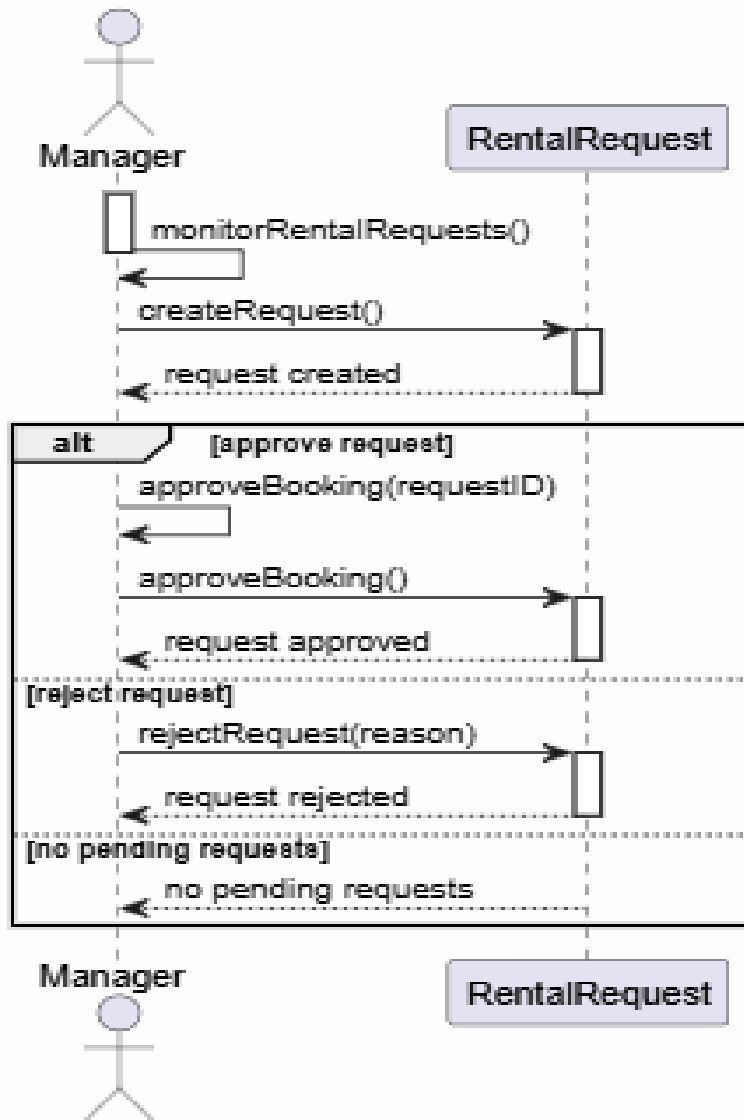
SC_PRS_28 – Suspend User Account (Done by: Matias Bali)



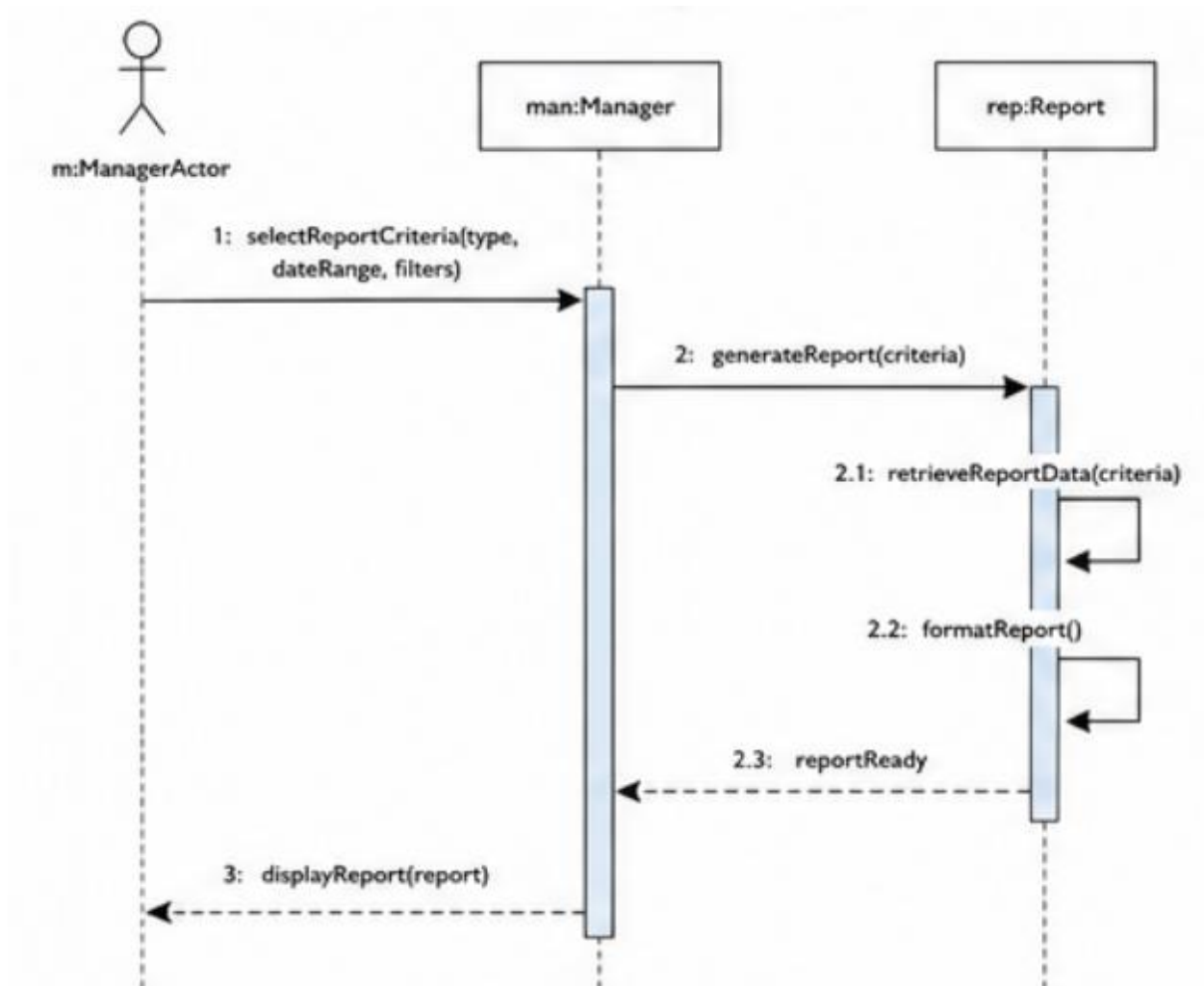
SC_PRS_29 – Verify and Approve Property Listings (Done by: Matias Bali)



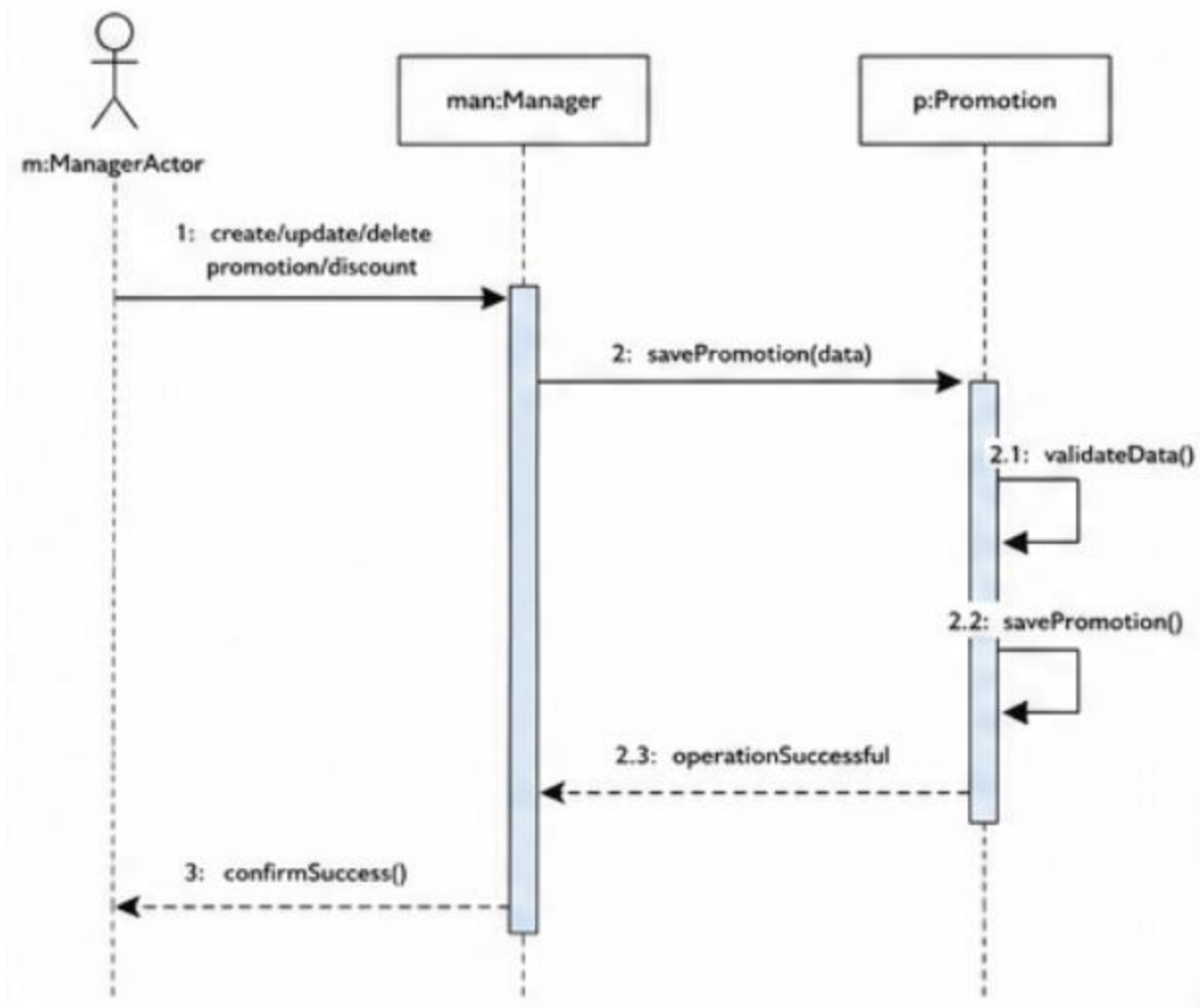
SC_PRS_30 – Monitor Rental Requests (Done by: Matias Bali)



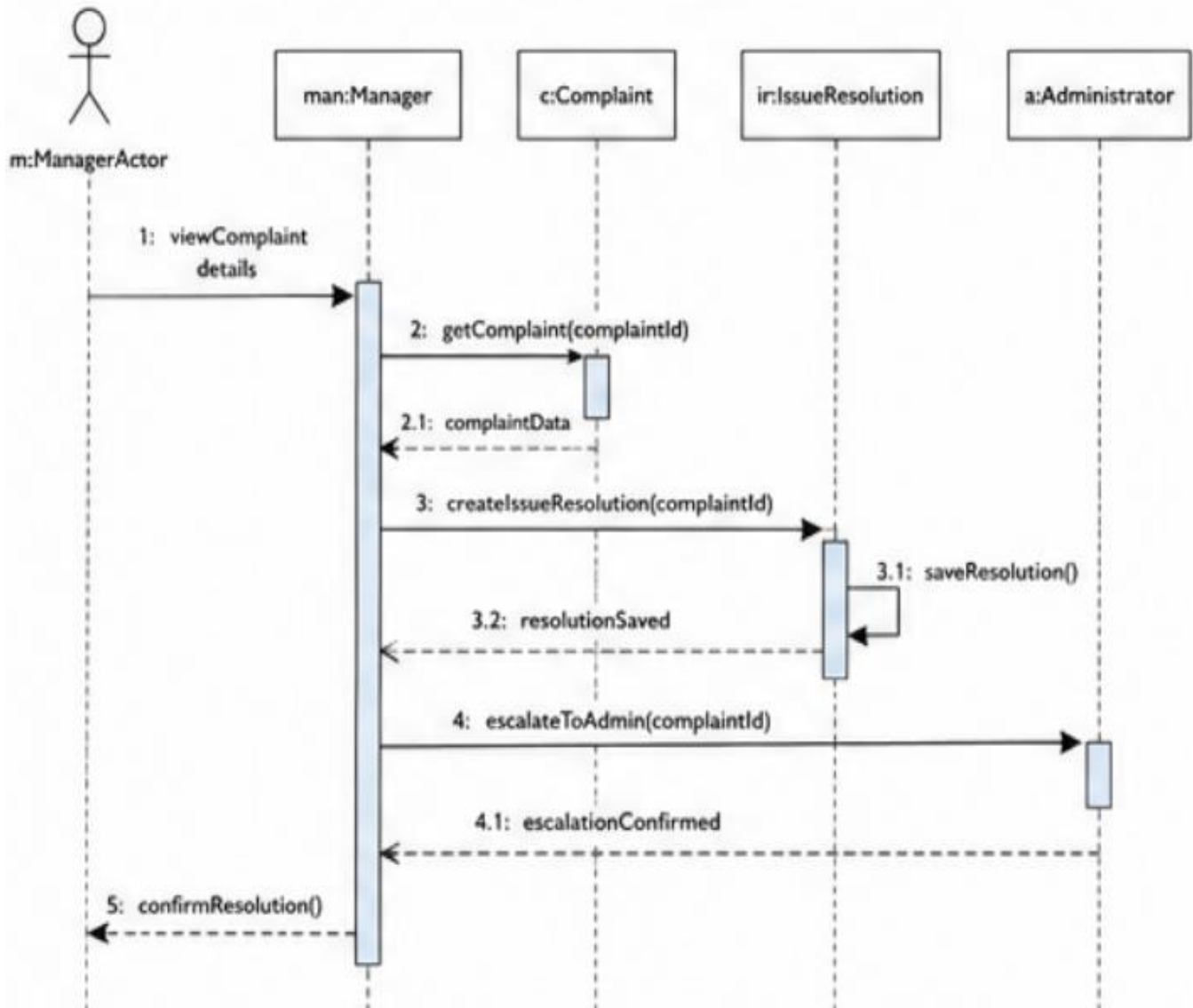
SC_PRS_31 – Generate Management Reports (Done by: Ornelt Lemellari)



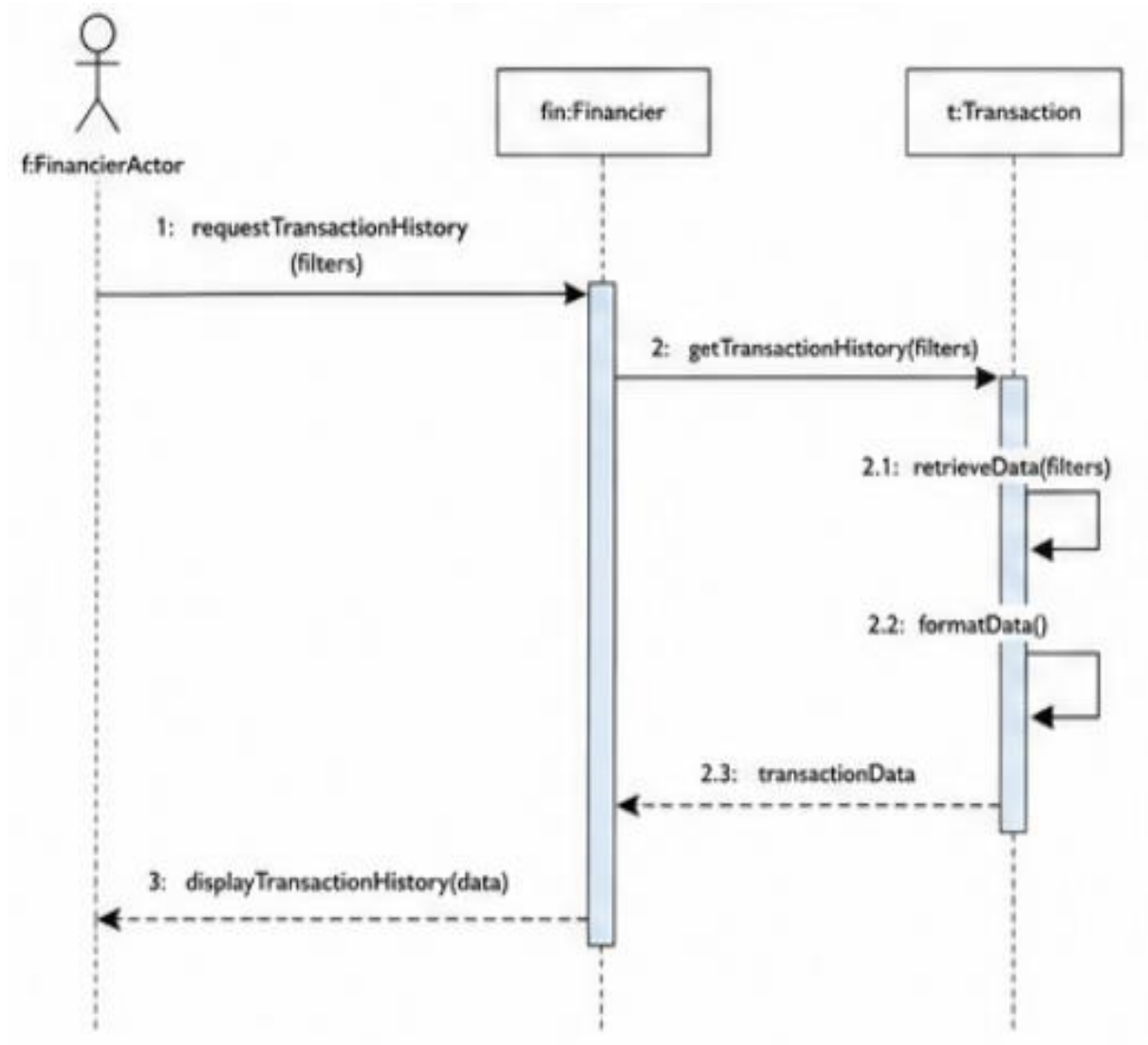
SC_PRS_32 – Approve Large Discounts and Promotions (Done by: Ornelt Lemellari)



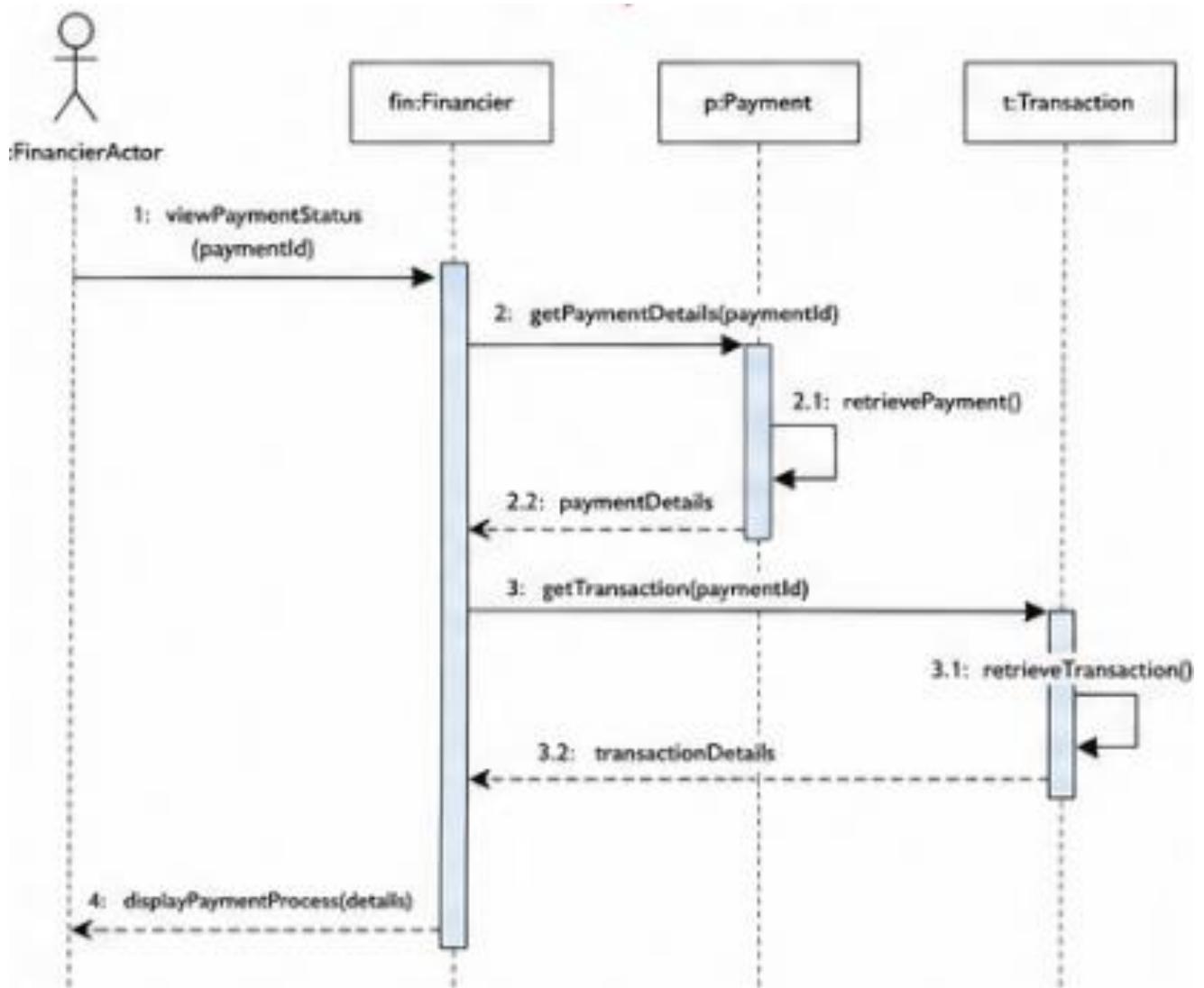
SC_PRS_33 – Review Complaints and Forward Unresolved Issues (Done by: Ornelt Lemellari)



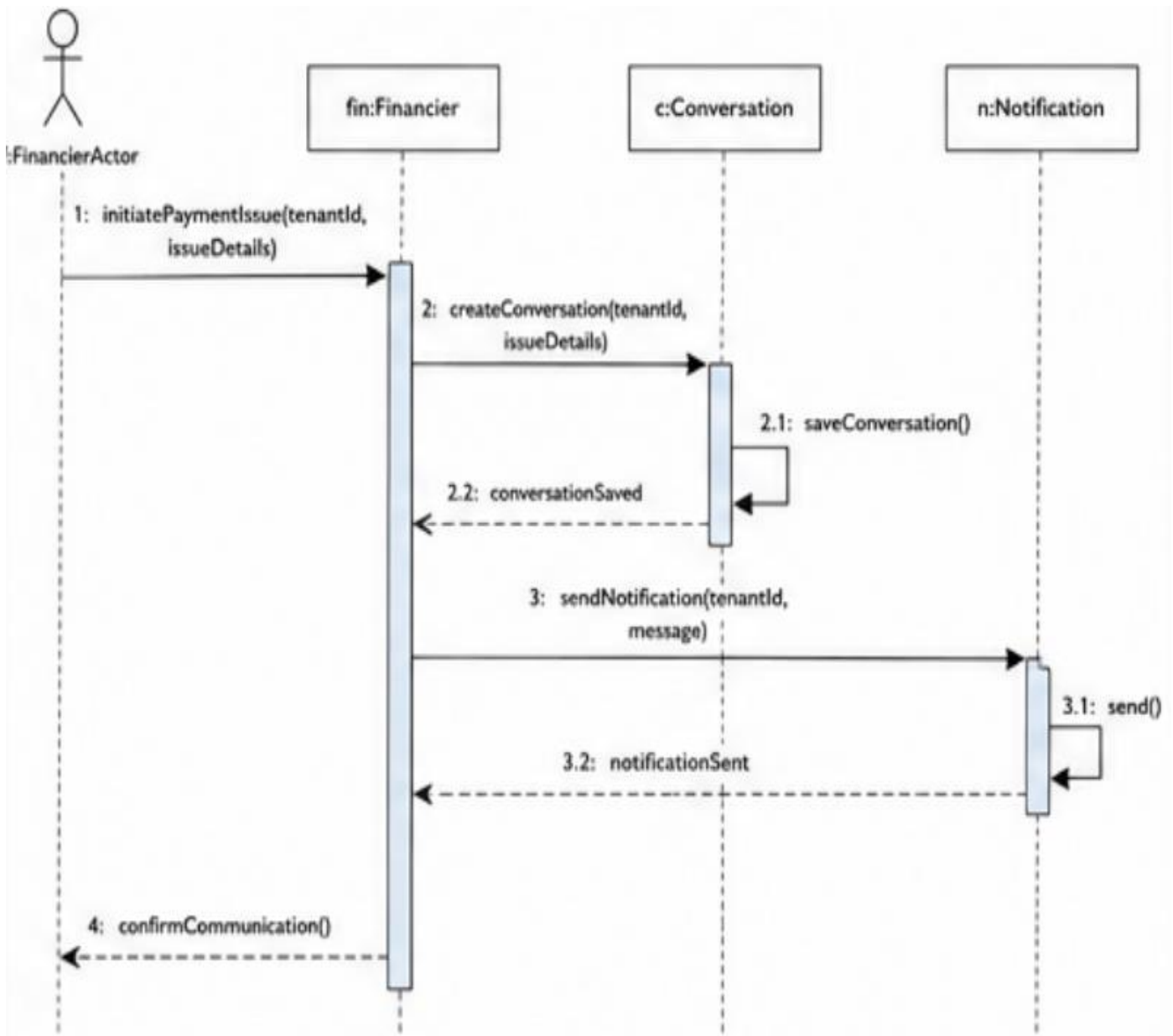
SC_PRS_34 – View Transaction History (Done by: Ornel Lemellari)



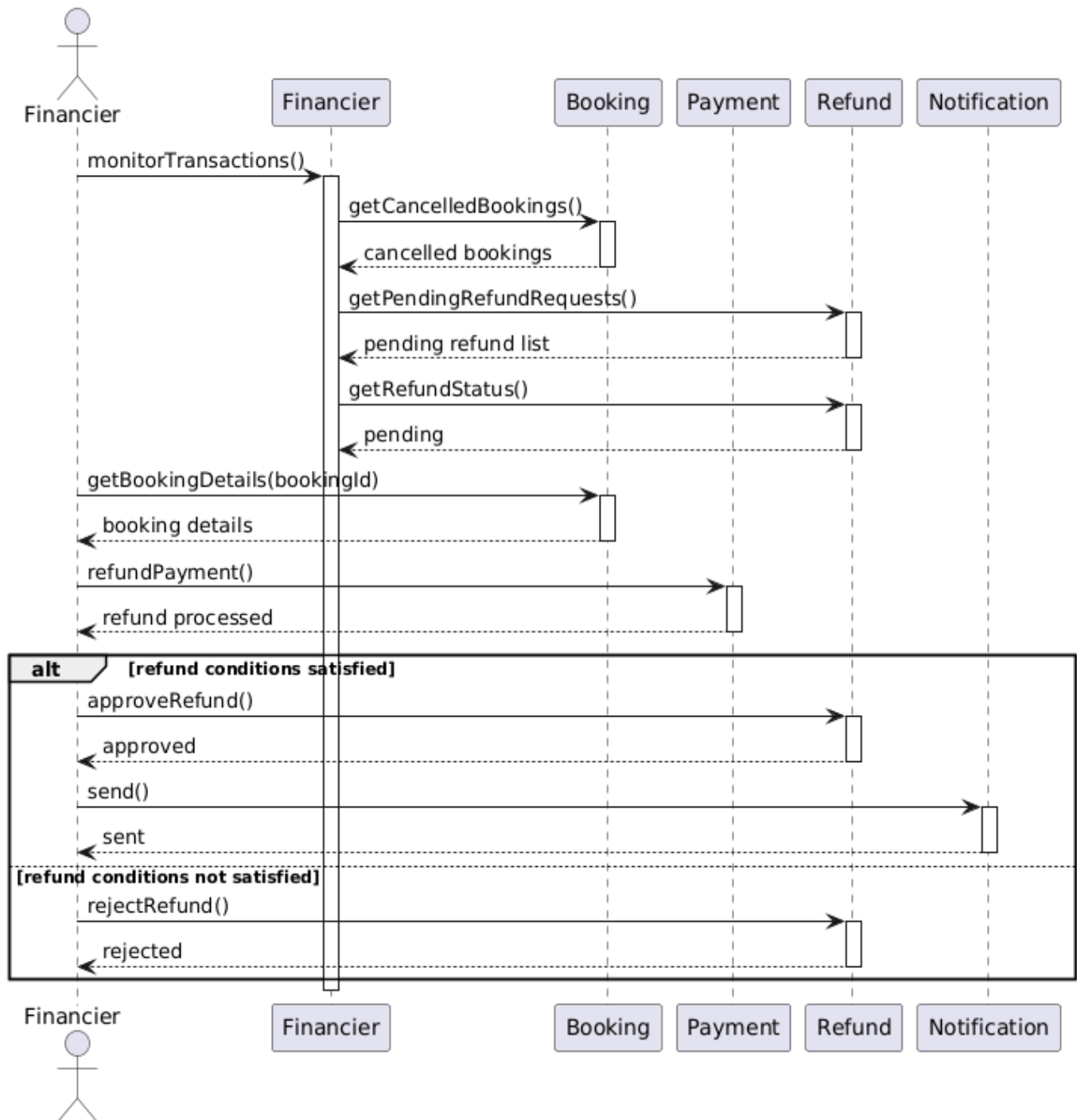
SC_PRS_35 – Monitor Payment Process and Security (Done by: Ornelt Lemellari)



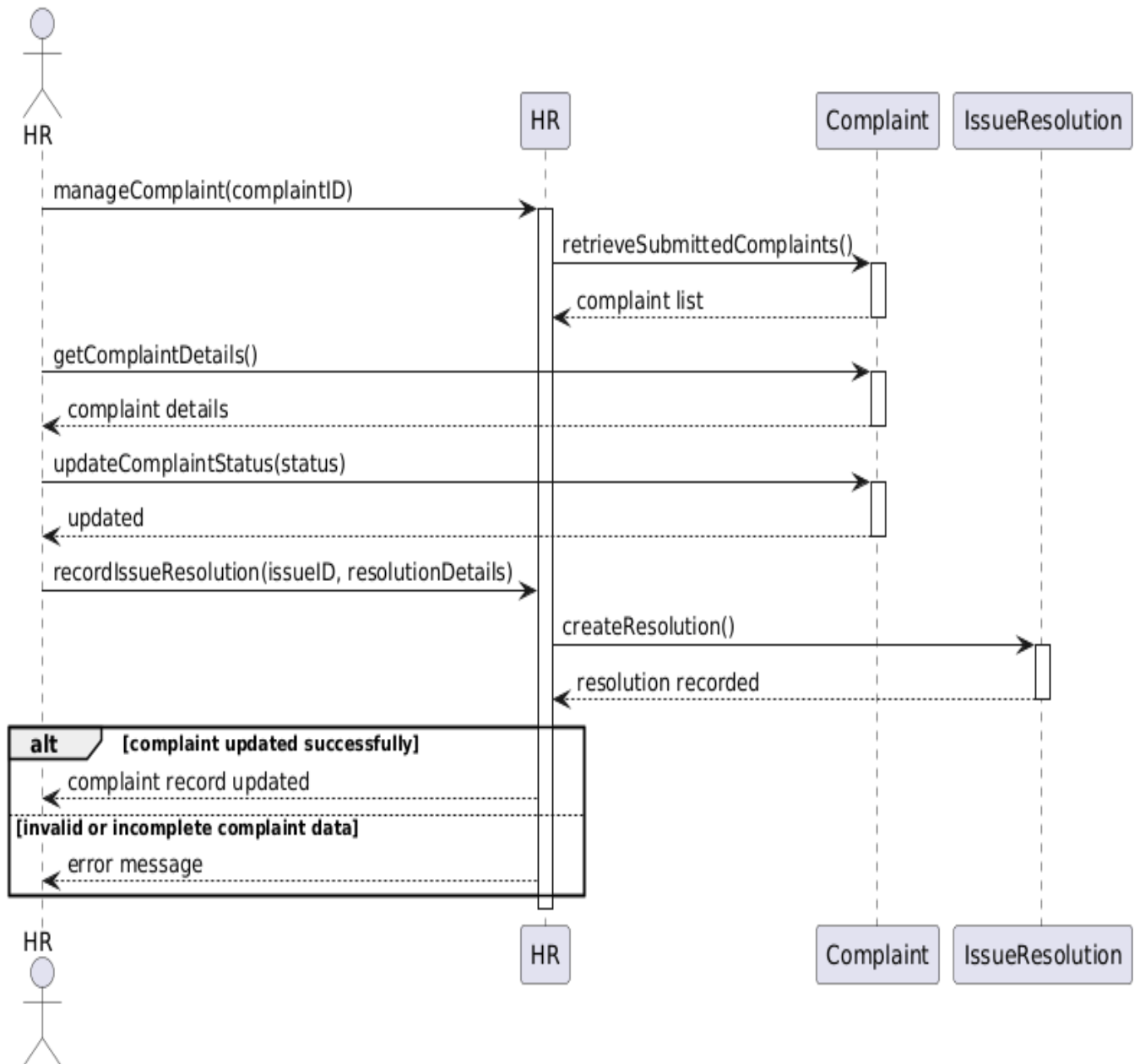
SC_PRS_36 – Communicate Payment Issues with Tenants and Owners
(Done by: Ornel Lemellari)



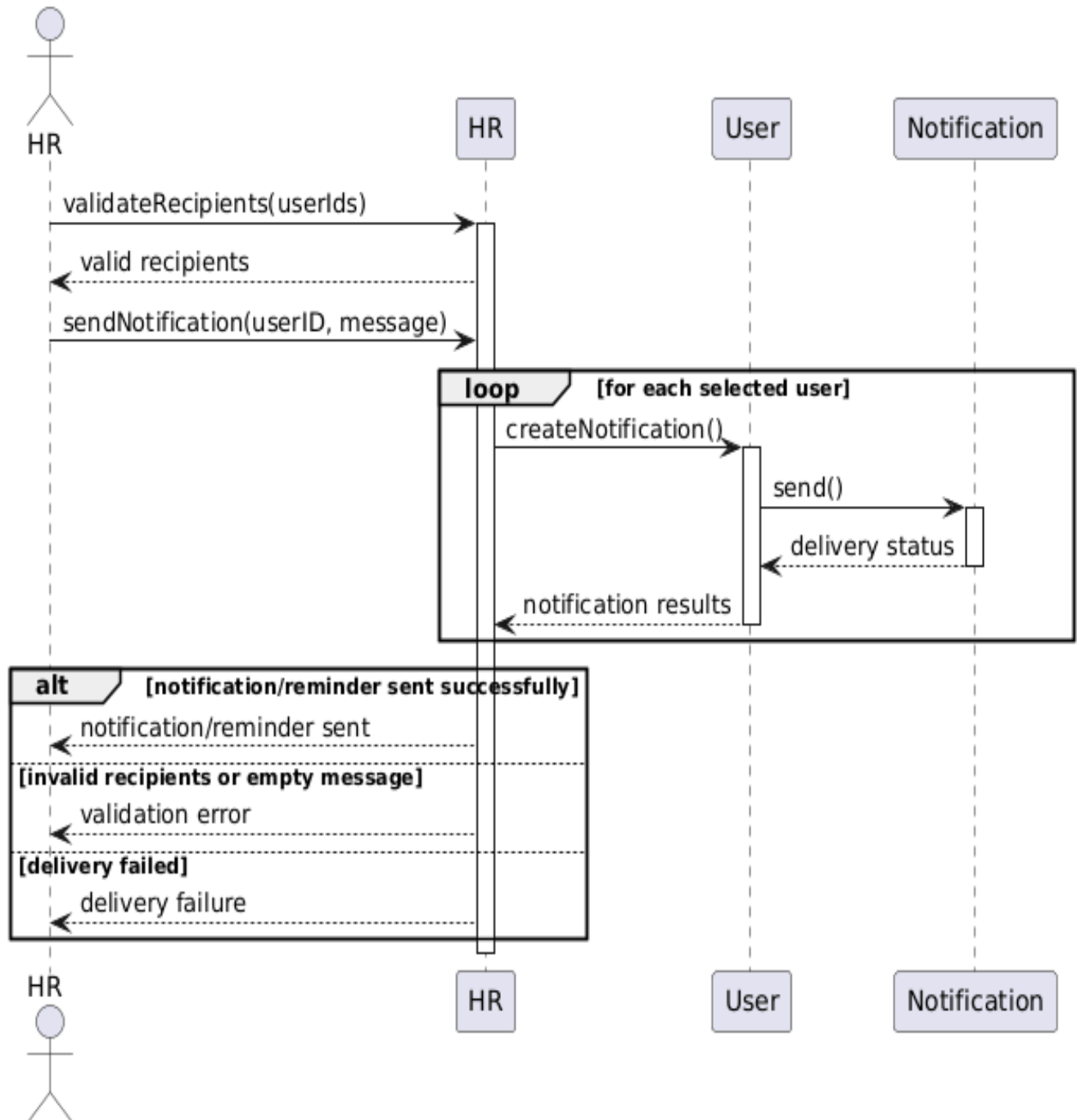
SC_PRS_37 – Approve Refunds for Cancelled Rental Transactions (Done by: Kevin Muho)



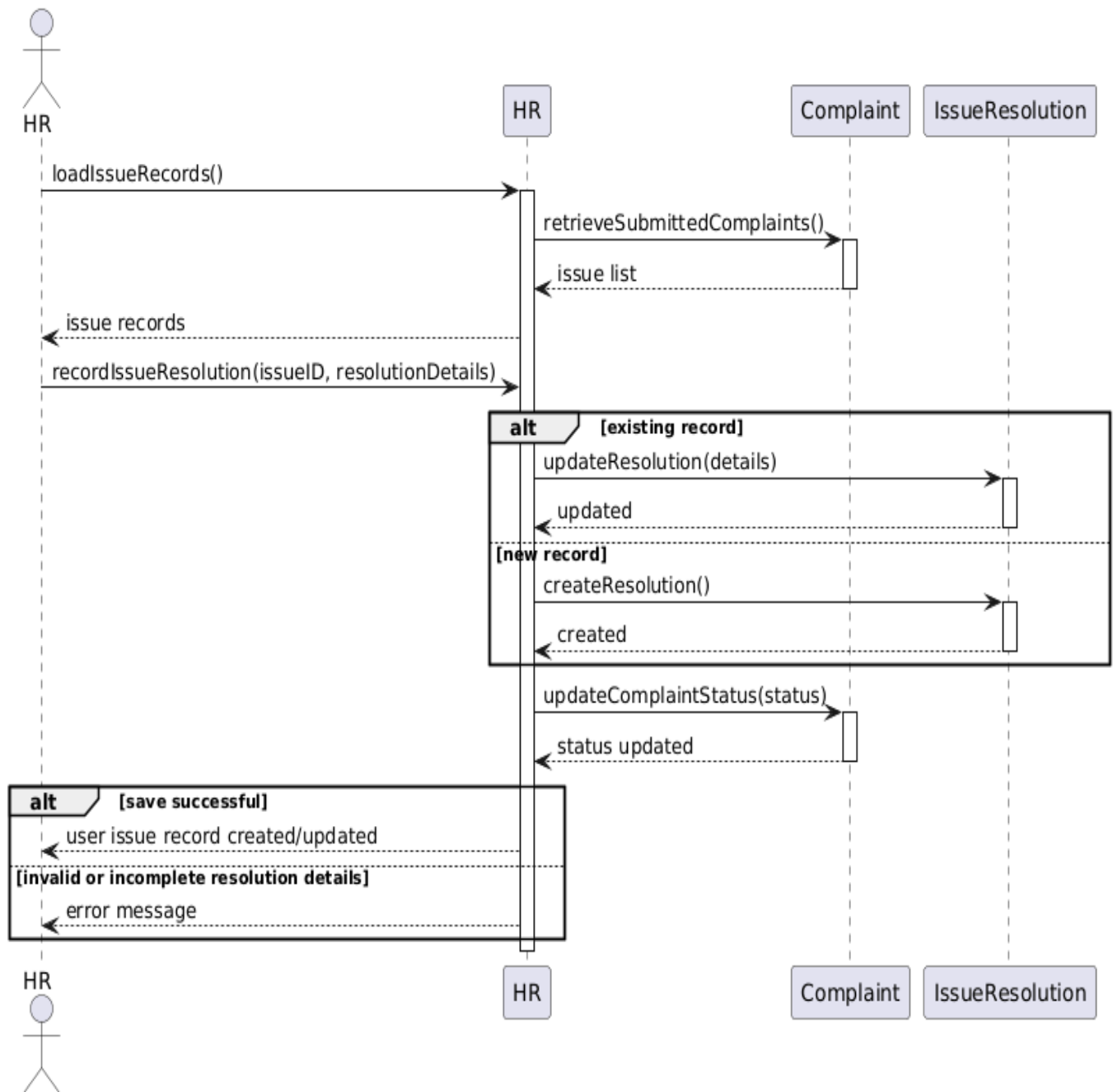
SC_PRS_38 – Manage Complaints (Done by: Kevin Muho)



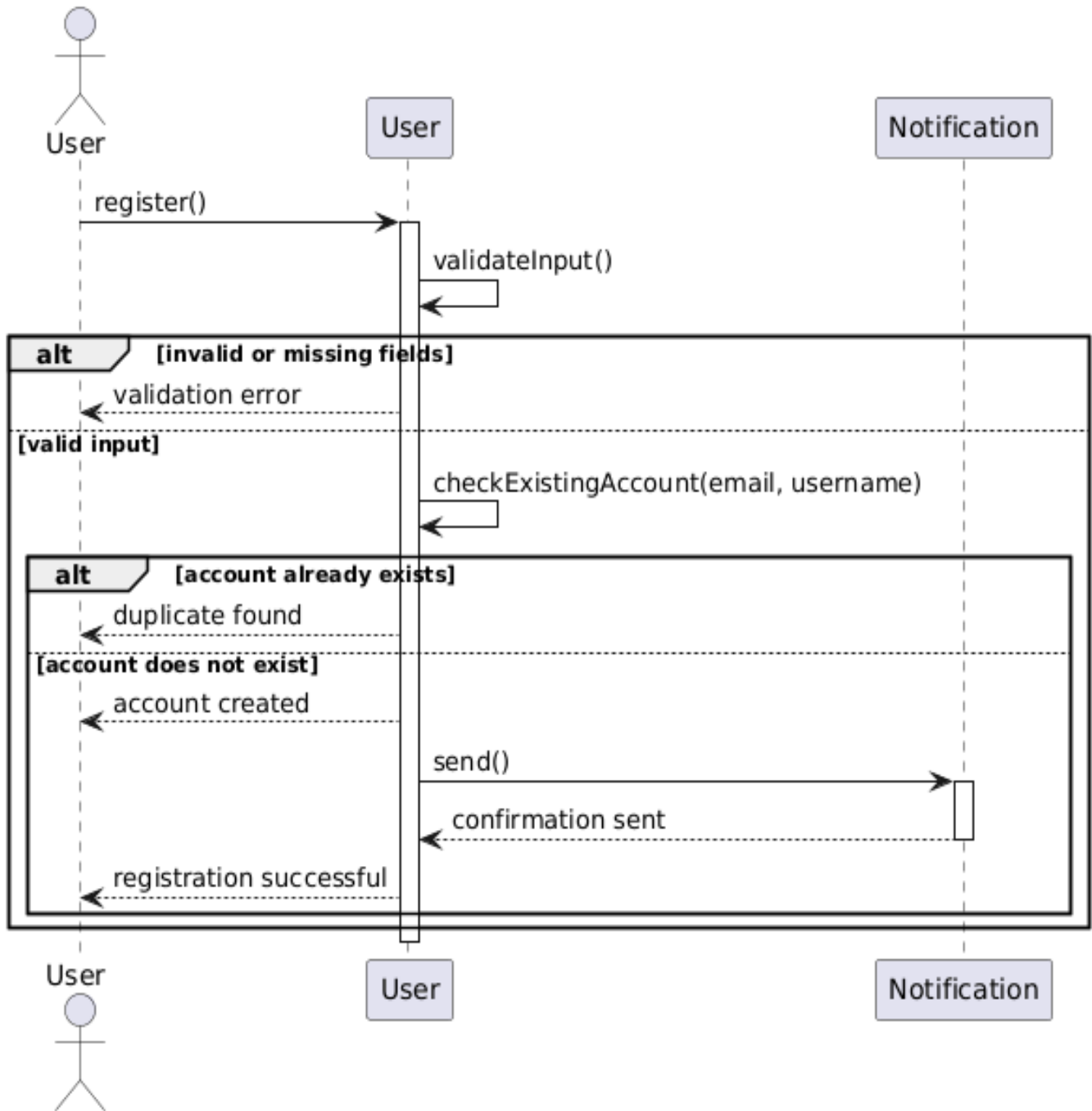
SC_PRS_39 – Send Notifications and Reminders (Done by: Kevin Muho)



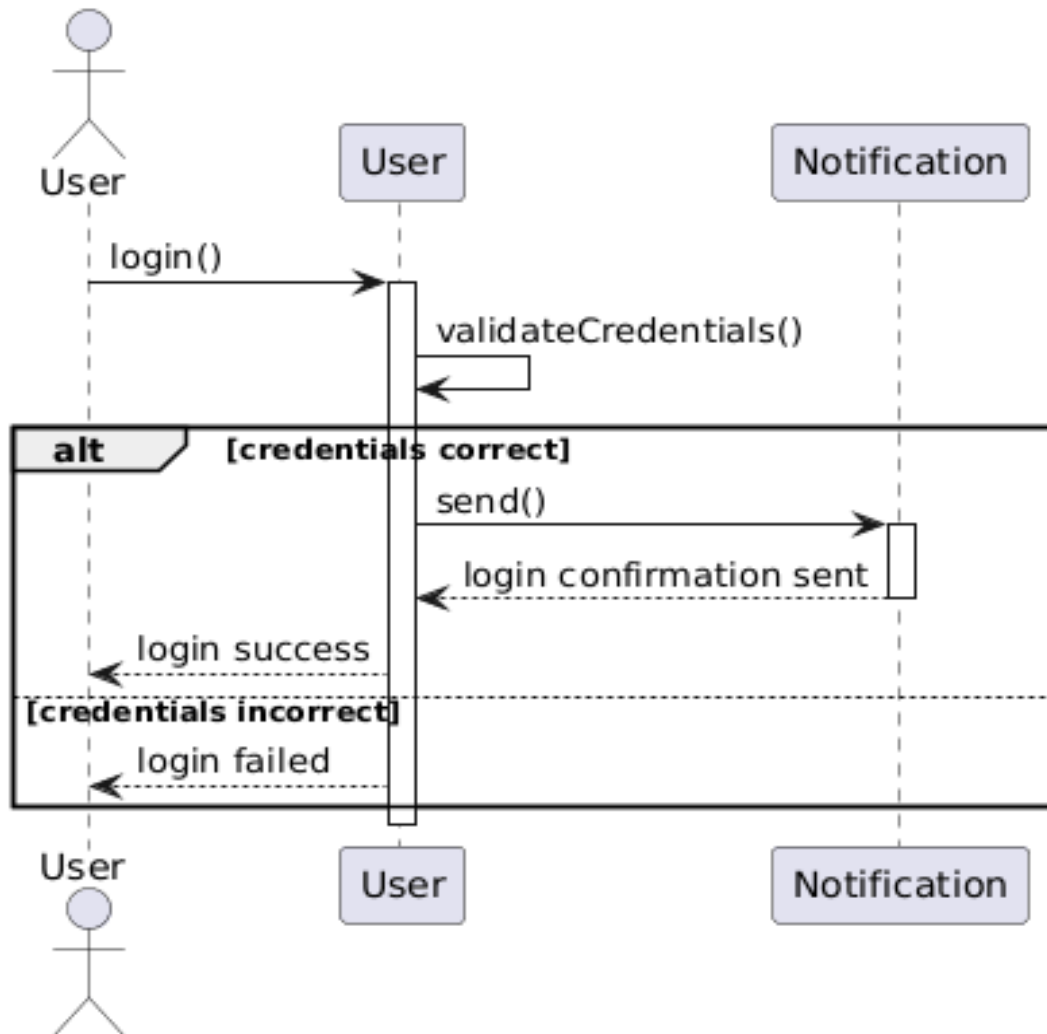
SC_PRS_40 – Maintain Records of User Issues and Resolutions (Done by: Kevin Muho)



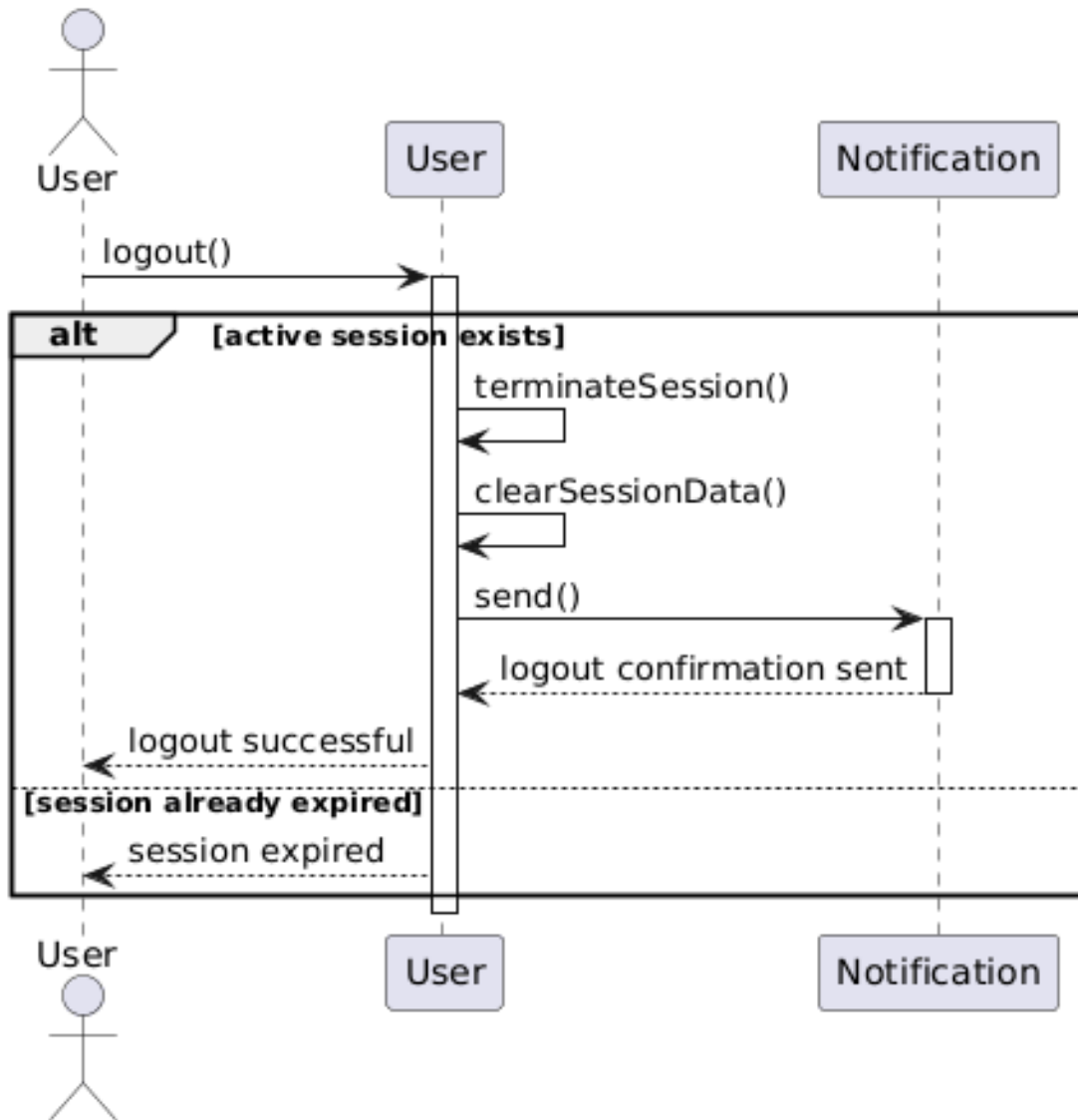
SC_PRS_41 – Register Account (Done by: Kevin Muho)



SC_PRS_42 – Log In (Done by: Kevin Muho)



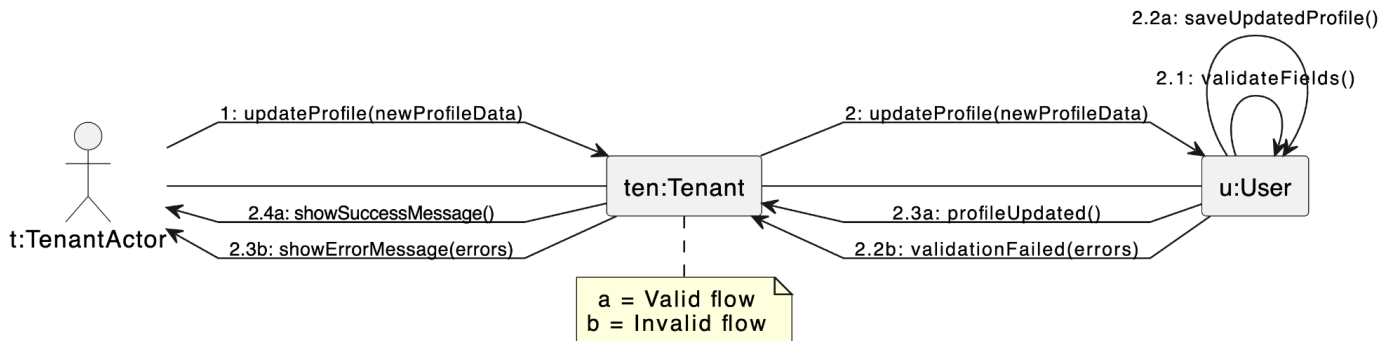
SC_PRS_43 – Log Out (Done by: Kevin Muho)



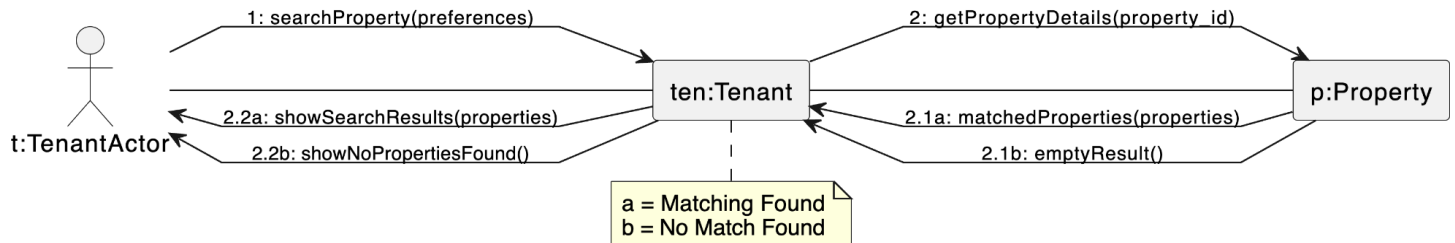
Property Rental System Requirement Specification

5.7. Collaboration diagram

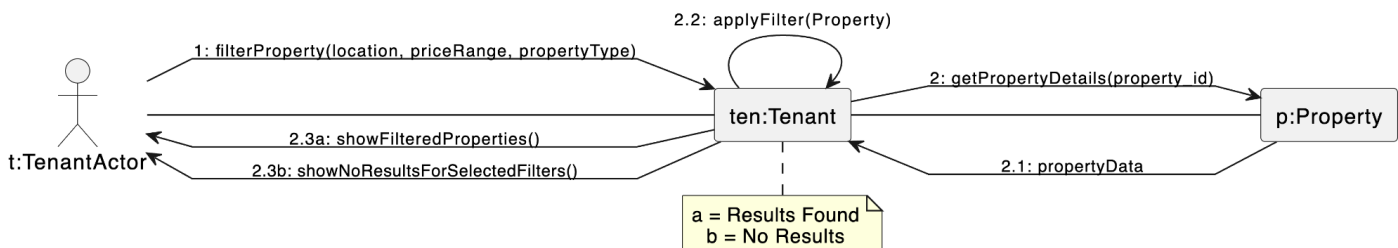
CC_PRS_1 - Update personal information (Done by:Anisa Domi)



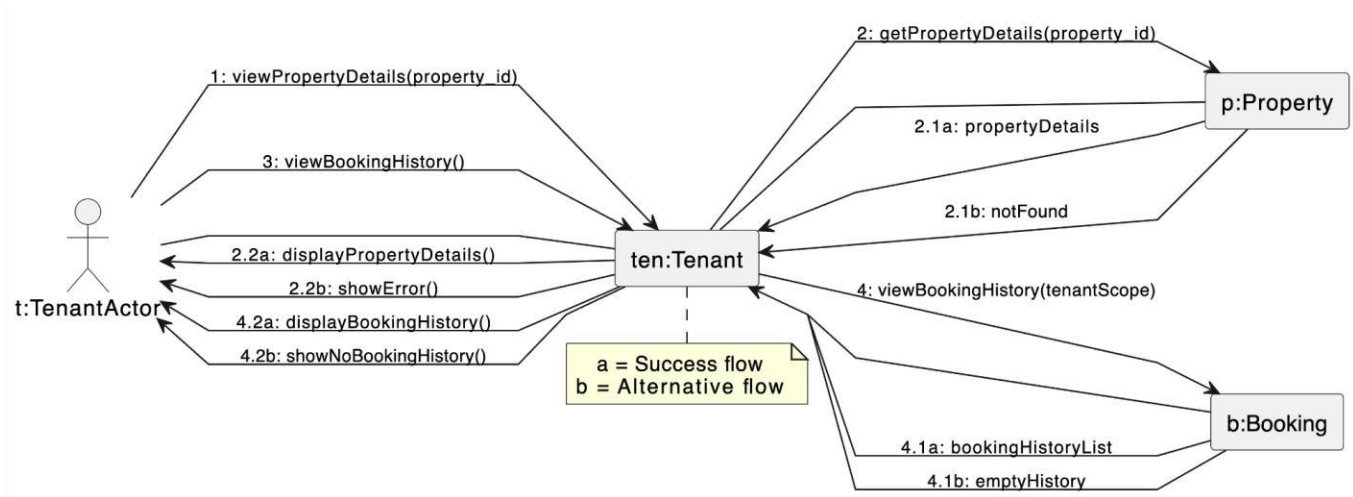
CC_PRS_2 - Search properties (Done by:Anisa Domi)



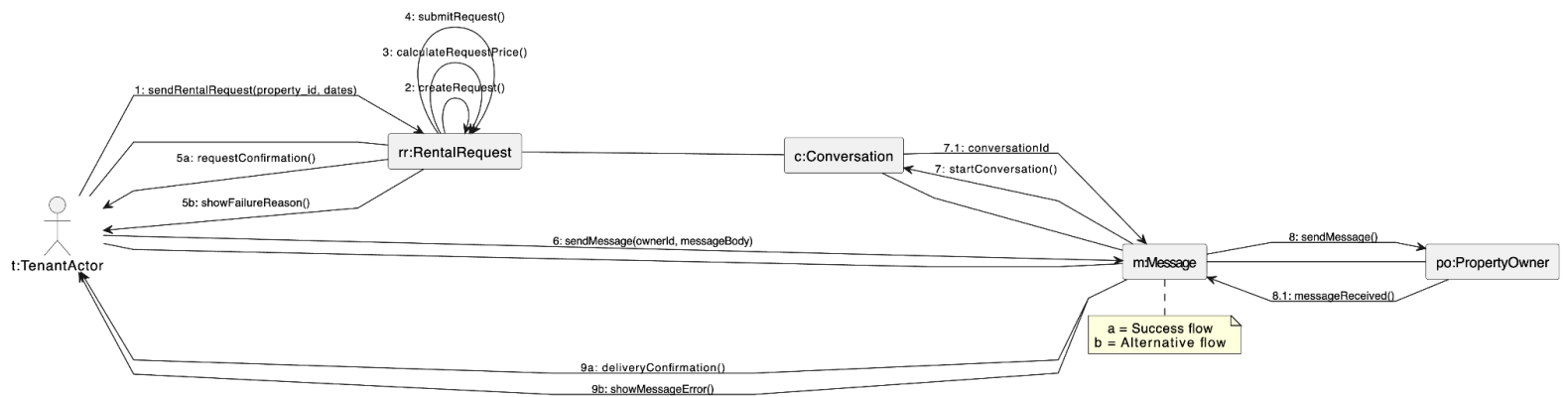
CC_PRS_3 - Filter properties (Done by:Anisa Domi)



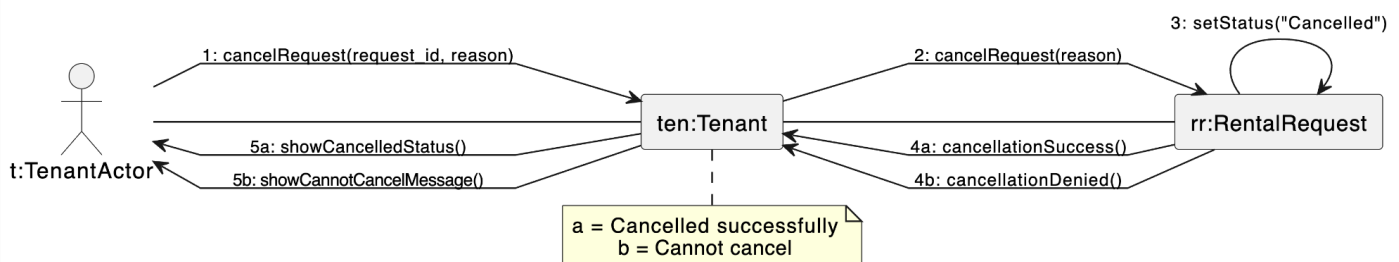
Property Rental System Requirement Specification
CC_PRS_4 - View property details and booking history (Done by:Anisa Domi)



CC_PRS_5 - Send rental requests and message to property owners (Done by:Anisa Domi)

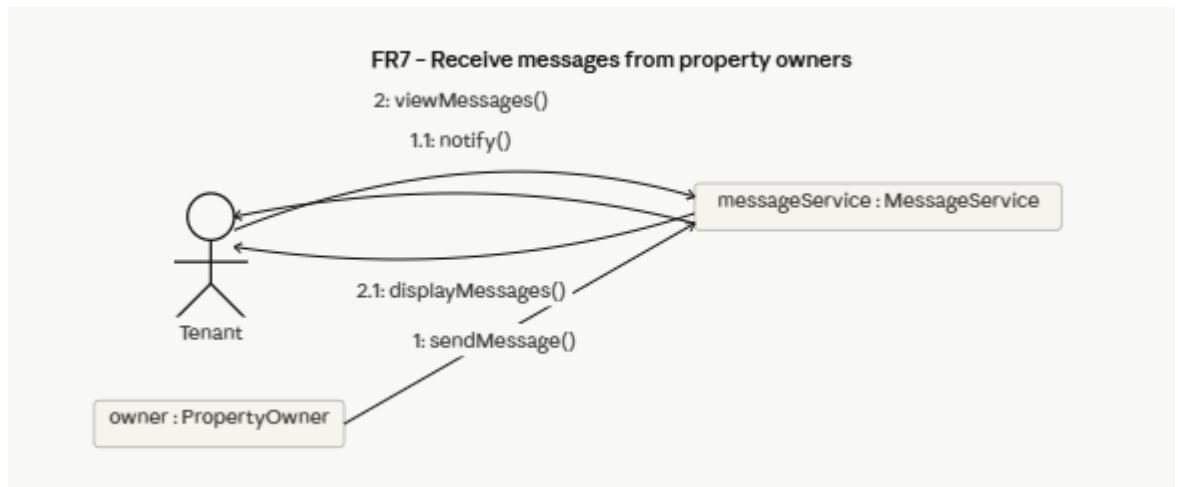


CC_PRS_6 - Cancel rental request (Done by:Anisa Domi)

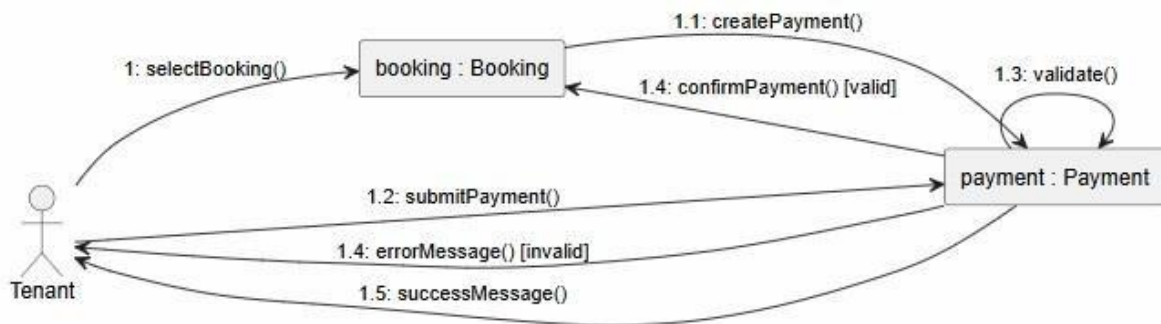


Property Rental System Requirement Specification

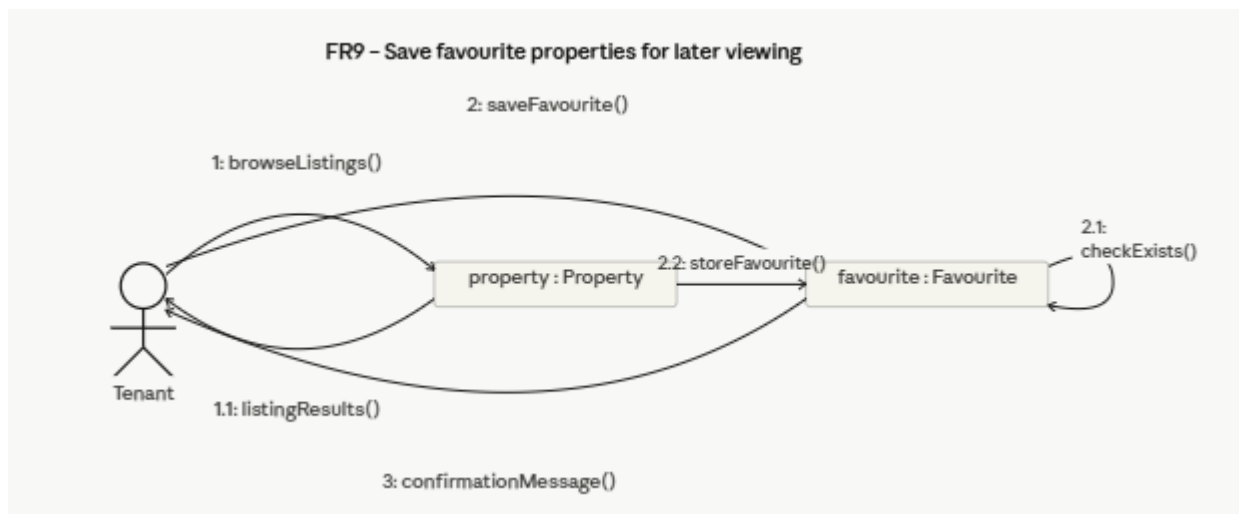
CC_PRS_7-Receive Messages from property owners (Done by: Haidi Toto)



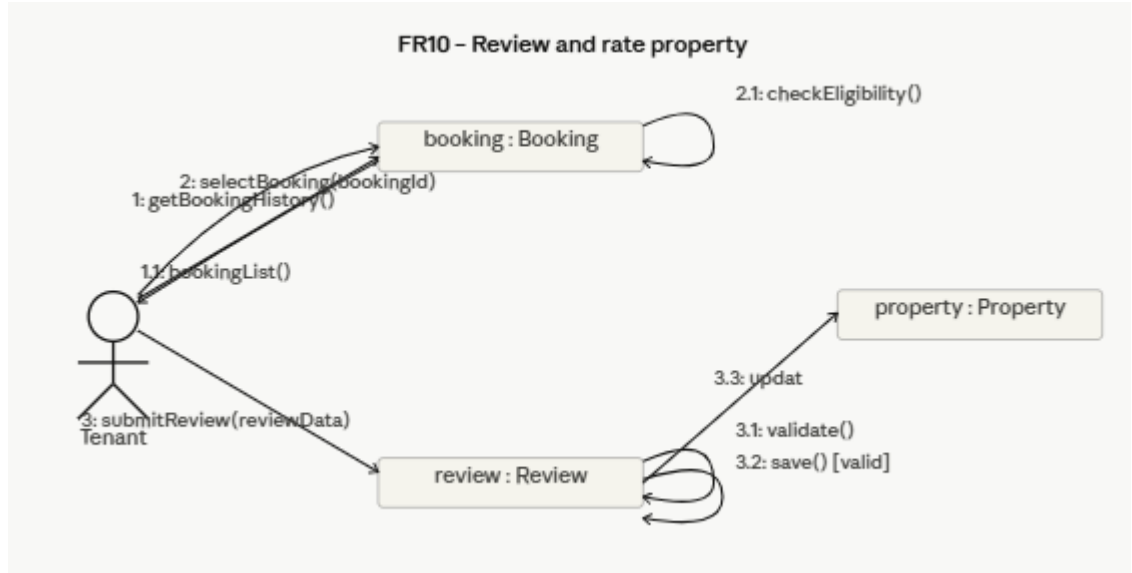
CC_PRS_8 – Submit Rental Payment (Done by: Haidi Toto)



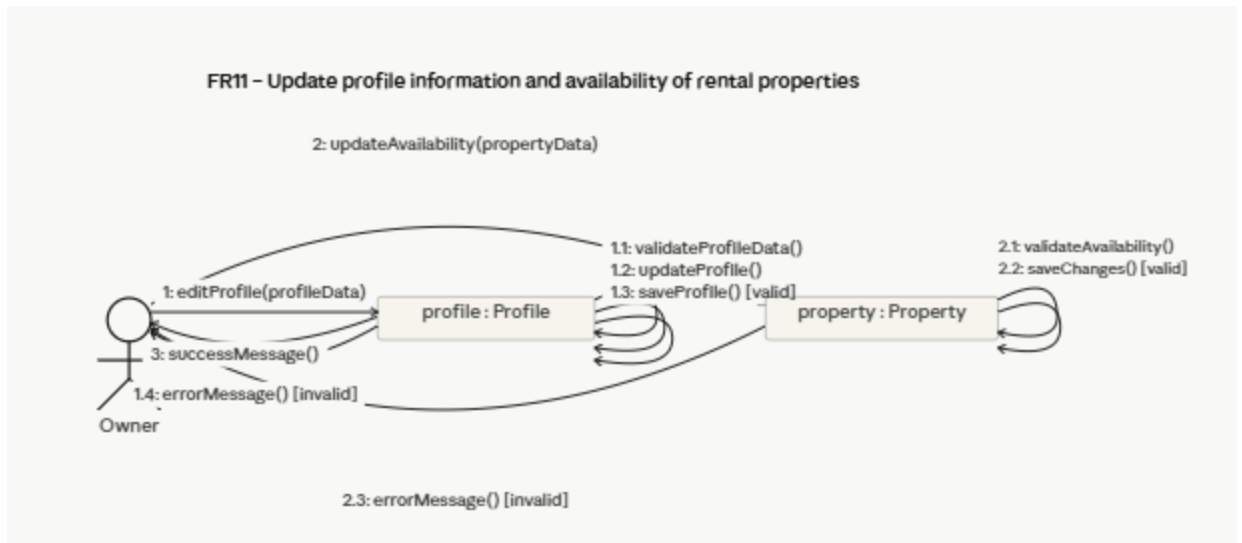
CC_PRS_9 – Save Favourite Properties for later viewing (Done by: Haidi Toto)



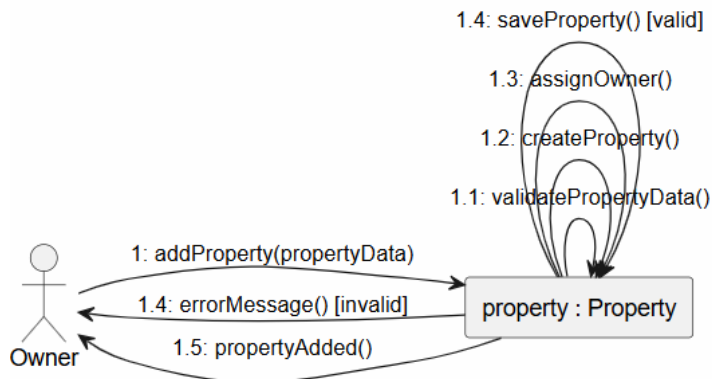
Property Rental System Requirement Specification
CC_PRS_10 – Review and Rate Property (Done by: Haidi Toto)



CC_PRS_11 – Update Profile information and Availability of Rental Properties (Done by: Haidi Toto)

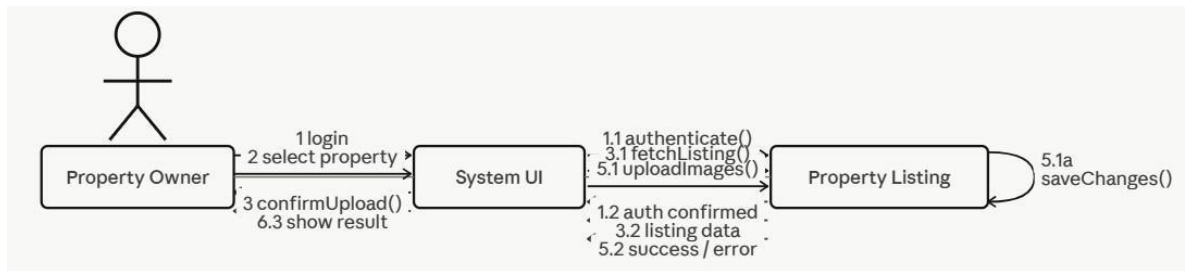


CC_PRS_12 – Add Property Listing (Done by: Haidi Toto)

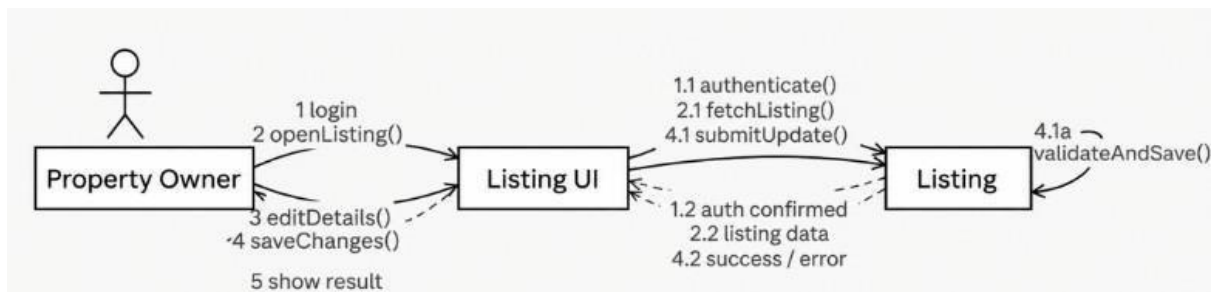


Property Rental System Requirement Specification

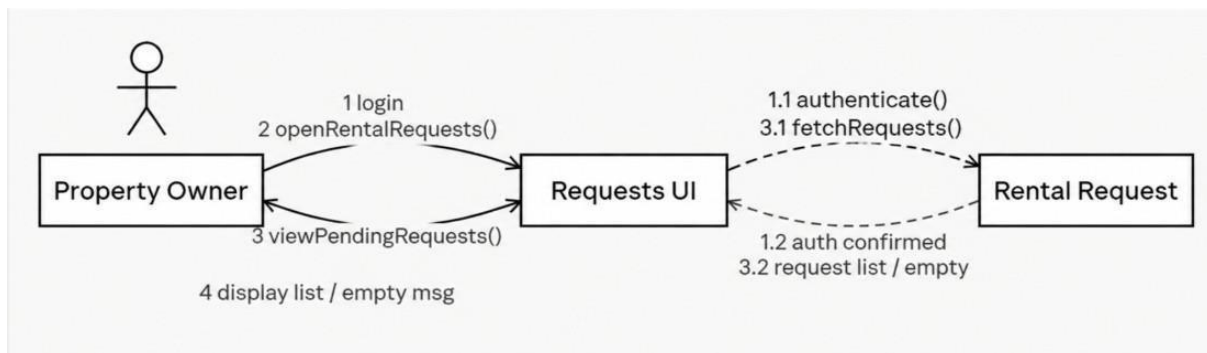
CC_PRS_13-Upload Property Images (Done by: Ledio Tafciu)



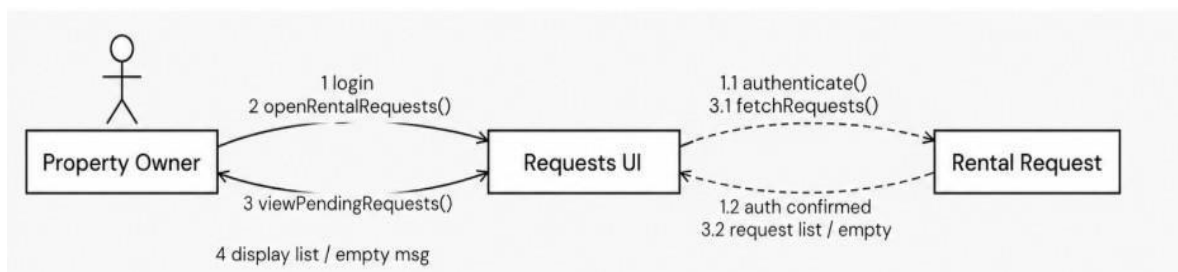
CC_PRS_14 - Edit Property Listing (Done by: Ledio Tafciu)



CC_PRS_15 - Delete Property Listing (Done by: Ledio Tafciu)

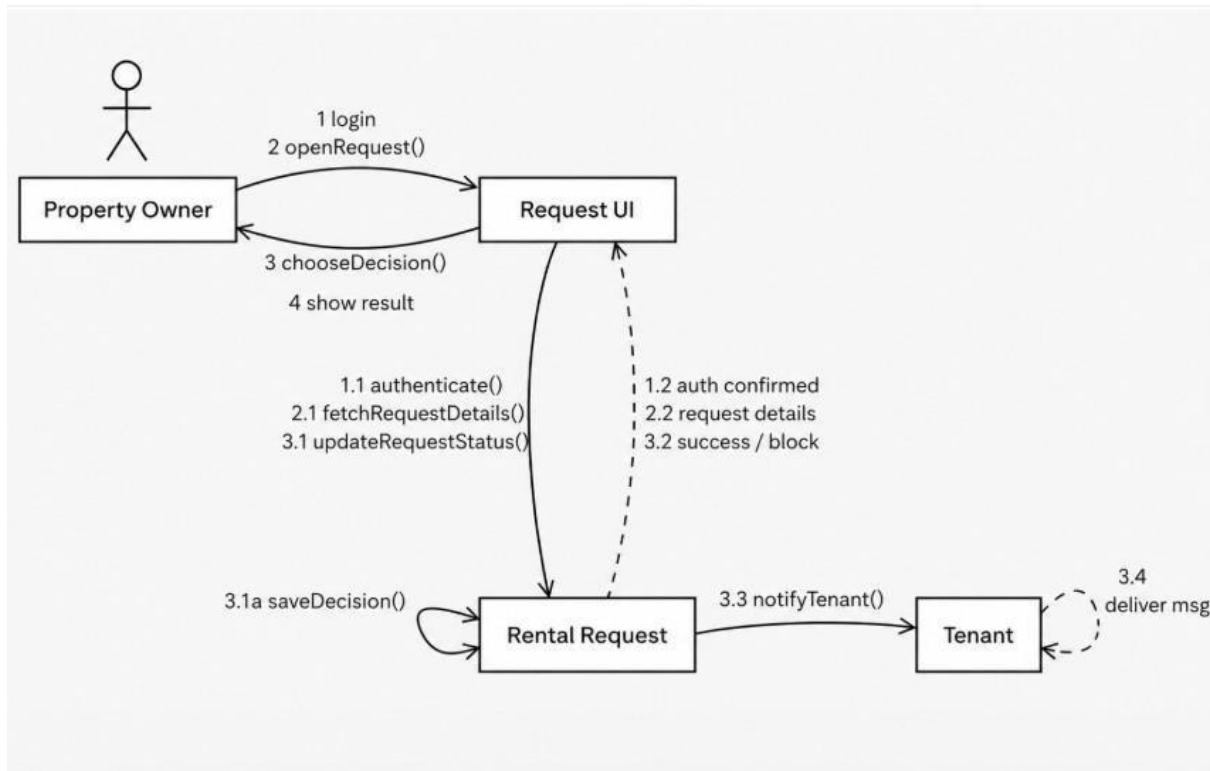


CC_PRS_16 - View Rental Requests (Done by: Ledio Tafciu)

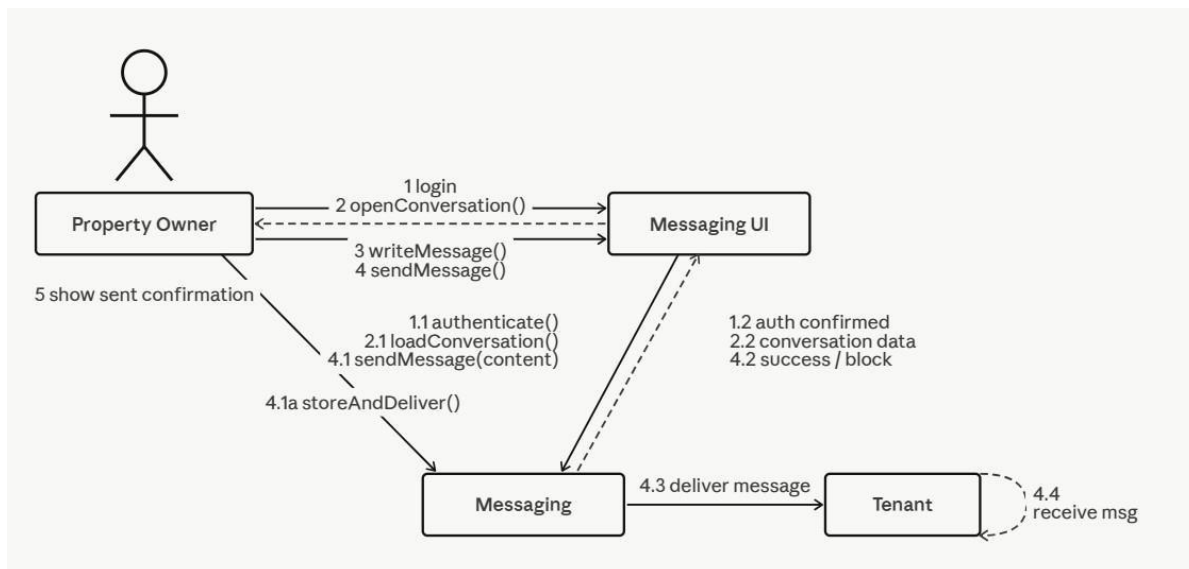


Property Rental System Requirement Specification

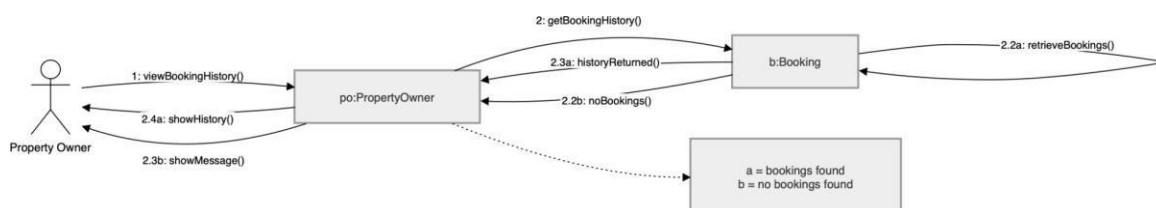
CC_PRS_17 - Accept or Reject Rental Request (Done by: Ledio Tafciu)



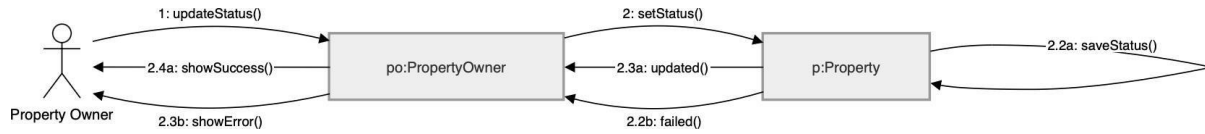
CC_PRS_18 - Send Message To Tenant (Done by: Ledio Tafciu)



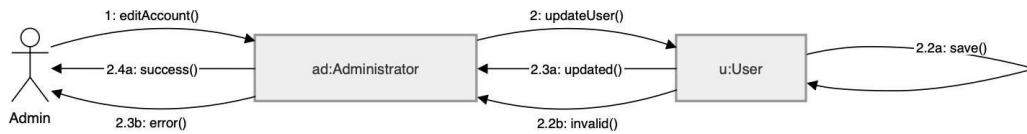
CC_PRS_19 - View Booking History (Done by: Desard Lybeshari)



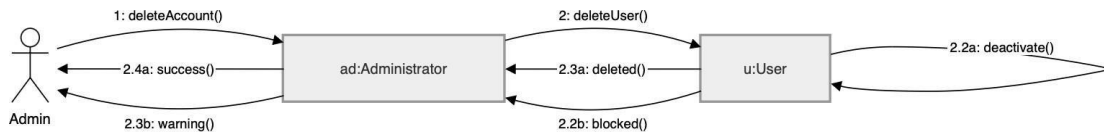
Property Rental System Requirement Specification
CC_PRS_20 - Mark Property as Unavailable or Rented (Done by:Desard Lybeshari)



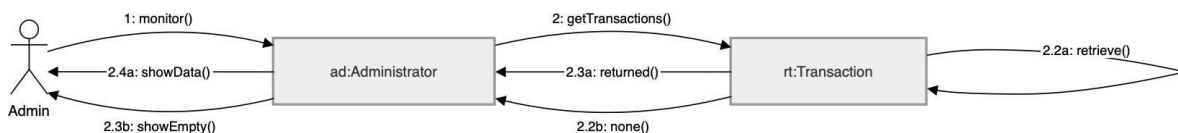
CC_PRS_21 - Edit User Accounts (Done by:Desard Lybeshari)



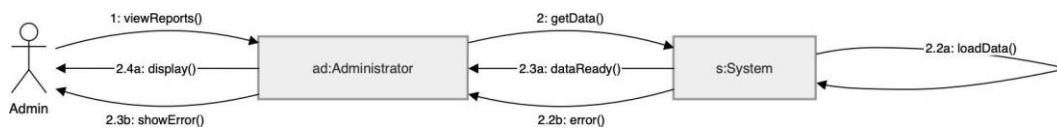
CC_PRS_22 - Delete User Accounts (Done by:Desard Lybeshari)



CC_PRS_23 - Monitor Rental Transactions (Done by:Desard Lybeshari)

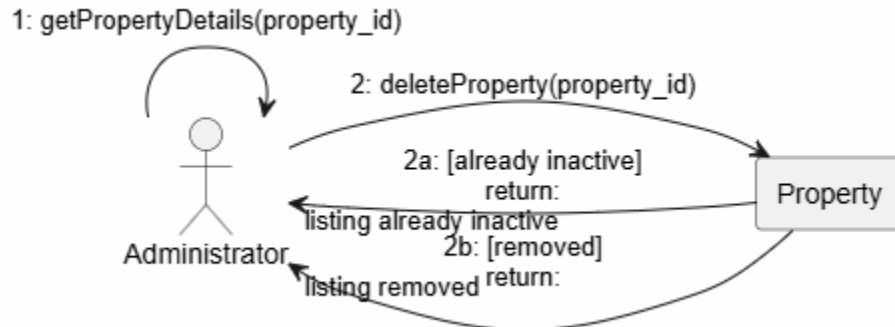


CC_PRS_24 - View System Reports and Property Listings (Done by:Desard Lybeshari)

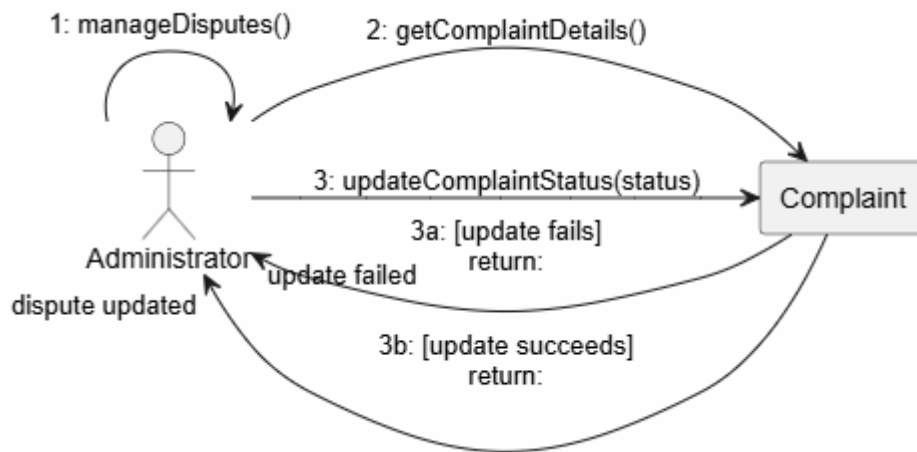


Property Rental System Requirement Specification

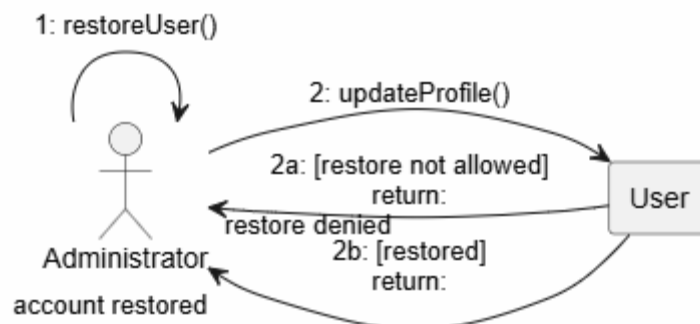
CC_PRS_25- Remove inappropriate listings (Done by:Matias Bali)



CC_PRS_26- Manage disputes & system settings (Done by:Matias Bali)

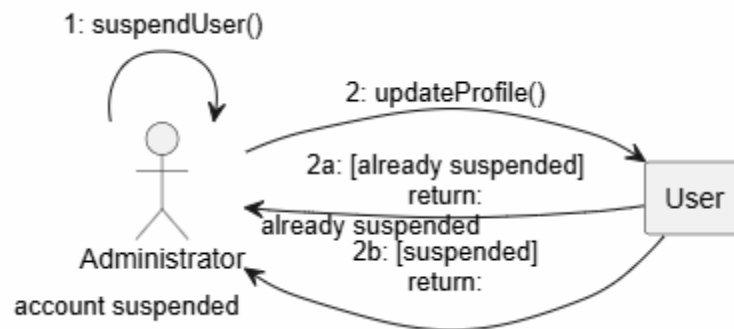


CC_PRS_27- Restore user accounts (Done by:Matias Bali)

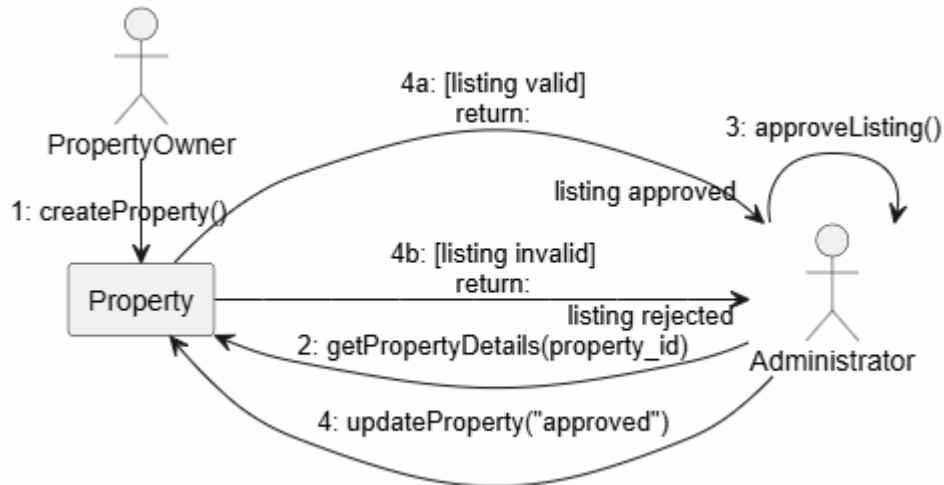


Property Rental System Requirement Specification

CC_PRS_28- Suspend user accounts (Done by:Matias Bali)

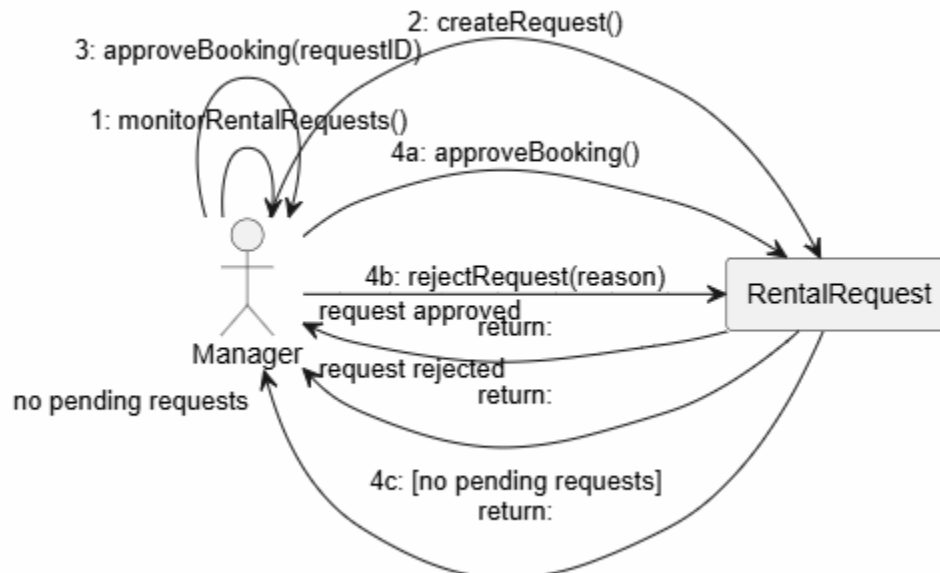


CC_PRS_29- Verify and Approve property listings (Done by:Matias Bali)

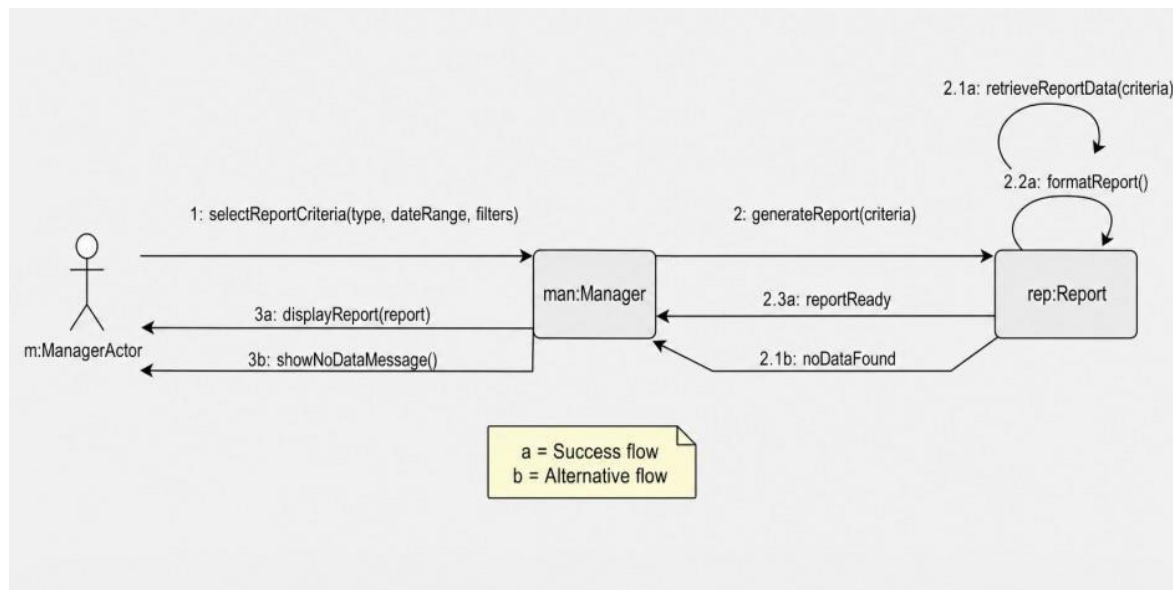


Bali)

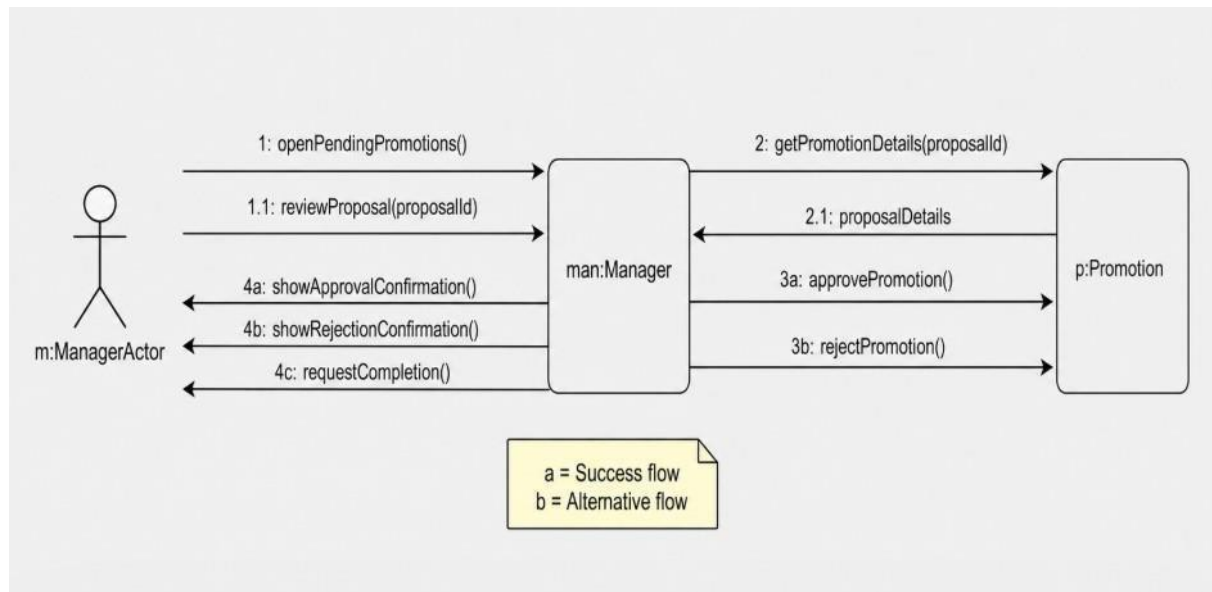
CC_PRS_30- Monitor rental requests (Done by:Matias Bali)



Property Rental System Requirement Specification
CC_PRS_31 – Generate Management Reports (Done by: Ornelt Lemellari)

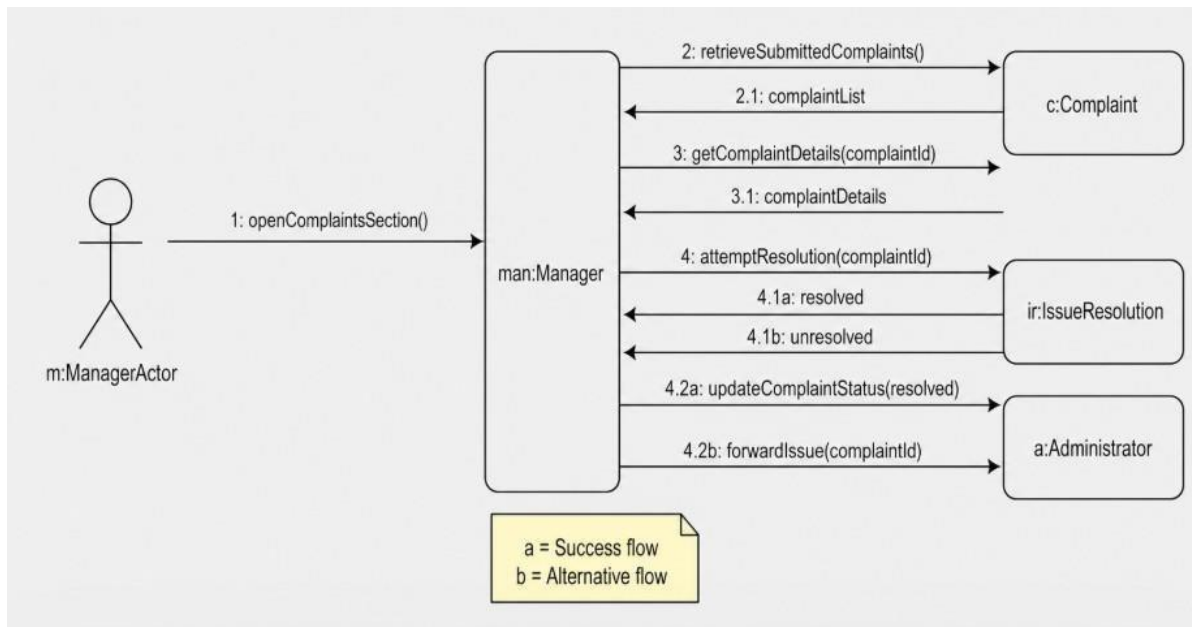


CC_PRS_32 – Approve Large Discounts (Done by: Ornelt Lemellari)

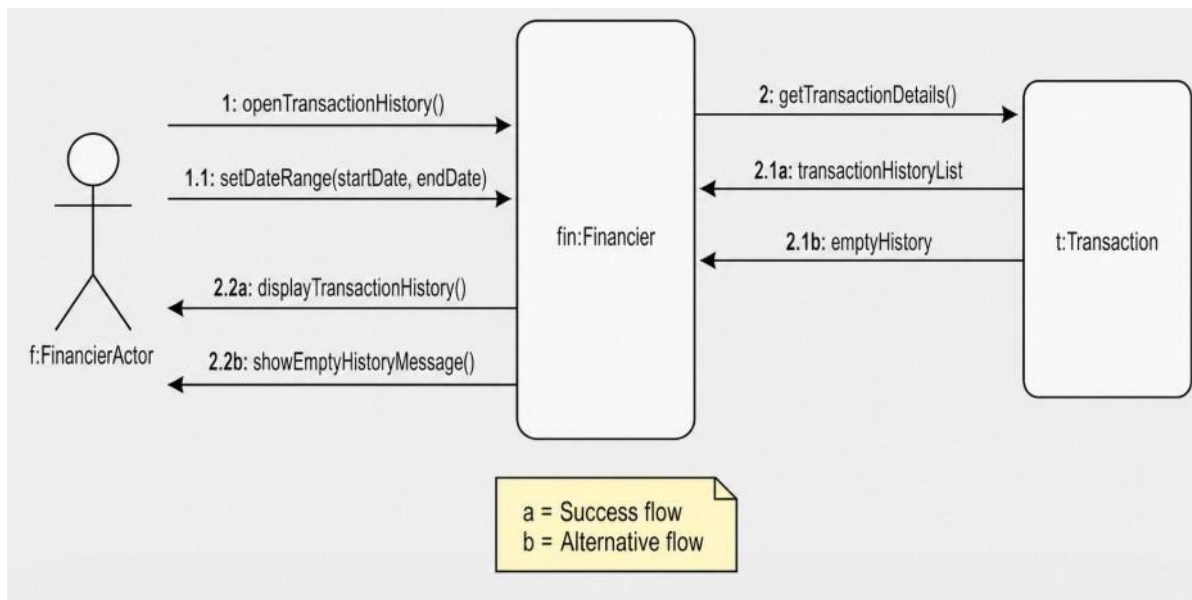


Property Rental System Requirement Specification

CC_PRS_33 – Review Complaints and forward unresolved issues (Done by: Ornelt Lemellari)

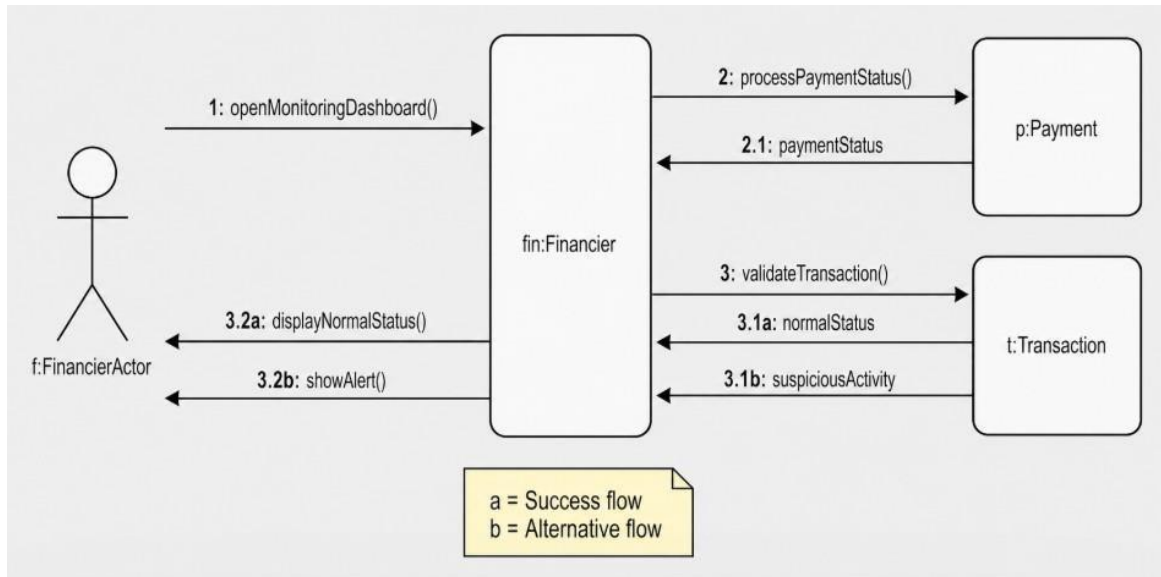


CC_PRS_34 – View Transaction History (Done by: Ornelt Lemellari)

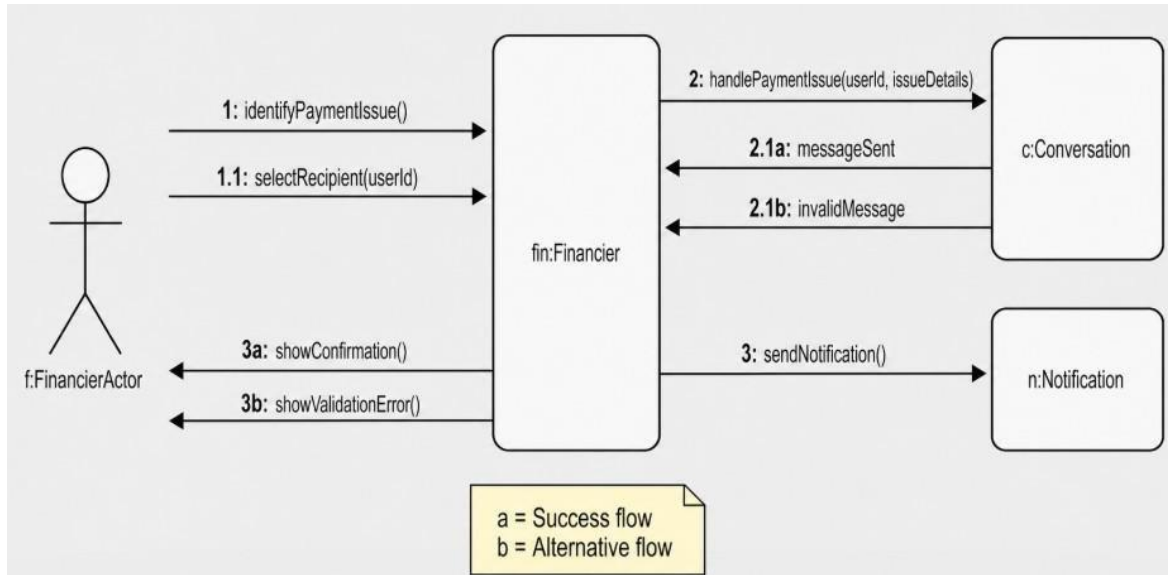


Property Rental System Requirement Specification

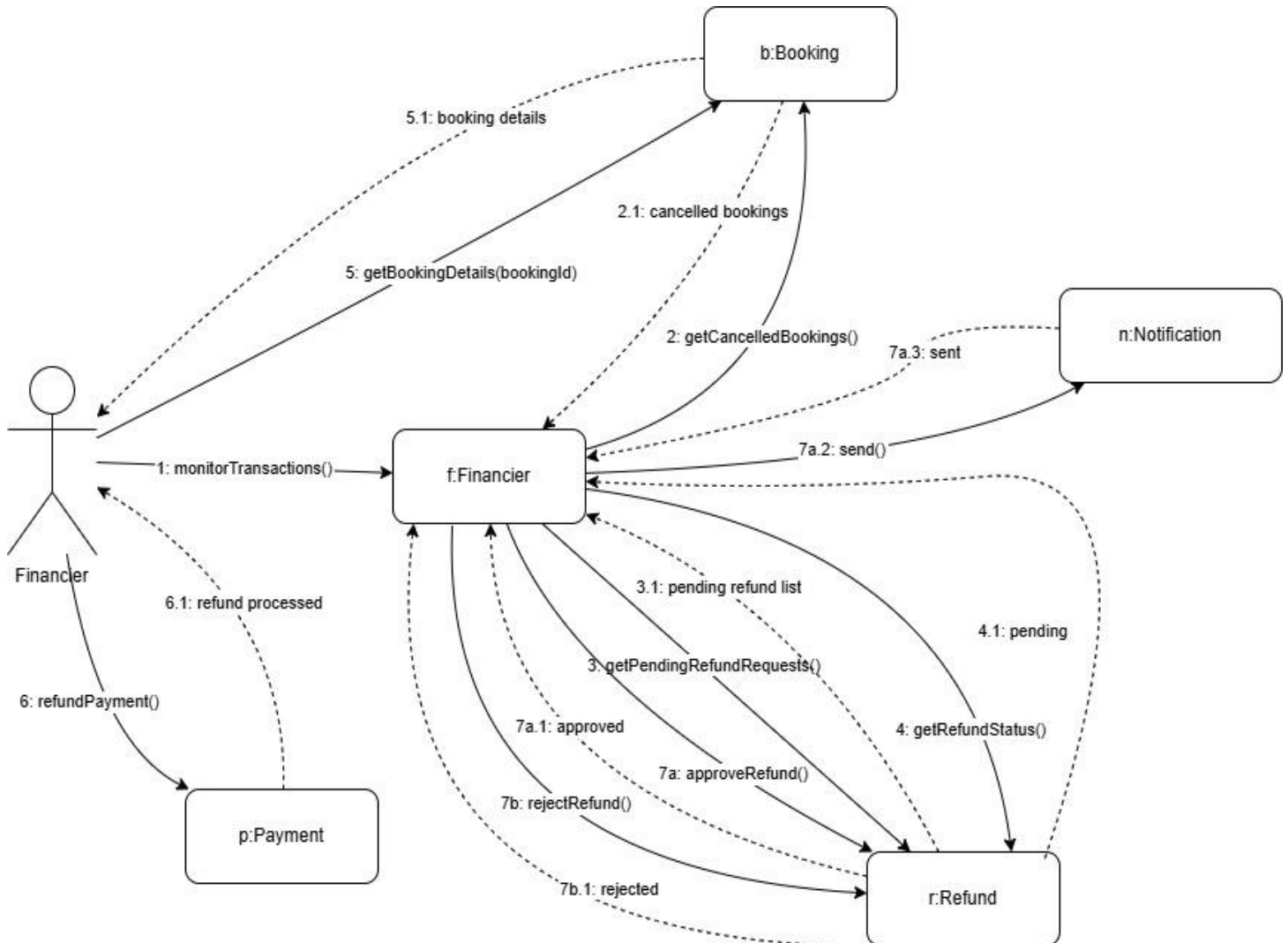
CC_PRS_35 – Monitor Payment process and Security (Done by: Ornel Lemellari)



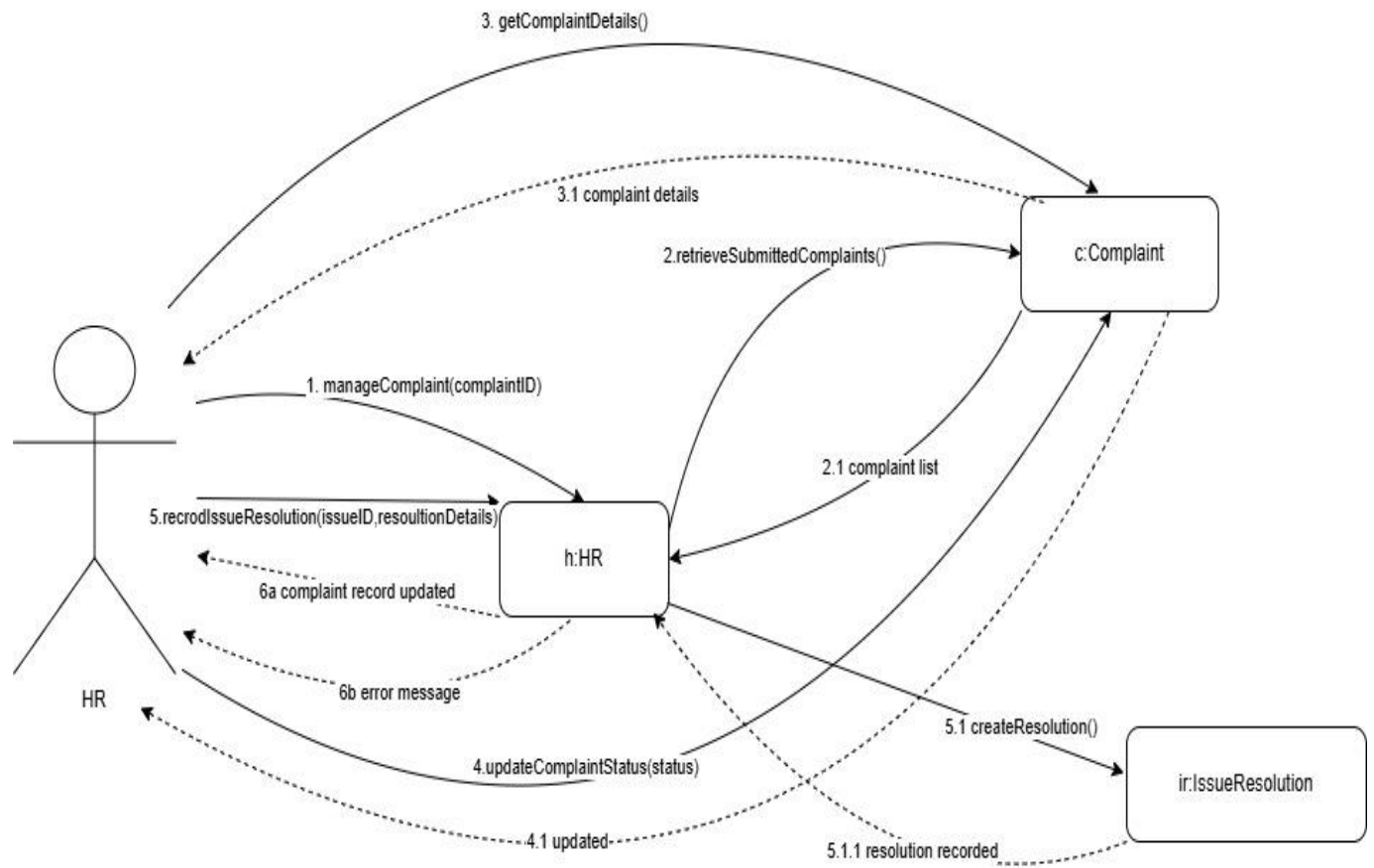
CC_PRS_36 – Communicate Payment Issues with tenants and owners (Done by: Ornel Lemellari)



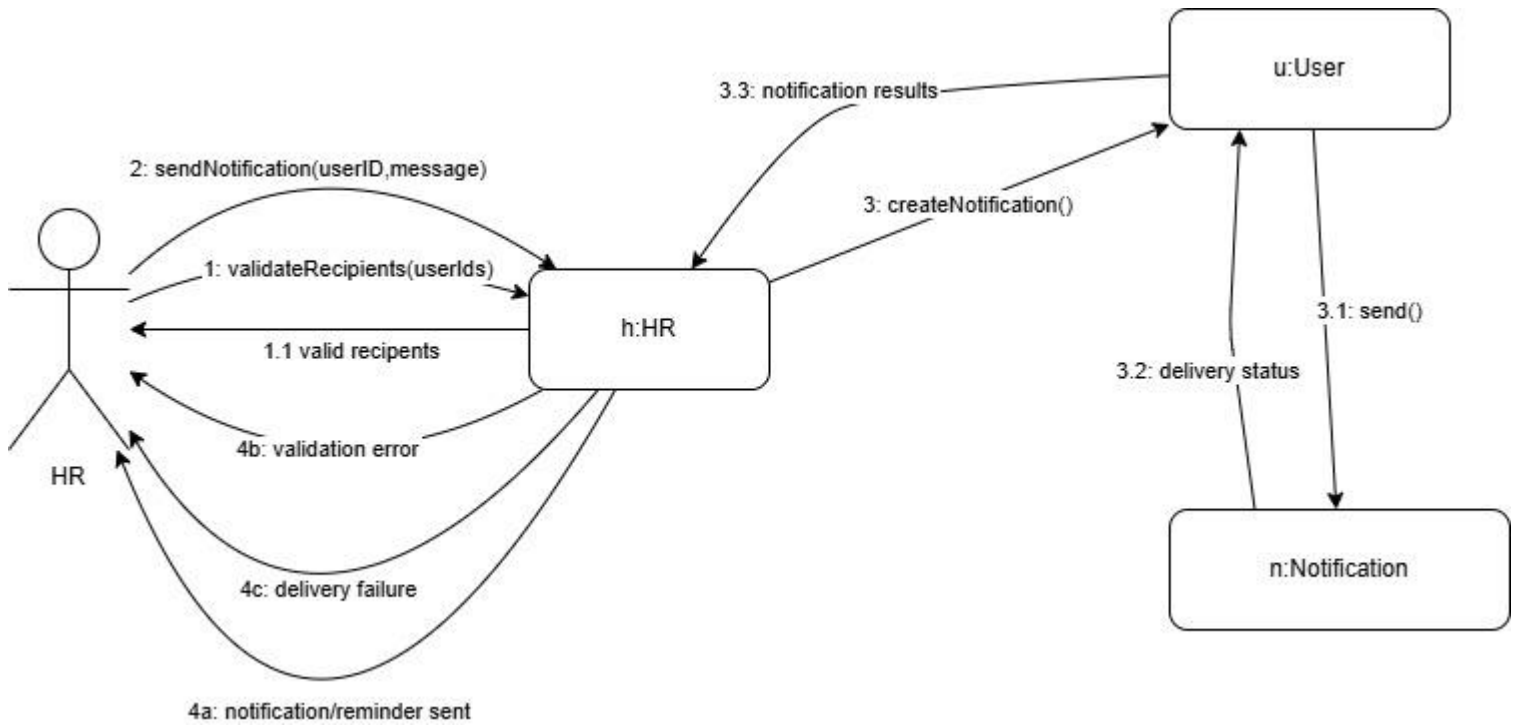
CC_PRS_37- Approve refunds for cancelled rental transactions (Done by:Kevin Muho)



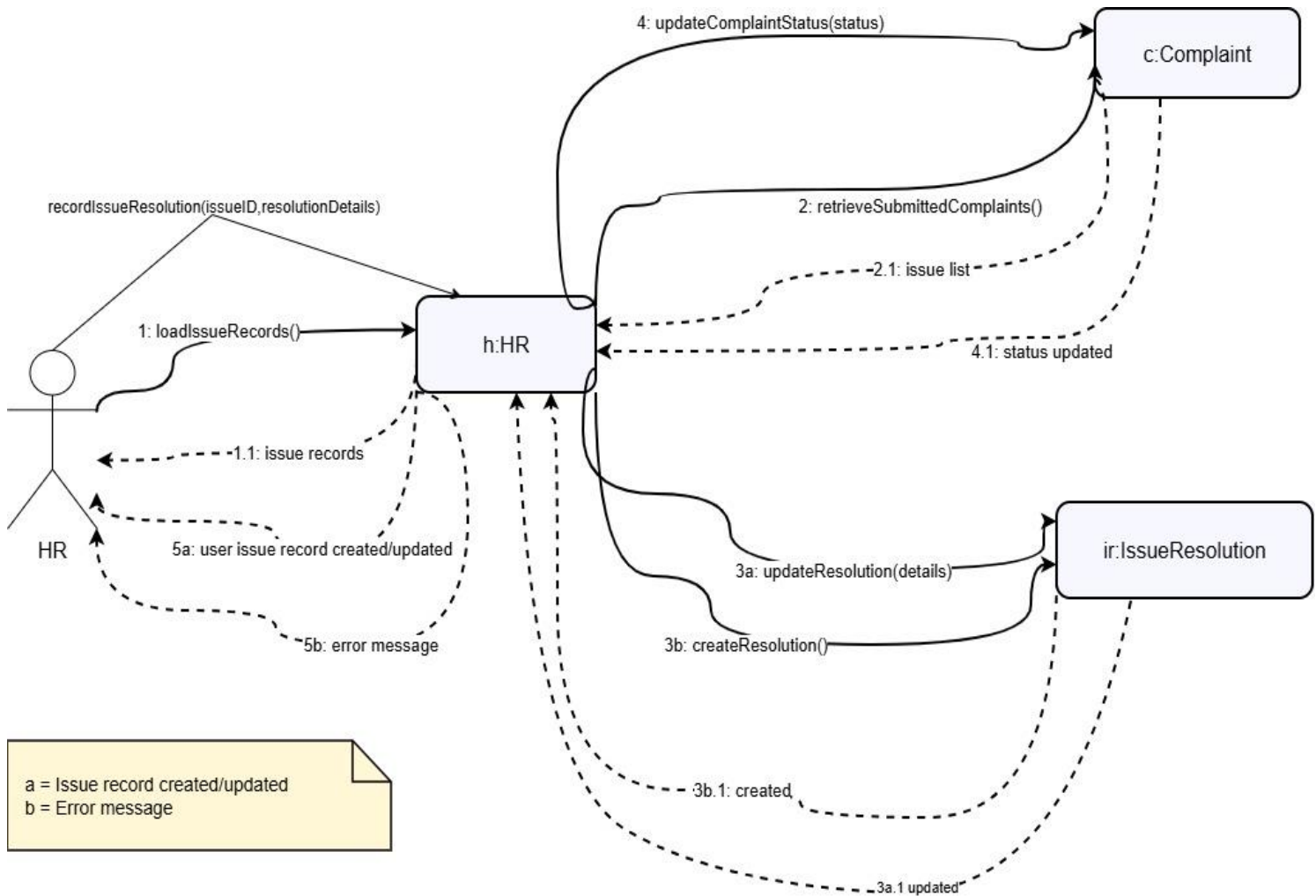
CC_PRS_38- Manage Complaints (Done by: Kevin Muho)



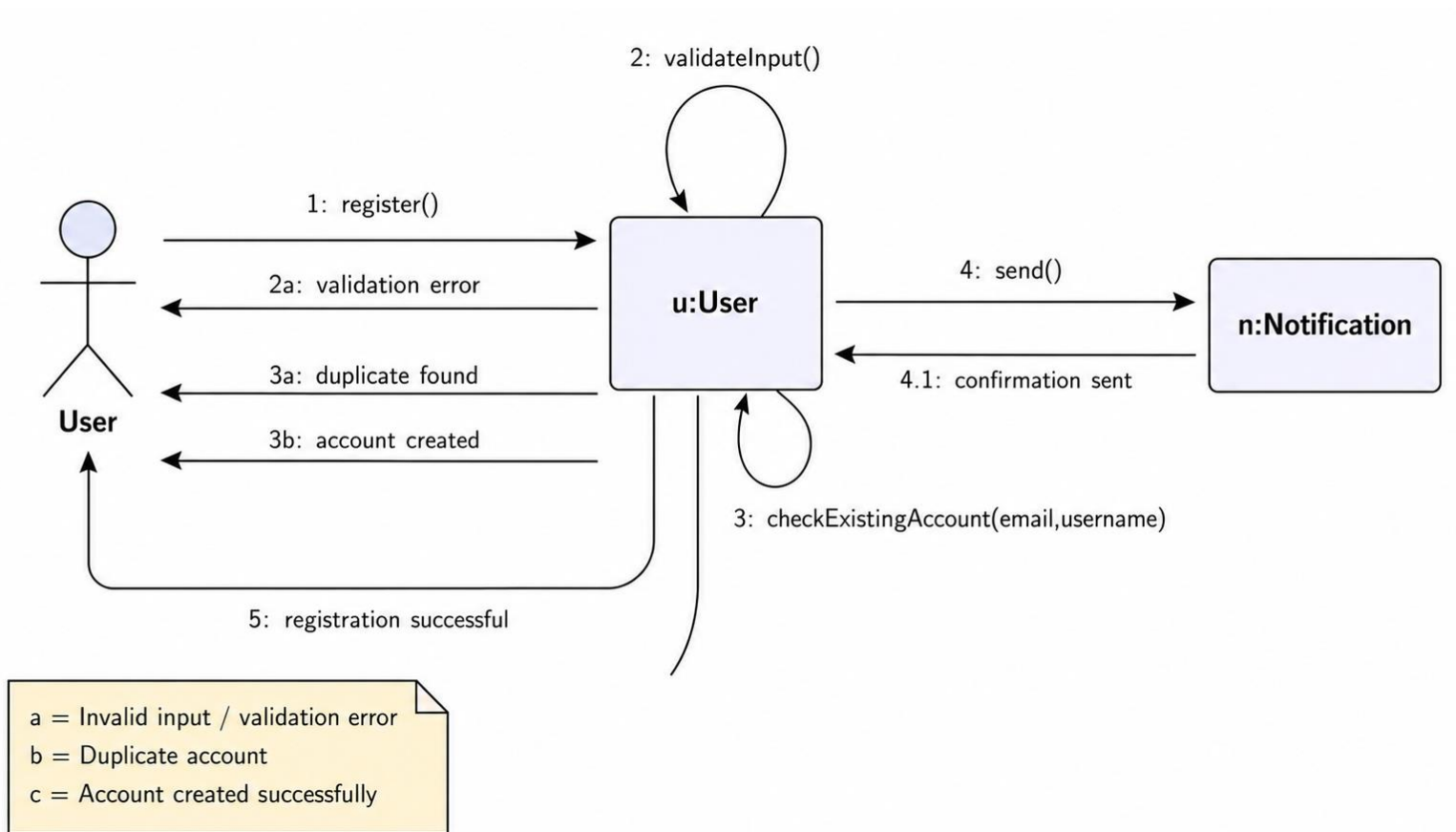
CC_PRS_39- Send notifications and reminders (Done by:Kevin Muho)



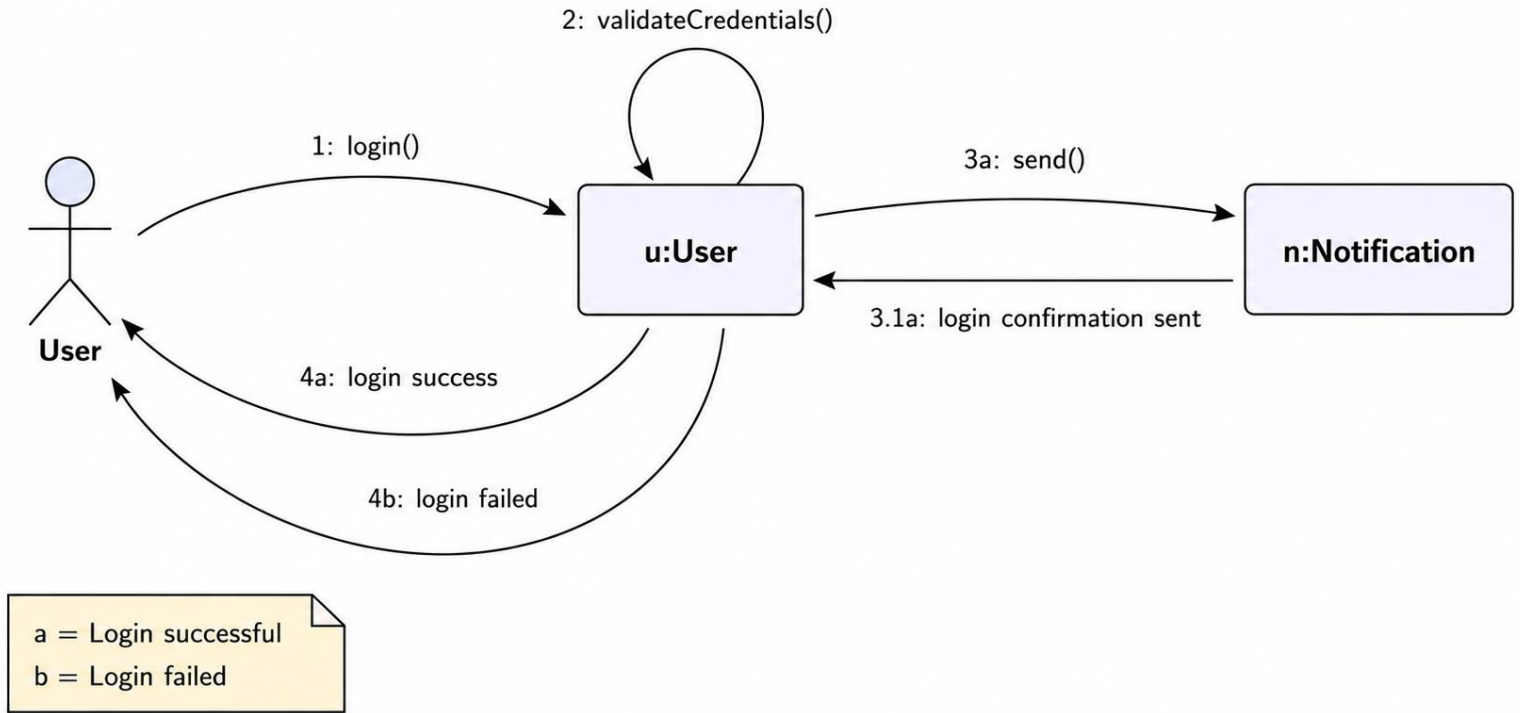
CC_PRS_40- Maintain Record of User Issues and Resolutions (Done by:Kevin Muho)



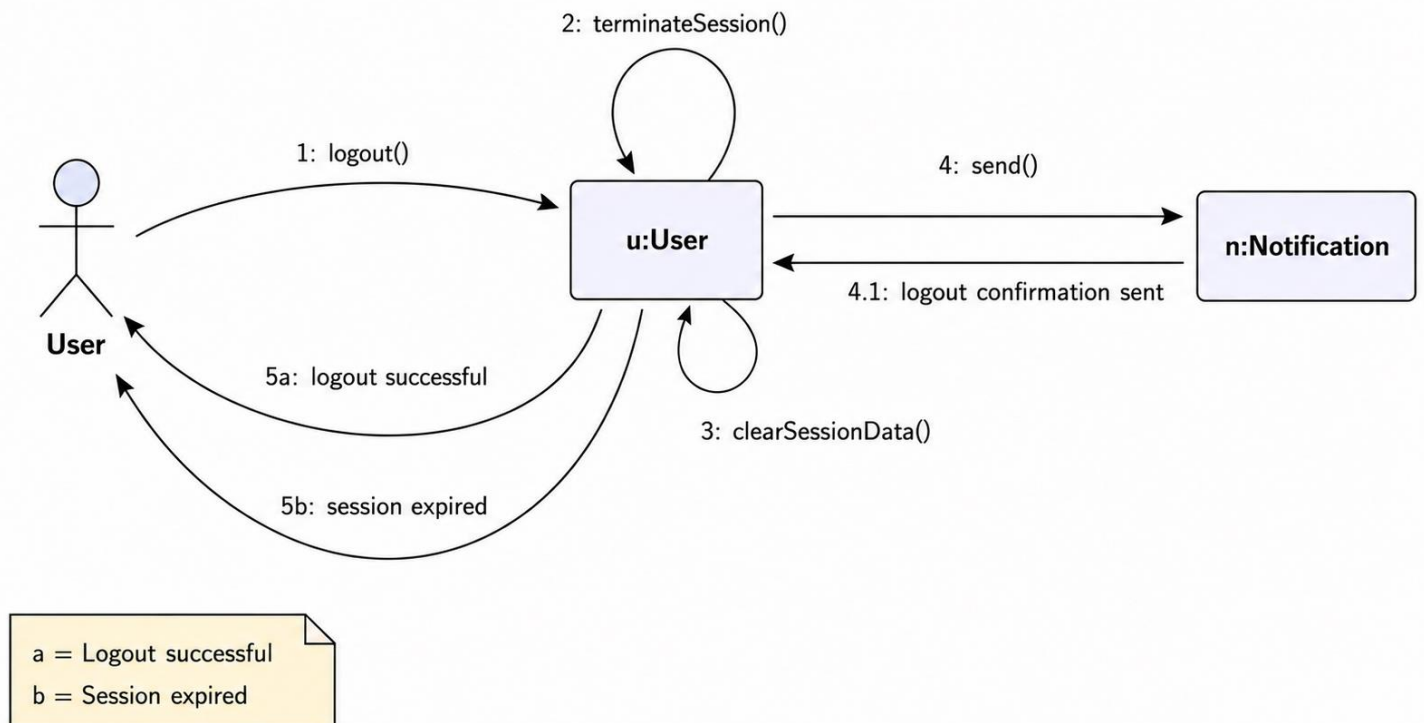
CC_PRS_41- Register Account (Done by:Kevin Muho)



CC_PRS_42 - Log In (Done by:Kevin Muho)

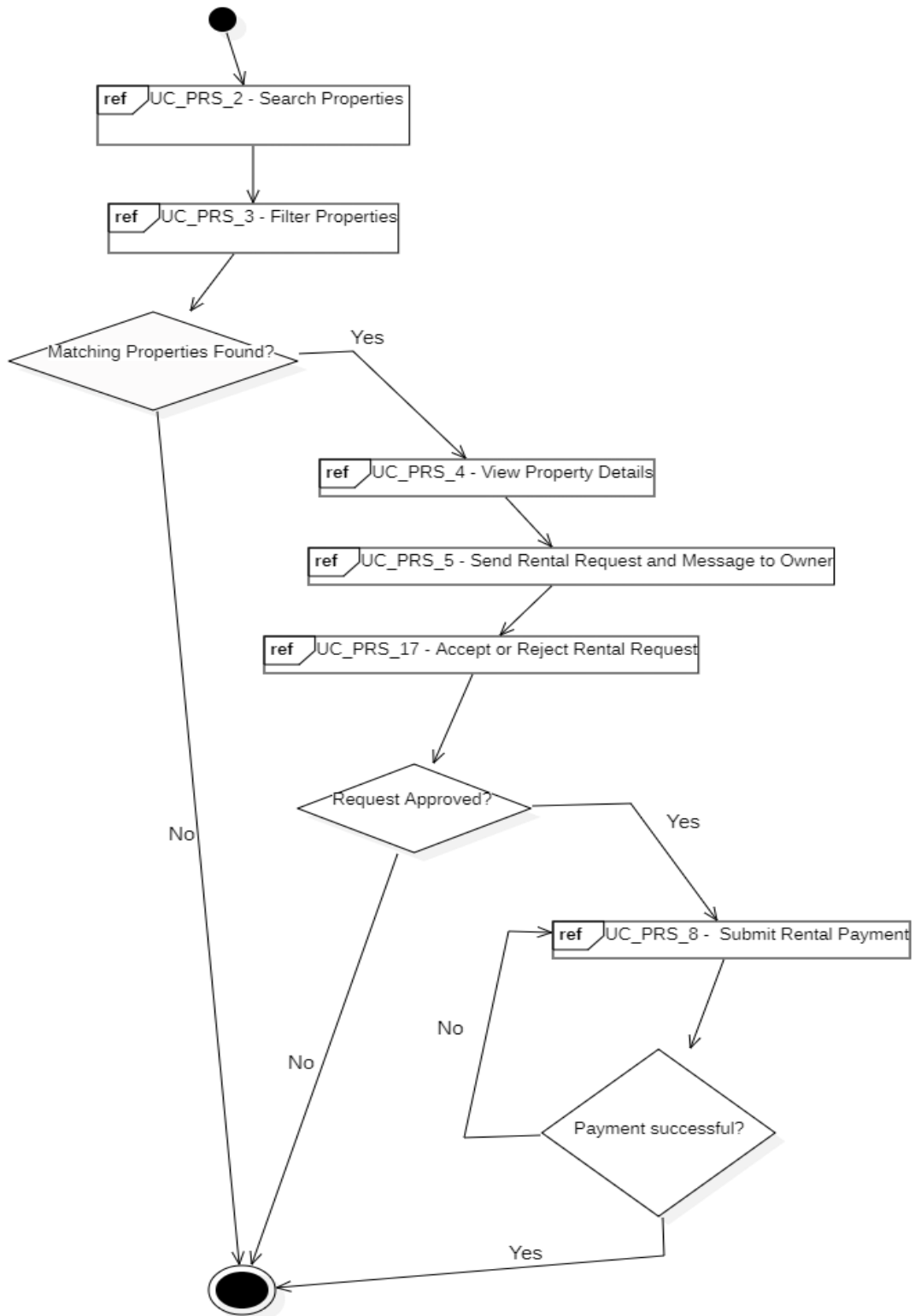


CC_PRS_43 – Log Out (Done by:Kevin Muho)

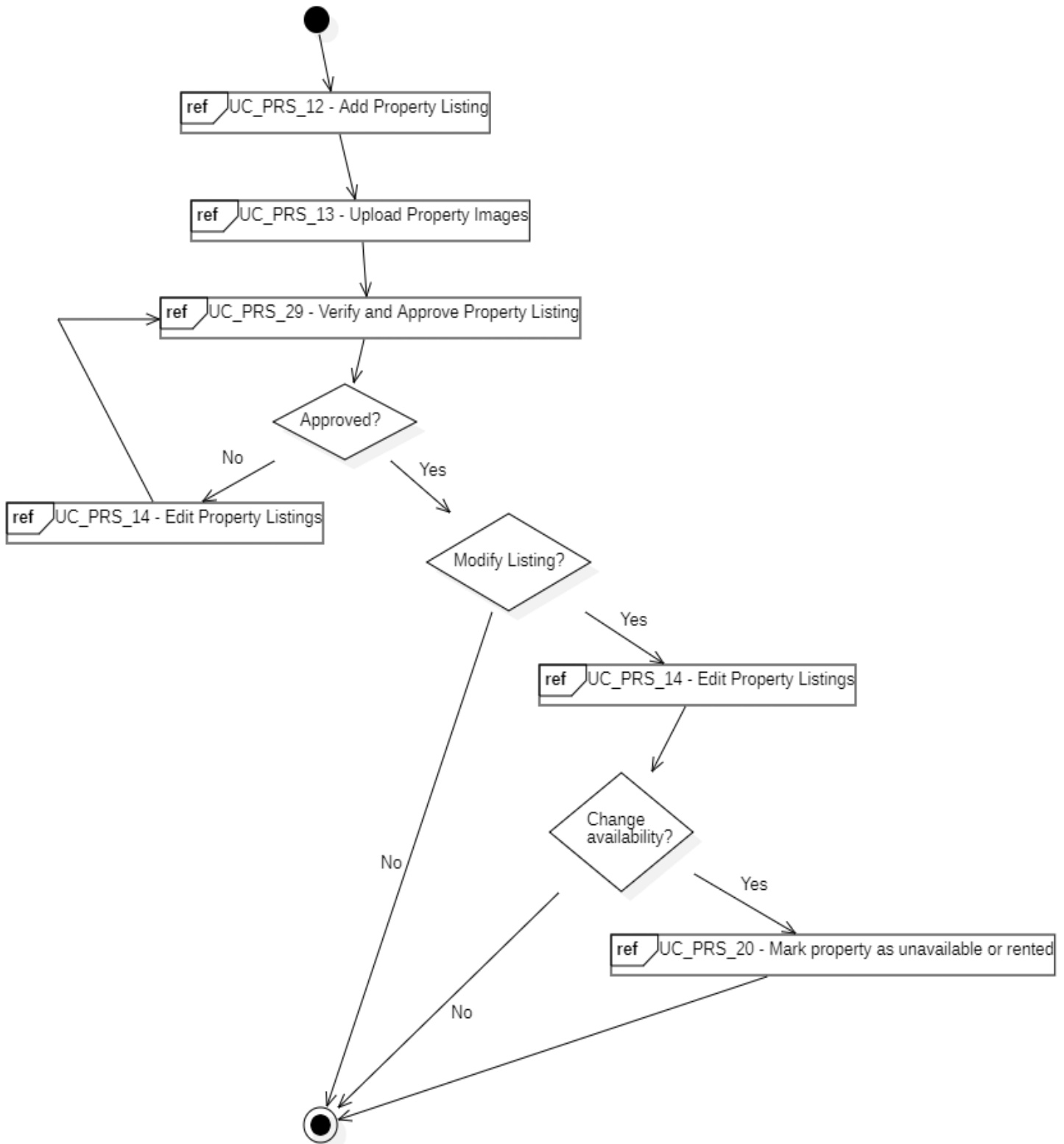


5.8IOD

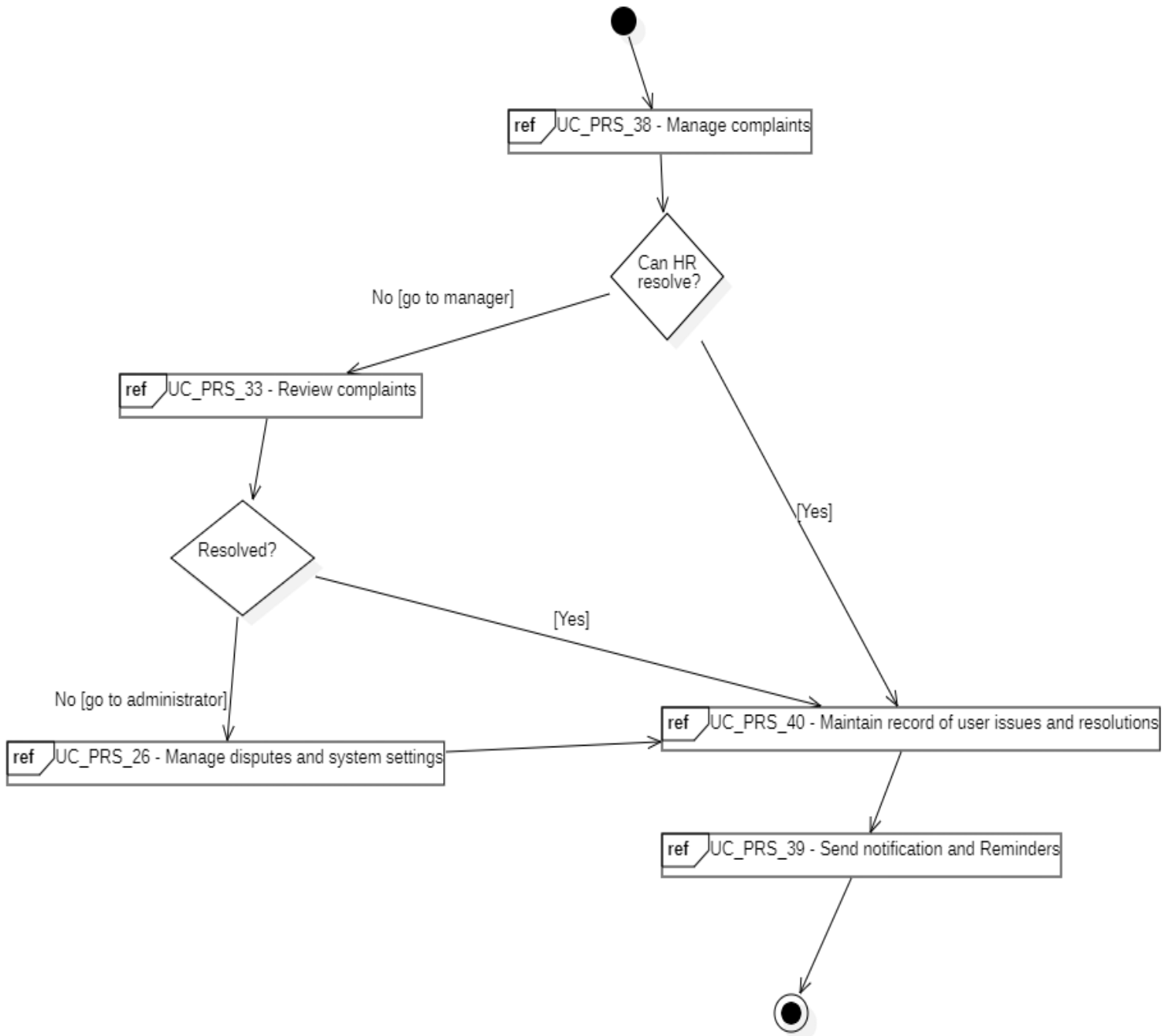
1. Property Booking Process (Done by:Kevin Muho)



2. Property Listing & Managment (Done by:Anisa Domi)

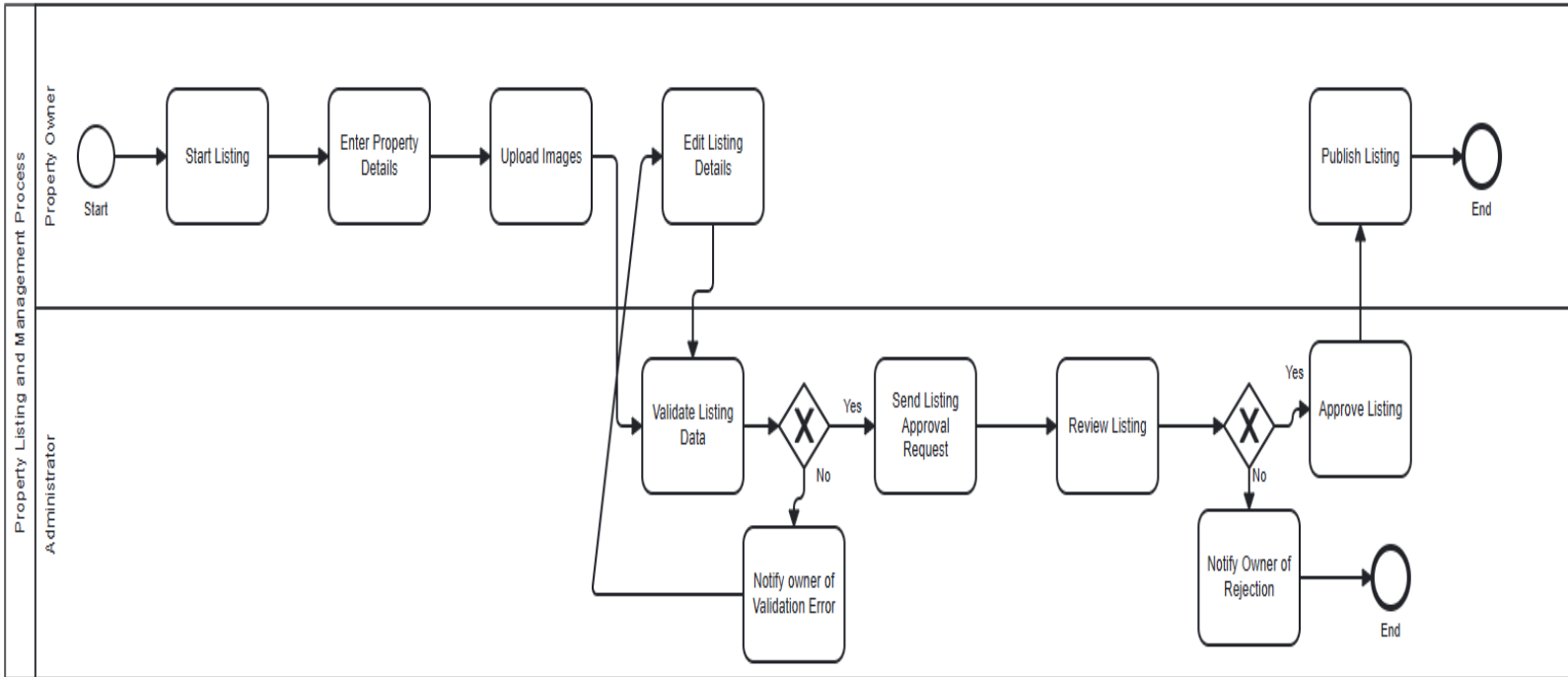


3. Complaint & Issue Management (Done by:Desard Lybeshari)



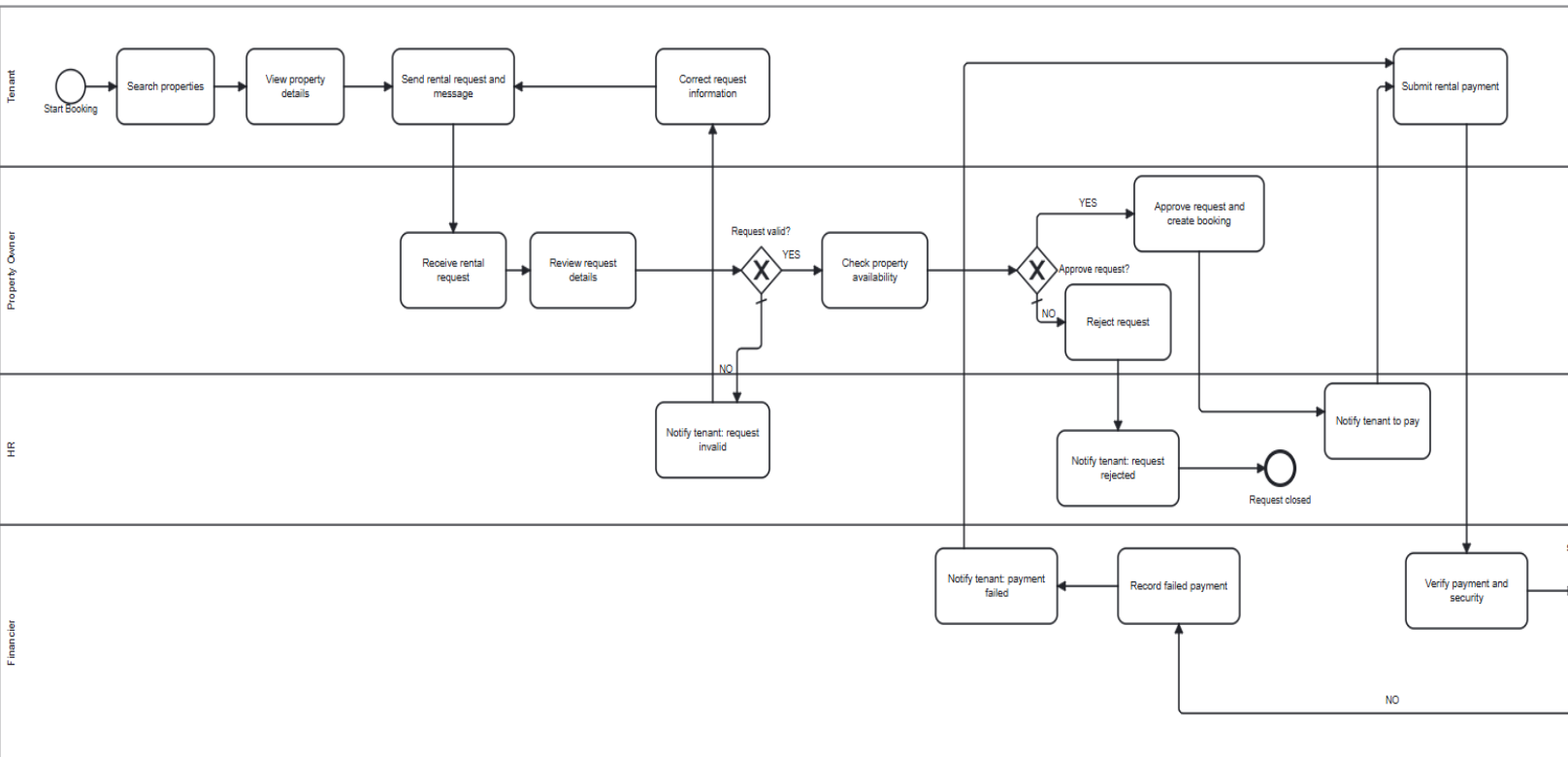
Property Rental System Requirement Specification
5.9 BPMN

BPMN - Property Listing & Management (Done by: Anisa Domi)

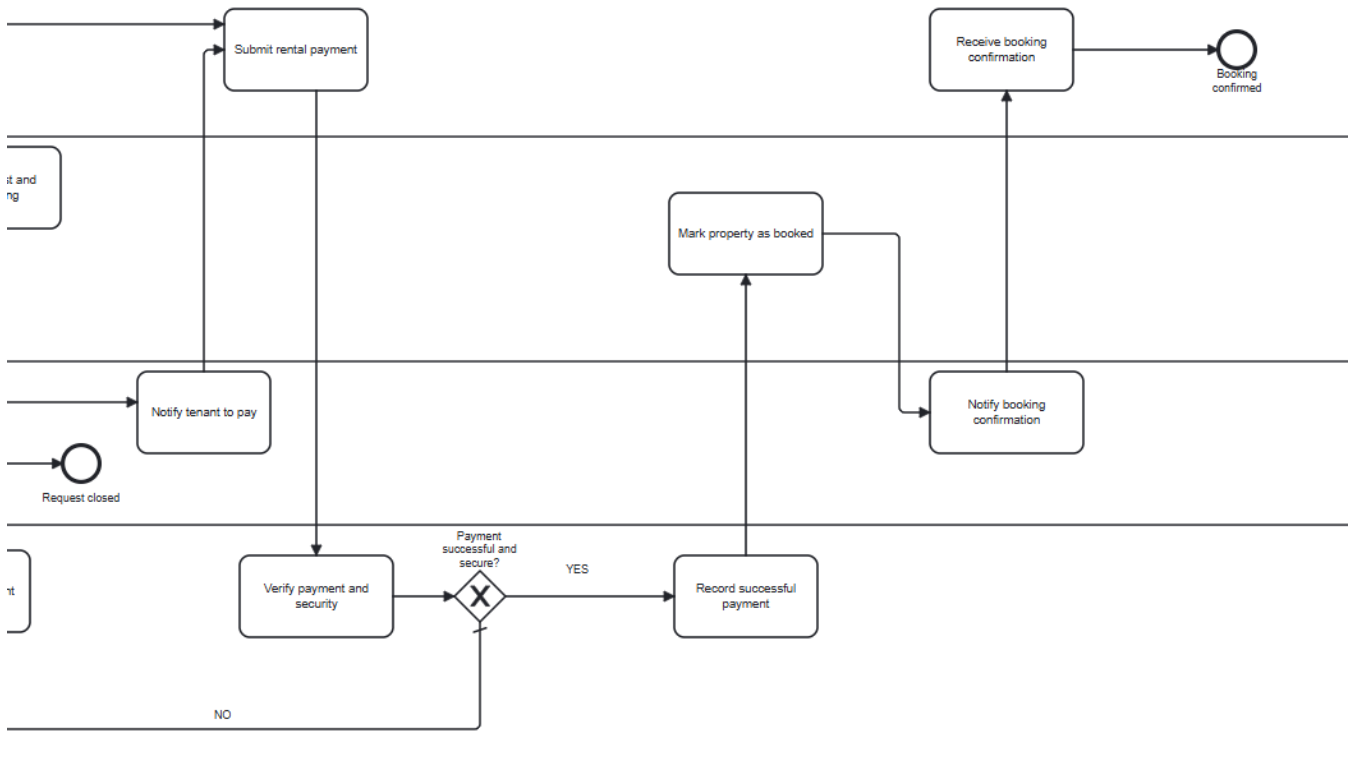


Property Rental System Requirement Specification

BPMN - Property Booking (Done by:Kevin Muho)

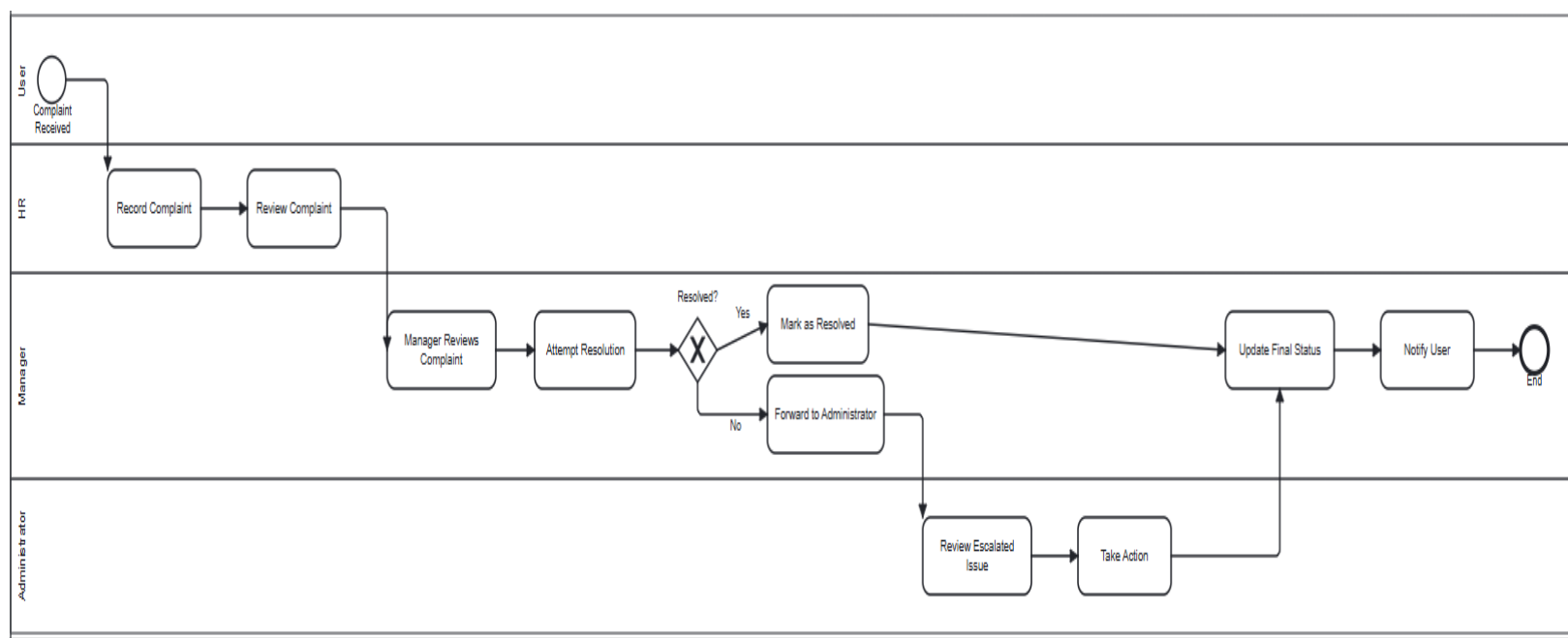


Continuation down here



Property Rental System Requirement Specification

BPMN – Complaint & Issue Management (Done by:Desard Lybeshari)



6.Design Patterns

Builder Pattern (Done by: Anisa Domi,Ledio Tafciu)

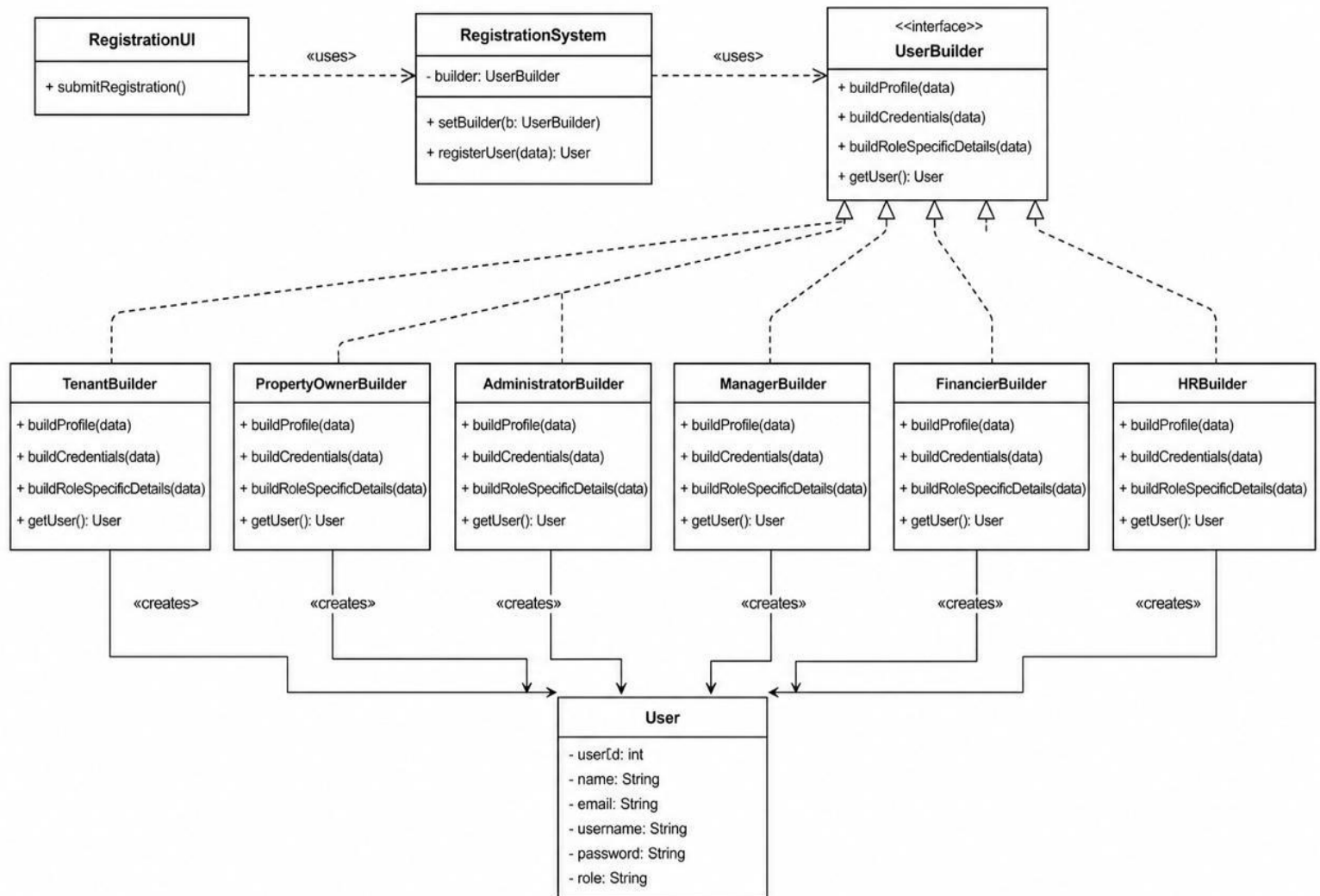
Justifying the design pattern presence → The Builder Pattern is used in the system to automatically construct different User objects during registration based on the selected role.

In this design:

The RegistrationSystem acts as the Director controlling the build sequence. UserBuilder defines the construction steps, each concrete builder assembles role-specific data, and the User product is returned fully built by the system.

For example:

In **FR41** when a user registers an account and system selects the correct builder for the chosen role, builds profile data, credentials, and role-specific details step by step and returns a complete User object without changing the registration flow.



Chain of Responsibility Pattern (Done by:Kevin Muho,Ornelt Lemellari, Matias Bali)

Justifying the design pattern presence→The Chain of Responsibility Pattern is used in the system to automatically process complaints through multiple handling stages without placing all complaint logic in one class.

In this design:

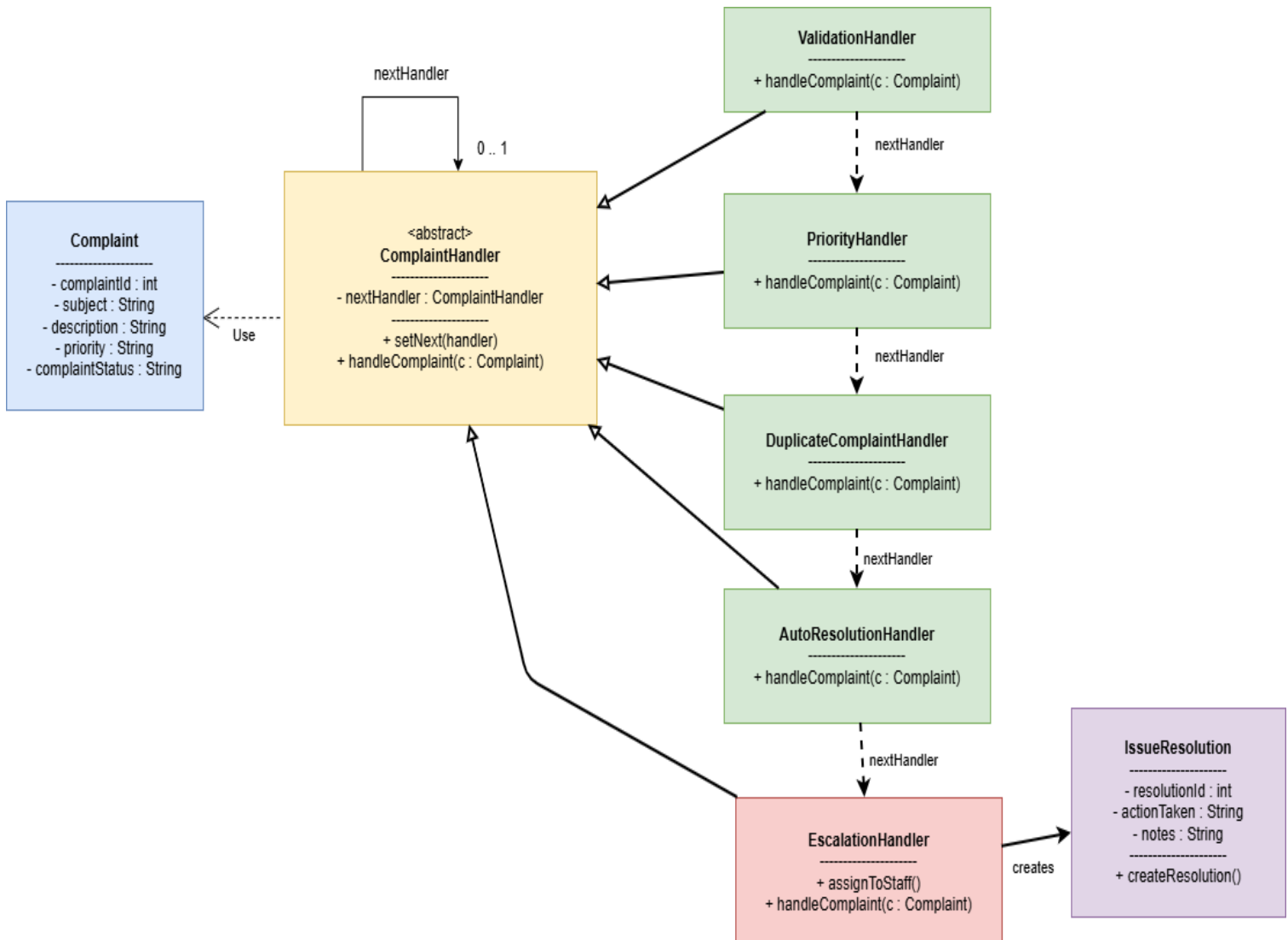
ComplaintHandler acts as the abstract handler and stores nextHandler.
ValidationHandler, PriorityHandler, DuplicateComplaintHandler, AutoResolutionHandler, and EscalationHandler act as concrete handlers. Complaint is the request object passed through the chain. IssueResolution stores the final resolution details.

For example:

In **FR38**, when HR manages complaints, the system first validates and categorizes the complaint automatically.

In **FR33**, if the complaint cannot be resolved automatically, it is escalated for manager review.

In **FR26**, unresolved or serious complaints can be escalated to the administrator as a dispute/system issue.



Observer Pattern (Done by:Desard Lybeshari,Haidi Toto)

Justifying the design pattern presence→ The Observer Pattern is used in the system to automatically notify users when important events occur, such as rental request updates or payment completion.

In this design:

RentalRequest and Payment act as Subjects because their status can change.

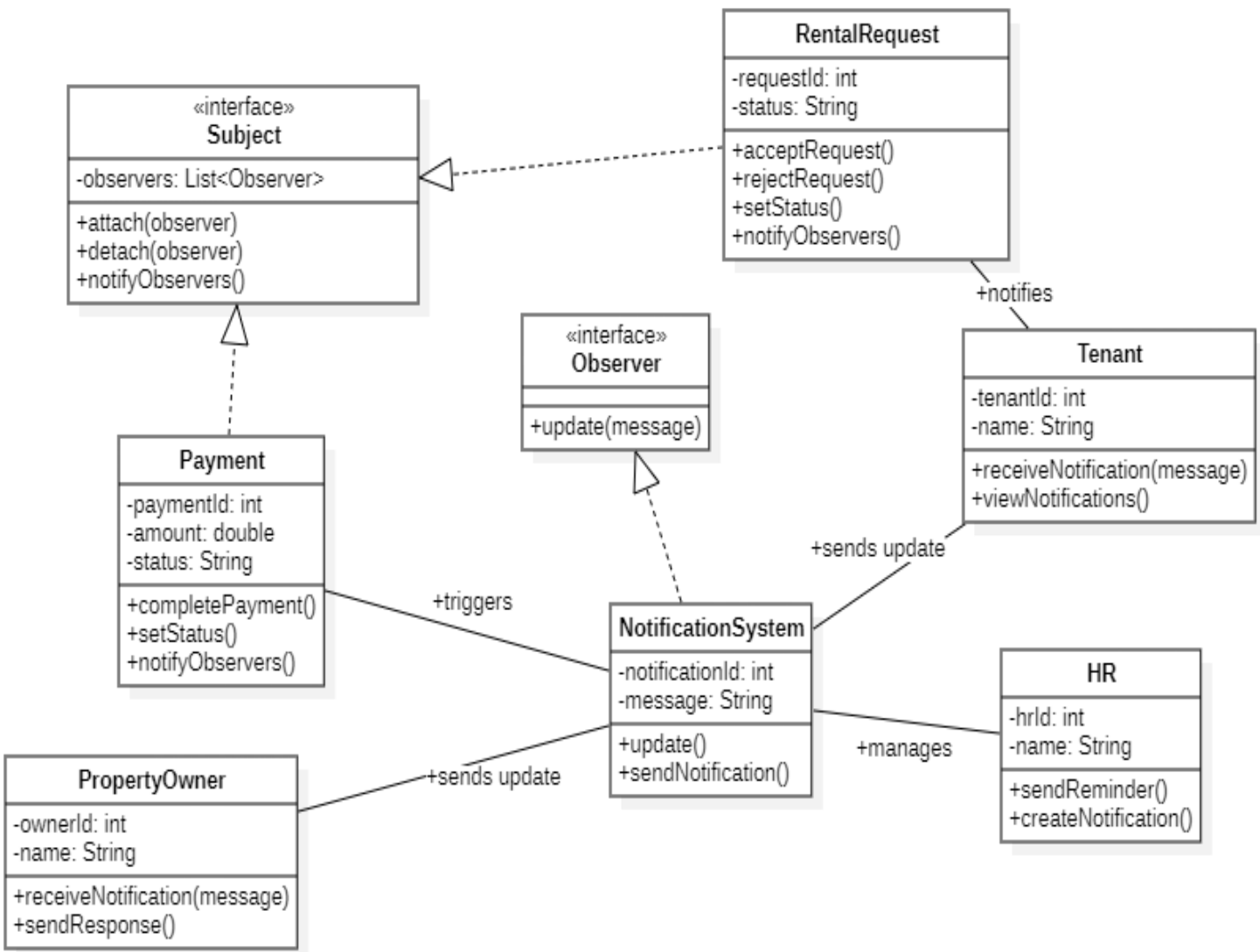
NotificationSystem acts as the Observer, listening for these changes.

Tenant and PropertyOwner receive notifications about approvals, rejections, payments, and reminders.

For example:

-In **FR17**, when a property owner accepts or rejects a rental request, the tenant is notified.

-In **FR8** and **FR39**, payment updates and reminders are also sent automatically.



Facade Pattern (Done by:Kevin Muho,Ornelt Lemellari,Matias Bali)

Justifying the design pattern presence→ The Facade Pattern is used in the system to simplify the property booking workflow by providing a unified interface for handling multiple booking-related operations through a single facade class.

In this design:

IBookingFacade acts as the facade interface that defines the booking operations.

BookingFacadeImpl acts as the concrete facade implementation that coordinates the subsystem classes.

Property, RentalRequest, Booking, Payment, and Notification act as subsystem classes responsible for specific booking operations.

Tenant acts as the client that interacts with the facade instead of directly communicating with all subsystem classes.

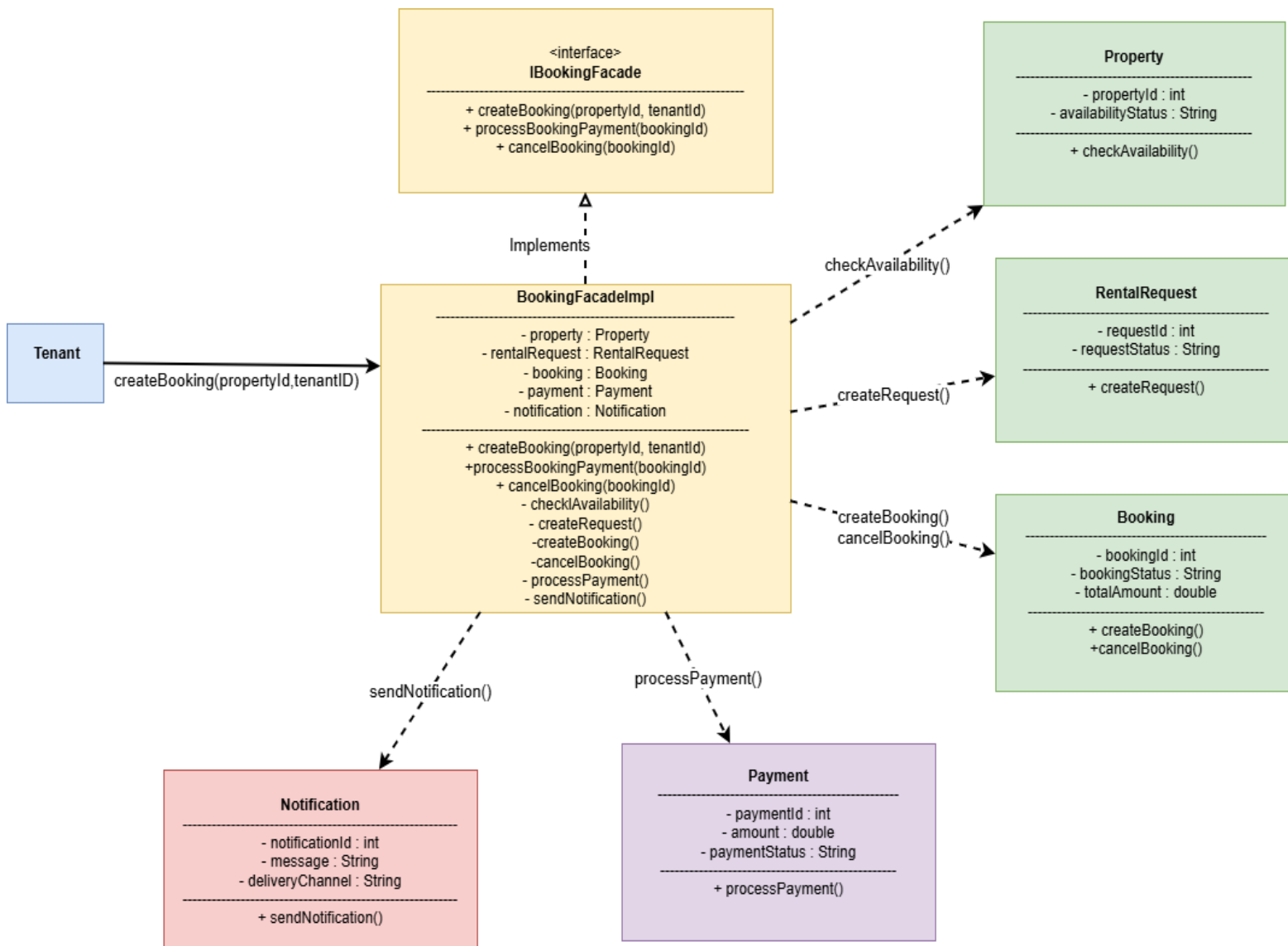
For example:

In **FR2** and **FR3**, the system checks property availability before creating a booking request.

In **FR5**, the system automatically creates a rental request through the facade.

In **FR17**, the system manages booking creation and status updates after request approval.

The facade also sends notifications automatically after important booking and payment events occur.



Adapter Pattern (Done by:Desard Lybeshari,Haidi Toto)

Justifying the design pattern presence→ The Adapter Pattern is used to connect the rental system with an external payment service without changing the internal system structure.

In this design:

PaymentServiceAdapter acts as a bridge between the system and ExternalPaymentService. The system sends payment data using the PaymentProcessor interface, while the adapter converts the request and response formats required by the external service.

This pattern focuses on *software structure and object collaboration*, since it shows how objects inside the system communicate during payment processing.

For example:

On **FR8 – Submit Rental Payment**, the system sends the payment request to the payment service. Also, the system receives the payment result and displays confirmation.

